NOTES FOR GEOMETRIC REPRESENTATION THEORY 1

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Contents

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The goal of this course is to study certain representations of complex semisimple Lie algebras, such as $\mathfrak{sl}_2(\mathbb{C})$, via geometric methods.

Unless otherwisely stated, all schemes and morphisms are defined over $\mathbb{C}$.

1. Introduction

1.1. **Recollection: semisimple Lie algebras and root systems.** Let $\mathfrak{g}$ be a complex semisimple Lie algebra. We exclude the case $\mathfrak{g} = 0$ to avoid some awkward words.

We fix a Borel subalgebra $\mathfrak{b} \subseteq \mathfrak{g}$ and a Cartan subalgebra $\mathfrak{h}$ contained in $\mathfrak{b}$. Let $\mathfrak{b}^-$ be the opposite Borel subalgebra such that $\mathfrak{b}^- \cap \mathfrak{b} = \mathfrak{h}$. Write $\mathfrak{n} := [\mathfrak{b}, \mathfrak{b}]$ and $\mathfrak{n}^- := [\mathfrak{b}^-, \mathfrak{b}^-]$.

Recall the following standard terminologies and facts (see [H1]).

- Elements in $\mathfrak{h}^* := \mathsf{Hom}_{\mathbb{C}}(\mathfrak{h}, \mathbb{C})$ are called *weights*.
- Nonzero weights that appear in $\mathfrak{g}$ are called *roots*. We have a *root decomposition*
$$\mathfrak{g} \simeq \mathfrak{h} \bigoplus (\bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha),$$
where Φ is the set of roots. Each root subspace $\mathfrak{g}_\alpha$ is 1-dimensional.
- Suppose that α, β and $\alpha + \beta$ are roots, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.
- Roots that appear in $\mathfrak{b}$ are called *positive roots*. We have $\Phi = \Phi^+ \sqcup -\Phi^+$.
- A positive root is *simple* if it is not the sum of other positive roots. Let $\Delta \subseteq \Phi$ be the subset of positive roots.
- Let $\mathfrak{h}^*_{\mathbb{R}} := \mathbb{R}\Phi \subseteq \mathfrak{h}^*$ be the $\mathbb{R}$-span of Φ. This gives a real structure on $\mathfrak{h}^*$. In other words, we have $\mathfrak{h}^*_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \xrightarrow{\simeq} \mathfrak{h}^*$.
- The *Killing form* of $\mathfrak{g}$ restricts to a non-degenerate symmetric bilinear form on $\mathfrak{h}$. Let $(-,-) : \mathfrak{h}^* \times \mathfrak{h}^* \to \mathbb{C}$ be the dual form. Its restriction on $\mathfrak{h}^*_{\mathbb{R}}$ is an inner form.
- The pair $(\mathfrak{h}^*_{\mathbb{R}}, \Phi)$ is a *root system*. Let W be the *Weyl group* of this root system, which is the reflection group acting on $\mathfrak{h}^*_{\mathbb{R}}$ with generators
$$s_\alpha(\lambda) := \lambda - 2\frac{(\alpha, \lambda)}{(\alpha, \alpha)}\alpha, \ \ \alpha \in \Phi$$
- Elements $s_\alpha \in W$ ($\alpha \in \Delta$) are called *simple reflections*. Recall that W is generated by simple reflections. For $w \in W$, the *length* $l(w)$ of w is the minimal length of presenting w as a product of simple reflections. Such a presentation is called a *reduced presentation* of w.
- There is a unique longest element in W, which we denote by w_0.
- For each root $\alpha \in \Phi$, we have the corresponding *coroot* $\check{\alpha} \in \mathfrak{h}$ characterized by the formula
$$\check{\alpha}(\lambda) = 2\frac{(\alpha, \lambda)}{(\alpha, \alpha)}, \ \ \forall \lambda \in \mathfrak{h}^*.$$
Moreover, $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}\check{\alpha}$.
- A weight $\lambda \in \mathfrak{h}^*$ is *integral* if $\check{\alpha}(\lambda) \in \mathbb{Z}$ for any root $\alpha \in \Phi$. Integral weights form a $\mathbb{Z}$-lattice $\Lambda \subseteq \mathfrak{h}^*$. In other words, we have $\Lambda \otimes_{\mathbb{Z}} \mathbb{C} \xrightarrow{\simeq} \mathfrak{h}^*$.
- An *integral* weight $\lambda \in \mathfrak{h}^*$ is *dominant* if $\check{\alpha}(\lambda) \geq 0$ for any positive root $\alpha \in \Phi^+$. Dominant integral weights form a semigroup $\Lambda^+ \subseteq \Lambda$.

- For weights $\lambda, \lambda' \in \Lambda$, we say $\lambda \le \lambda'$ if $\lambda' - \lambda$ is *positive*. In other words, $\lambda' - \lambda$ is a finite sum of positive roots.
- Let
$$\rho := \Big(\sum_{\alpha \in \Phi^+} \alpha\Big)/2$$
be the half sum of all positive roots. Recall that $\check{\alpha}(\rho) = 1$ for any simple positive root $\alpha \in \Delta$.
- The involution $\lambda \mapsto -\lambda$ on the root system $(\mathfrak{h}^*_{\mathbb{R}}, \Phi)$ determines (up to inner automorphisms) an involution τ on the semisimple Lie algebra $\mathfrak{g}$, which is called the *Cartan involution*. We have $\tau|_{\mathfrak{h}} = -1$ and $\tau(\mathfrak{b}) = \mathfrak{b}^-$.

1.2. **From finite dimentional representations to Verma modules.** Recall the following well-known result.

Theorem 1.2.1 (Weyl)**.** *The category $\mathfrak{g}-\mathsf{mod}_{\mathsf{fd}}$ of finite dimensional $\mathfrak{g}$-modules is semisimple. Moreover, there is a canonical bijection*

$$\Lambda^+ \xrightarrow{\simeq} \mathsf{Irr}(\mathfrak{g}-\mathsf{mod}_{\mathsf{fd}}),\ \lambda \mapsto L_\lambda \tag{1.1}$$

such that L_λ has highest weight λ.

The $\mathfrak{g}$-module L_λ is called the *Weyl module with highest weight λ.*

As an $\mathfrak{h}$-modules, we have

$$\bigoplus_{\mu \le \lambda, \mu \in \Lambda} \mathbb{C}_\mu \otimes \mathsf{Hom}_{\mathfrak{h}}(\mathbb{C}_\mu, L_\lambda) \xrightarrow{\simeq} L_\lambda,$$

where each direct summand in the LHS is a *weight subspace* of L_λ. The following question is natural.

Question 1.2.2. How to calculate the dimension of the μ-weight subspace of L_λ, i.e., $\dim(\mathsf{Hom}_{\mathfrak{h}}(\mathbb{C}_\mu, L_\lambda))$?

Example 1.2.3. For $\mathfrak{g} = \mathfrak{sl}_2$, let $\mathfrak{b}$ and $\mathfrak{h}$ be its standard Borel and Cartan subalgebra. We have short exact sequences

$$0 \to \mathfrak{h} \to \mathbb{C}^2 \xrightarrow{\Sigma} \mathbb{C} \to 0$$

and

$$0 \to \mathbb{C} \xrightarrow{\Delta} \mathbb{C}^2 \to \mathfrak{h}^* \to 0.$$

We identify $\mathfrak{h}^*$ with $\mathbb{C}$ such that $(\lambda_1, \lambda_2) \in \mathbb{C}^2$ is sent to $l := \lambda_2 - \lambda_1$ by the map $\mathbb{C}^2 \to \mathfrak{h}^* \simeq \mathbb{C}$. Note that:

- A weight $\lambda \in \mathfrak{h}^*$ is integral iff the corresponding number l is integral;
- A weight $\lambda \in \mathfrak{h}^*$ is dominant iff the corresponding number l is non-negative;
- For weights $\lambda, \lambda' \in \mathfrak{h}^*$ and the corresponding numbers l, l', we have $\lambda \le \lambda'$ iff $l' - l \in 2\mathbb{Z}^+$.

The vector space L_λ can be identified with the space of homogeneous polynomials $f \in \mathbb{C}[x_1, x_2]$ with degree l, and the $\mathfrak{sl}_2$-action on this vector space is given by changing coordinates. It follows that weights of L_λ correspond to numbers $l - 2d$, $d = 0, \cdots, l$, and each weight subspace is 1-dimensional. Namely, $x_1^{l-d} x_2^d$ is a generator to the weight subspace corresponding to $l - 2d$.

To describe the solution for general $\mathfrak{g}$, we introduce the *character* of $V \in \mathfrak{g}\text{-}\mathsf{mod}_{\mathsf{fd}}$ by the formula

$$\mathsf{Ch}(V) := \sum_{\mu \in \mathfrak{h}^*} \dim(\mathsf{Hom}_{\mathfrak{h}}(\mathsf{C}_\mu, V)) T^\mu,$$

which is an element in the polynomial algebra. Note that we have

$$\mathsf{Ch}(V \bigoplus V') = \mathsf{Ch}(V) + \mathsf{Ch}(V'), \ \ \mathsf{Ch}(V \otimes V') = \mathsf{Ch}(V) \cdot \mathsf{Ch}(V').$$

Theorem 1.2.4 (Weyl character formula)**.** *We have*

$$\mathsf{Ch}(L_\lambda) \cdot \sum_{w \in W} (-1)^{l(w)} T^{w(\rho)} = \sum_{w \in W} (-1)^{l(w)} T^{w(\lambda+\rho)}.$$

In other words, we can calculate $\mathsf{Ch}(L_\lambda)$ by expanding

$$\sum_{w \in W} (-1)^{l(w)} \frac{T^{w(\lambda+\rho)}}{\sum_{w' \in W} (-1)^{l(w')} T^{w'(\rho)}}, \tag{1.2}$$

as a *formal infinite* sum of T^μ ($\mu \in \Lambda$), which turns out to be a finite sum.

Remark 1.2.5. Note that the denominator does not depend on λ.

The following example shows that each term in the alternating sum (1.2) is a genuine infinite sum.

Example 1.2.6. For $\mathfrak{g} = \mathfrak{sl}_2$, we have

$$\mathsf{Ch}(L_\lambda) = \frac{T^{l+1} - T^{-l-1}}{T - T^{-1}} = T^l + T^{l-2} + \cdots T^{-l}.$$

However, both

$$\frac{T^{l+1}}{T - T^{-1}} = T^l + T^{l-2} + \cdots$$

and

$$\frac{T^{-l-1}}{T - T^{-1}} = T^{-l-2} + T^{-l-4} + \cdots$$

are infinite sums.

This raises the following natural question.

Question 1.2.7. Is there a representation theoretic intepretation for the term

$$\frac{T^{w(\lambda+\rho)}}{\sum_{w \in W} (-1)^{l(w)} T^{w(\rho)}}?$$

For example, for $\mathfrak{g} = \mathfrak{sl}_2$, we can ask the following question.

Question 1.2.8. Can we find (infinite dimensional) $\mathfrak{sl}_2$-modules $N \subseteq M$ with

$$\mathsf{Ch}(M) = \frac{T^{l+1}}{T - T^{-1}}, \ \ \mathsf{Ch}(N) = \frac{T^{-l-1}}{T - T^{-1}}$$

such that $L_\lambda \simeq M/N$?

It turns out there is a positive answer to Question 1.2.7.

Theorem 1.2.9 (Bernstein–Gelfand–Gelfand resolution)**.** *There is a canonical resolution*

$$0 \to M_{w_0 \cdot \lambda} \cdots \to \bigoplus_{\ell(w)=2} M_{w \cdot \lambda} \to \bigoplus_{\ell(w)=1} M_{w \cdot \lambda} \to M_\lambda \to L_\lambda \to 0.$$

In above, $M_{w\cdot\lambda}$ is the *Verma module* with highest weight

$$w \cdot \lambda := w(\lambda + \rho) - \rho,$$

which is an *infinite-dimensional* $\mathfrak{g}$-module.

The upshot is: we need to study infinite-dimensional representations to understand the finite-dimensional ones better.

Warning 1.2.10 (or features)**.** The category $\mathfrak{g}\text{-}\mathsf{mod}$ of (all) $\mathfrak{g}$-modules is not semisimple. A $\mathfrak{g}$-module may fail to be integrable.

Example 1.2.11. As we will soon see, for $\mathfrak{g} = \mathfrak{sl}_2$, the short exact sequence

$$0 \to M_{w_0\cdot\lambda} \to M_\lambda \to L_\lambda \to 0$$

does not split. Also, M_λ is not an (algebraic) representation for SL_2.

1.3. **Recollection: enveloping algebra.** Recall that for any Lie algebra $\mathfrak{g}$, we have a canonical equivalence

$$\mathfrak{g}\text{-}\mathsf{mod} \simeq U(\mathfrak{g})\text{-}\mathsf{mod}, \tag{1.3}$$

where $U(\mathfrak{g})$ is the *universal enveloping algebra* of $\mathfrak{g}$, which is an associative (but not commutative) $\mathbb{C}$-algebra. Namely, let

$$T(\mathfrak{g}) := \bigoplus_{i=0}^{\infty} \mathfrak{g}^{\otimes i}$$

be the tensor algebra for the underlying vector space $\mathfrak{g}$. Consider the two-sided ideal $I \subseteq T(\mathfrak{g})$ generated by elements

$$X \cdot Y - Y \cdot X - [X, Y] \in T(\mathfrak{g}).$$

One can check that

$$U(\mathfrak{g}) := T(\mathfrak{g})/I$$

fits into the desired equivalence (1.3).

For future reference, let us recall that the (non-negative) grading on $T(\mathfrak{g})$ induces the *PBW filtration* on $U(\mathfrak{g})$:

$$\mathsf{F}^{\le n}U(\mathfrak{g}) := \mathrm{Im}(\bigoplus_{i=0}^{n} \mathfrak{g}^{\otimes i} \to U(\mathfrak{g})).$$

One can check that this makes $U(\mathfrak{g})$ an *almost commutative filtered associative algebra*, i.e.,

$$\mathsf{F}^{\le m}U(\mathfrak{g}) \cdot \mathsf{F}^{\le n}U(\mathfrak{g}) \subseteq \mathsf{F}^{\le m+n}U(\mathfrak{g})$$

and

$$[\mathsf{F}^{\le m}U(\mathfrak{g}), \mathsf{F}^{\le n}U(\mathfrak{g})] \subseteq \mathsf{F}^{\le m+n-1}U(\mathfrak{g}).$$

In particular,

$$\mathsf{gr}^\bullet U(\mathfrak{g}) := \bigoplus_{n=0}^{\infty} \mathsf{gr}^n U(\mathfrak{g}) := \bigoplus_{n=0}^{\infty} \mathsf{F}^{\le n}U(\mathfrak{g})/\mathsf{F}^{\le n-1}U(\mathfrak{g})$$

is a graded commutative algebra.

Theorem 1.3.1 (Poincare–Birkhoff–Witt)**.** *The map*

$$\mathfrak{g} \to \mathsf{F}^{\le 1}U(\mathfrak{g}) \to \mathsf{gr}^1 U(\mathfrak{g}) \to \mathsf{gr}^\bullet U(\mathfrak{g})$$

induces an isomorphism

$$\mathsf{Sym}^\bullet(\mathfrak{g}) \xrightarrow{\simeq} \mathsf{gr}^\bullet U(\mathfrak{g}).$$

The construction $\mathfrak{g} \mapsto U(\mathfrak{g})$ is functorial. In other words, for any Lie algebra homomorphism $\mathfrak{g} \to \mathfrak{g}'$, we have a ring homomorphism $U(\mathfrak{g}) \to U(\mathfrak{g}')$ such that the following diagram commute

$$\begin{tikzcd} \mathfrak{g}'\text{–}\mathsf{mod} \arrow[r, "\mathsf{res}"] \arrow[d, "\simeq"] & \mathfrak{g}\text{–}\mathsf{mod} \arrow[d, "\simeq"] \\ U(\mathfrak{g}')\text{–}\mathsf{mod} \arrow[r, "\mathsf{res}"] & U(\mathfrak{g})\text{–}\mathsf{mod}. \end{tikzcd}$$

It follows that the top horizontal functor admits a left adjoint

$$\mathsf{Ind}_{\mathfrak{g}}^{\mathfrak{g}'} : \mathfrak{g}\text{–}\mathsf{mod} \to \mathfrak{g}'\text{–}\mathsf{mod}, \quad M \mapsto U(\mathfrak{g}') \underset{U(\mathfrak{g})}{\otimes} M$$

Example 1.3.2. When $\mathfrak{g}' = 0$, we obtain a functor

$$\mathfrak{g}\text{–}\mathsf{mod} \to \mathsf{Vect}, \quad M \mapsto \mathbb{C} \underset{U(\mathfrak{g})}{\otimes} M =: M_{\mathfrak{g}}.$$

The vector space $M_{\mathfrak{g}}$ is called the *coinvariance* of M.

1.4. **Verma modules.**

Construction 1.4.1. Consider the following diagram of Lie algebras

$$\begin{tikzcd} & \mathfrak{b} \arrow[dl] \arrow[dr] & \\ \mathfrak{g} & & \mathfrak{b}/[\mathfrak{b},\mathfrak{b}] & \mathfrak{h}. \arrow[l, "\simeq"'] \end{tikzcd}$$

For any $\lambda \in \mathfrak{h}^*$, by restriction, we view $\mathbb{C}_\lambda$ as a 1-dimensional $\mathfrak{b}$-module. By induction, we obtain a $\mathfrak{g}$-module

$$M_\lambda := \mathsf{Ind}_{\mathfrak{b}}^{\mathfrak{g}}(\mathbb{C}_\lambda) = U(\mathfrak{g}) \underset{U(\mathfrak{b})}{\otimes} \mathbb{C}_\lambda.$$

Exercise 1.4.2. (Problem 1, Homework 1) Show that

$$U(\mathfrak{g}) \simeq U(\mathfrak{n}^-) \otimes U(\mathfrak{b})$$

as $(U(\mathfrak{n}^-), U(\mathfrak{b}))$-bimodules. Deduce that

$$M_\lambda \simeq U(\mathfrak{n}^-)$$

as $\mathfrak{n}^-$-module.

Exercise 1.4.3. (Problem 2, Homework 1) Show that M_λ is a semisimple $\mathfrak{h}$-module and each weight subspace is finite-dimensional.

In particular, we can talk about *weights* and *weight subspaces* of M_λ.

Exercise 1.4.4. (Problem 3, Homework 1) Show that λ is the unique highest weight of M_λ, and the image of the composition

$$\mathbb{C} \xrightarrow{\subseteq} U(\mathfrak{n}^-) \xrightarrow{\simeq} M_\lambda$$

is the λ-weight subspace.

Definition 1.4.5. We call M_λ the *Verma module* with highest weight λ.

The goal of this course is to study Verma modules and the subcategory $\mathcal{O} \subseteq \mathfrak{g}\text{–}\mathsf{mod}$ generated by them.

Exercise 1.4.6. (Problem 4, Homework 1) Deduce Theorem 1.2.4 from Theorem 1.2.9. You can achieve this *without* using the Weyl denominator formula. Hint: find $\mathsf{Ch}(U(\mathfrak{n}^-))$ by studying the case $\lambda = 0$.

1.5. **Geometric incarnations.** Let G be an algebraic group with Lie algebra $\mathfrak{g}$. Let $H \subseteq B \subseteq G$ be the Cartan and Borel subgroups corresponding to $\mathfrak{h} \subseteq \mathfrak{b}$. The quotient G/B is a complex variety, called the *flag variety*.

In future lectures, for each character λ of H (which in particular is an integral weight of $\mathfrak{h}$), we will construct a canonical G-equivariant line bundle $\mathcal{O}(\lambda)$ on G/B. Roughly speaking, we consider the trivial line bundle $\mathcal{O}_G$ on G, equipped with the trivial (G, B)-equivariant structure, but twist the right B-equivariant structure by the character λ. By descent theory, this indeed gives a G-equivariant line bundle on G/B.

Example 1.5.1. For $G = \mathsf{SL}_2$, the standard action of G on $\mathbb{A}^2 \smallsetminus 0$ induces a transitive action of G on $\mathbb{P}^1$. The stabilizer of $\infty \in \mathbb{P}^1$ for this action is the standard Borel subgroup $B \subseteq G$. It follows that $G/B \simeq \mathbb{P}^1$.

Via this isomorphism, for a character λ with

$$\lambda(\mathsf{diag}(t, t^{-1})) = t^n,$$

the line bundle $\mathcal{O}(\lambda)$ on G/B corresponds to the line bundle $\mathcal{O}(n)$ on $\mathbb{P}^1$.

Theorem 1.5.2 (Borel–Weil)**.** *For dominant λ, we have a canonical isomorphism of G-modules*

$$\Gamma(G/B, \mathcal{O}(\lambda)) \simeq L_\lambda^* \simeq L_{-w_0(\lambda)}. \tag{1.4}$$

This motivates the following question.

Question 1.5.3. Is it possible to realize the Verma module M_λ as the global section of some objects over G/B?

We should not restrict ourselves to vector bundles on G/B because there are very few of them. Experiences in algebraic geometry suggest we should try (quasi-coherent) sheaves. However, we should first answer the following preliminary question.

Question 1.5.4. For a quasi-coherent $\mathcal{O}$-module $\mathcal{F}$ over G/B, is there some additional structure on $\mathcal{F}$ that makes $\Gamma(G/B, \mathcal{F})$ a $\mathfrak{g}$-module?

Note that in (1.4), the G-action on the LHS comes from the G-equivariant structure on $\mathcal{O}(\lambda)$. However, the Verma module M_λ is *never* a G-module.

Exercise 1.5.5. (Problem 5, Homework 1) Show that M_λ is not an algebraic representation of G.

Nevertheless, there is a beautiful answer Qusetion 1.5.4, motivated by the following observation.

Theorem 1.5.6 (Bott)**.** *Let $\mathcal{T}_{G/B}$ be the tangent bundle on G/B. We have a canonical isomorphism of Lie algebras:*

$$\Gamma(G/B, \mathcal{T}_{G/B}) \simeq \mathfrak{g}.$$

In other words, the Lie algebra of global vector fields on G/B is isomorphic to $\mathfrak{g}$.

We will prove Theorem 1.5.6 in future lectures. For now, please convince yourself using the SL_2-case.

Exercise 1.5.7. (Problem 6, Homework 1) Find an isomorphism of Lie algebras

$$\Gamma(\mathbb{P}^1, \mathcal{T}_{\mathbb{P}^1}) \simeq \mathfrak{sl}_2.$$

Theorem 1.5.6 suggests that we should consider sheaves $\mathcal{F}$ over G/B equipped with an action of the tangent sheaf $\mathcal{T}_{G/B}$. One can call them *sheaves with connections*, but the standard terminology is *$\mathcal{D}$-modules*. Here the letter "D" stands for "differential".

In future lectures, for each Verma module M_λ, we will attach a (twisted) $\mathcal{D}$-module $\mathcal{F}_\lambda$ on G/B such that

$$\Gamma(G/B, \mathcal{F}_\lambda) \simeq M_\lambda$$

as a $\mathfrak{g}$-modules. Moreover, the BGG resolution (Theorem 1.2.9) will be proved by finding a resolution of $\mathcal{O}(\lambda)$ by the (twisted) $\mathcal{D}$-modules $\mathcal{F}_{w\cdot\lambda}$, which itself is a sheaf theoretic incarnation of the Bruhat decomposition

$$G/B = \bigcup_{w\in W} BwB/B.$$

As a consequence, we obtain a *geometric* intepretation of the Weyl character formula.

The ultimate goal of this course is to relate $\mathfrak{g}$-modules with $\mathcal{D}$-modules on G/B. This is known as the *localization theory* for semisimple Lie algebras.

2. Category $\mathcal{O}$

2.1. Definition of category $\mathcal{O}$.

Definition 2.1.1. We define BGG's category $\mathcal{O}$ to be the full subcategory of $\mathfrak{g}\text{-}\mathsf{mod}$ that contains $\mathfrak{g}$-modules M satisfying the following conditions:

(i) As a $\mathfrak{g}$-module, M is finitely generated.
(ii) As an $\mathfrak{h}$-module, M is semisimple. In other words, M is a *weight module*.
(iii) As an $\mathfrak{n}$-module, M is a union of finite-dimensional submodules.

Lemma 2.1.2. *The Verma module M_λ is contained in $\mathcal{O}$.*

Proof. The Verma module M_λ satisfies condition (i) and (ii) respectively by Exercise 1.4.2 and Exercise 1.4.3.

It remains to prove that M_λ satisfies condition (iii). Let v_λ be a nonzero highest weight vector of M_λ. By Exercise 1.4.4,

$$M_\lambda = \bigcup_i \mathsf{F}^i U(\mathfrak{g}) \cdot v_\lambda.$$

Each $\mathsf{F}^i U(\mathfrak{g}) \cdot v_\lambda$ is finite dimensional. Hence we only need to show these subspaces are $\mathfrak{n}$-stable. For $u \in \mathsf{F}^i U(\mathfrak{g})$ and $x \in \mathfrak{n}$ we have

$$x \cdot (u \cdot v_\lambda) = u \cdot (x \cdot v_\lambda) + [x, u] \cdot v_\lambda.$$

Note that $\mathfrak{h}$ acts on $x \cdot v_\lambda$ with weight strictly greater than λ. Hence $x \cdot v_\lambda = 0$. This implies the claim because

$$[x, u] \in [\mathfrak{g}, \mathsf{F}^i U(\mathfrak{g})] \subset \mathsf{F}^i U(\mathfrak{g}) \tag{2.1}$$

□

In fact, we have the following stronger result.

Lemma 2.1.3. *The $\mathfrak{n}$-action on M_λ is locally nilpotent.*

Proof. We only need to show $\mathfrak{n}$ acts nilpotently on each weight vector v of M_λ. This follows from the fact that for $x \in \mathfrak{n}$, the element $x \cdot v$ is a weight vector with weight strictly greater than the weight of v.

□

The following results are obvious.

Lemma 2.1.4. *The subcategory $\mathcal{O} \subseteq \mathfrak{g}\text{-}\mathsf{mod}$ is stable under taking finite direct sums and subquotients.*

Lemma 2.1.5. *Let $\mathfrak{g}\text{-}\mathsf{mod}^{\mathsf{wt}} \subseteq \mathfrak{g}\text{-}\mathsf{mod}$ be the full subcategory of weight modules. Then the subcategory $\mathcal{O} \subseteq \mathfrak{g}\text{-}\mathsf{mod}^{\mathsf{wt}}$ is stable under taking extensions.*

Corollary 2.1.6. *The category $\mathcal{O}$ is an abelian category.*

Warning 2.1.7. The subcategory $\mathcal{O} \subseteq \mathfrak{g}\text{-}\mathsf{mod}$ is *not* stable under taking extensions.

Exercise 2.1.8. (Problem 7, Homework 1) For $\mathfrak{g} = \mathfrak{sl}_2$, find an extension of two Verma modules inside $\mathfrak{g}\text{-}\mathsf{mod}$ that is not a weight module. Hint: $\mathsf{Ind}_{\mathfrak{b}}^{\mathfrak{g}}$.

Proposition 2.1.9. *Each object in $\mathcal{O}$ is a quotient of a finite successive extension of some Verma modules.*

Proof. Let $M \in \mathcal{O}$ be an object. By condition (i), M is generated as a $\mathfrak{g}$-module by a finite-dimensional subspace $M_0 \subseteq M$. We may enlarge M_0 such that it is a finite dimensional $\mathfrak{h}$-module. In particular, there are finitely many weights in M_0.

Note that each weight in $U(\mathfrak{h}) \cdot M_0 = U(\mathfrak{n}) \cdot M_0$ is greater or equal to a weight in M_0. Since M is a highest weight module, there are finitely many weights satisfying the above property. Since each weight subspace of M is finite-dimensional, we see that $U(\mathfrak{h}) \cdot M_0$ is finite-dimensional. Therefore we can replace M_0 by $U(\mathfrak{h}) \cdot M_0$ and assume M_0 is a $\mathfrak{b}$-module.

Now consider the natural $\mathfrak{g}$-linear map

$$\mathsf{Ind}_{\mathfrak{b}}^{\mathfrak{g}}(M_0) \to M \tag{2.2}$$

induced by the $\mathfrak{b}$-linear map $M_0 \to M$. Since M is generated by M_0, the map (2.2) is surjective. Hence we only need to show $\mathsf{Ind}_{\mathfrak{b}}^{\mathfrak{g}}(M_0)$ is a successive extension of the Verma modules. For this purpose, we only need to show M_0 is a successive extension of 1-dimensional representations. This follows from the following exercise.

Exercise 2.1.10. Let M_0 be a finite-dimensional $\mathfrak{b}$-module such that the $\mathfrak{h}$-action is semisimple. Show that M_0 is a successive extension of 1-dimensional representations.

□

Corollary 2.1.11. *Let M be an object in $\mathcal{O}$.*

(1) *As an $\mathfrak{n}^-$-module, M is finitely generated.*
(2) *The $\mathfrak{n}$-action on M is locally nilpotent.*
(3) *Each weight subspace of M is finite-dimensional.*

2.2. **Irreducible objects.** We can classify the irreducible objects in $\mathcal{O}$.

Proposition-Definition 2.2.1. *The Verma module M_λ admits a unique irreducible quotient module, which we denote by L_λ. The highest weight of L_λ is λ.*

Proof. Recall that M_λ is generated by a highest weight vector. It follows that any proper submodule $N \subseteq M_\lambda$ is a weight module such that λ is not a weight of N. It follows that the union N_{max} of all the proper submodules satisfies the same property. By construction, this is the unique maximal proper submodule of M_λ. Therefore $M_\lambda / N_{\mathsf{max}}$ is the unique irreducible quotient of M_λ.

□

Remark 2.2.2. When λ is integral and dominant, *a priori* we do not know L_λ defined above coincides with the finite-dimensional Weyl module in Theorem 1.2.1. This is because we have not yet proved that L_λ in Proposition-Definition 2.2.1 is finite-dimensional.

Exercise 2.2.3 (Problem 1, Homework 2)**.** For $\mathfrak{g} = \mathfrak{sl}_2$, let λ be a weight and l be the corresponding complex number. Show that the Verma module M_λ is irreducible iff $l \notin \mathbb{Z}^+$. For $l \in \mathbb{Z}^+$, show that there is a non-split short exact sequence

$$0 \to M_{\lambda'} \to M_\lambda \to L_\lambda \to 0 \tag{2.3}$$

such that L_λ is the finite-dimensional irreducible $\mathfrak{sl}_2$-module with highest weight λ. Moreover, the complex number l' that corresponds to λ' is equal to $-l-2$.

Theorem 2.2.4. *We have*

$$\mathsf{Irr}(\mathcal{O}) \simeq \{L_\lambda \,|\, \lambda \in \mathfrak{h}^*\}.$$

Proof. By definition, L_λ and $L_{\lambda'}$ are isomorphic iff $\lambda = \lambda'$. It remains to show that any irreducible object in $\mathcal{O}$ is isomorphic to L_λ for some weight λ.

Let $L \in \mathcal{O}$ be an irreducible object. By Proposition 2.1.9, there exists a surjective morphism $M \to L$ such that M can be written as a successive extension of Verma modules. We have a short exact sequence

$$0 \to M_\lambda \to M \to M' \to 0$$

in $\mathcal{O}$ such that M' can be written as a successive extension of strictly fewer Verma modules. Consider the composition

$$M_\lambda \to M \to L \tag{2.4}$$

If (2.4) is nonzero, it must be a surjection because L is irreducible. Then Proposition–Definition 2.2.1 implies $L \simeq L_\lambda$ as desired.

If (2.4) is zero, we obtain a surjection $M' \to L$. We can replace M with M' and finish the argument using induction.

□

Corollary 2.2.5. *When λ is integral and dominant, L_λ is finite-dimensional and isomorphic to the Weyl module with highest weight λ.*

Proof. This Weyl module is an irreducible object in $\mathcal{O}$ with highest weight λ. Theorem 2.2.4 implies it is isomorphic to L_λ.

□

2.3. **Harish–Chandra isomorphism.** We are going to study the center of the associative ring $U(\mathfrak{g})$. This is motivated by the following general paradigm in representation theory. Let U be an associative ring and $Z \subseteq U$ be its center. For an U-module M, we can view it as a Z-module and consider its *(set theoretic) support*

$$\mathsf{supp}_Z(M) := \{\mathfrak{p} \in \mathsf{Spec}(Z) \,|\, M_\mathfrak{p} \neq 0\}.$$

Let M and N are U-modules such that $\mathsf{supp}_Z(M) \bigcap \mathsf{supp}_Z(N) = \varnothing$. One can easily show that $\mathsf{Hom}_U(M, N) = 0$.

Notation 2.3.1. We write $Z(\mathfrak{g}) := Z(U(\mathfrak{g}))$ for the center of the associative ring $U(\mathfrak{g})$.

We are going to show that $Z(\mathfrak{g})$ is *non-canonically* isomorphic to a polynomial algebra in $\mathsf{rank}(\mathfrak{g}) := \dim(\mathfrak{h})$ variables. Let us first look at the $\mathfrak{sl}_2$-case.

Exercise 2.3.2 (Problem 2, Homework 2)**.** For $\mathfrak{g} = \mathfrak{sl}_2$, let $e := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $f := \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $h := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ be the standard basis. Show that the *Casimir element*

$$\Omega := ef + fe + h^2/2 \in U(\mathfrak{sl}_2)$$

is contained in the center $Z(\mathfrak{sl}_2)$.

Exercise 2.3.3 (Problem 3, Homework 2)**.** For $\mathfrak{g} = \mathfrak{sl}_2$, show that the Casimir element Ω acts on the Verma module M_λ by the scaler $l + l^2/2$, where $l := \lambda(h)$.

Remark 2.3.4. We will see that $Z(\mathfrak{sl}_2)$ is isomorphic to $\mathbb{C}[\Omega]$. In particular, $Z(\mathfrak{sl}_2)$ acts on M_λ via a character that depends *algebraically* on λ. In other words, there is a *canonical* homomorphism

$$\phi : Z(\mathfrak{sl}_2) \to \mathcal{O}(\mathfrak{h}^*) \simeq \mathsf{Sym}(\mathfrak{h})$$

such that z acts on M_λ via the scaler $\phi(z)(\lambda)$. Namely, if we identify $\mathsf{Sym}(\mathfrak{h})$ with $\mathbb{C}[h]$, Exercise 2.3.3 implies $\phi(\Omega) = h + h^2/2$. Note that the homeomorphism ϕ is injective. Moreover, the image of ϕ is the subring preserved by the reflection $h \leftrightarrow -h-2$. Note that the last observation is compatible with Exercise 2.2.3.

With the above example in mind, let us start to study the general case.

Lemma 2.3.5. *The commutative ring $Z(\mathfrak{g})$ acts on the Verma module M_λ via a character, which we denote by $\xi_\lambda : Z(\mathfrak{g}) \to \mathbb{C}$.*

Proof. For $z \in Z(\mathfrak{g})$, the map $z \cdot - : M_\lambda \to M_\lambda$ is $U(\mathfrak{g})$-linear. In particular, it preserves the weight subspaces of M_λ. Let $v_\lambda \in M_\lambda$ be a nonzero highest weight vector. It follows that $\mathbb{C} \cdot v_\lambda \subseteq M_\lambda$ is preserved by the action of z, hence there exists a unique scaler $\xi_\lambda(z) \in \mathbb{C}$ such that

$$z \cdot v_\lambda = \xi_\lambda(z) v_\lambda.$$

Moreover, $\xi_\lambda(-) : Z(\mathfrak{g}) \to \mathbb{C}$ is a character by the axioms of an action.

It remains to show that $Z(\mathfrak{g})$ acts on any vector v of M_λ via the character ξ_λ constructed as above. Recall that v can be written as $u \cdot v_\lambda$ for some element $u \in U(\mathfrak{g})$. Then we have

$$z \cdot v = z \cdot (u \cdot v_\lambda) = u \cdot (z \cdot v_\lambda) = u \cdot (\xi_\lambda(z) v_\lambda) = \xi_\lambda(z)(u \cdot v_\lambda) = \xi_\lambda(z) v$$

as desired.

□

In the proof of the above lemma, we constructed ξ_λ by studying the action of $Z(\mathfrak{g})$ on a highest weight vector $v_\lambda \in \mathbb{C}_\lambda \subseteq M_\lambda$. In other words, it is equal to the composition

$$Z(\mathfrak{g}) \to U(\mathfrak{g}) \xrightarrow{- \cdot v_\lambda} M_\lambda \to \mathbb{C}_\lambda \simeq \mathbb{C},$$

where the third map is the projection to the highest weight vector, while the last isomorphism sends $v_\lambda \in \mathbb{C}_\lambda$ to $1 \in \mathbb{C}$. Using the identification $M_\lambda \simeq U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$, we can rewrite the above composition as

$$Z(\mathfrak{g}) \to U(\mathfrak{g}) \to U(\mathfrak{g}) \underset{U(\mathfrak{b})}{\otimes} \mathbb{C}_\lambda \simeq (U(\mathfrak{n}^-) \otimes U(\mathfrak{b})) \underset{U(\mathfrak{b})}{\otimes} \mathbb{C}_\lambda \to \mathbb{C}_\lambda \simeq \mathbb{C},$$

where the fourth map is given by the augmentation map $U(\mathfrak{n}^-) \to \mathbb{C}$ that sends any element $f \in \mathfrak{n}^-$ to zero. We can further rewrite the above composition as

$$Z(\mathfrak{g}) \to U(\mathfrak{g}) \to \mathbb{C} \underset{U(\mathfrak{n}^-)}{\otimes} U(\mathfrak{g}) \underset{U(\mathfrak{n})}{\otimes} \mathbb{C} \simeq U(\mathfrak{h}) \simeq \mathsf{Sym}(\mathfrak{h}) \xrightarrow{- \cdot \lambda} \mathbb{C},$$

where the second map is given by the augmentation maps of $U(\mathfrak{n}^-)$ and $U(\mathfrak{n})$.

This motivates the following result.

Proposition 2.3.6. *The composition*

$$\phi : Z(\mathfrak{g}) \to U(\mathfrak{g}) \to \mathbb{C} \underset{U(\mathfrak{n}^-)}{\otimes} U(\mathfrak{g}) \underset{U(\mathfrak{n})}{\otimes} \mathbb{C} \simeq U(\mathfrak{h}) \simeq \mathsf{Sym}(\mathfrak{h})$$

is a ring homomorphism.

Proof. By the previous discussion, for any $\lambda \in \mathfrak{h}^*$, the composition

$$Z(\mathfrak{g}) \xrightarrow{\phi} \mathsf{Sym}(\mathfrak{h}) \xrightarrow{-\cdot\lambda} \mathbb{C}$$

is equal to the character ξ_λ. In particular, it is a ring homomorphism. This implies that ϕ is a ring homomorphism because the homomorphism

$$\mathsf{Sym}(\mathfrak{h}) \to \prod_{\lambda \in \mathfrak{h}^*} \mathbb{C},\ r \mapsto r(\lambda)$$

is injective.

□

Warning 2.3.7. One can view $\mathsf{Sym}(\mathfrak{h})$ as a *subring* of $U(\mathfrak{g})$, but $Z(\mathfrak{g})$ is not contained in this subring. This can be seen from the $\mathfrak{sl}_2$-case.

Remark 2.3.8. By definition, $\phi : Z(\mathfrak{g}) \to \mathsf{Sym}(\mathfrak{h})$ is compatible with the natural filtrations on both sides.

Let

$$\varpi : \mathfrak{h}^* \simeq \mathsf{Spec}(\mathsf{Sym}(\mathfrak{h})) \to \mathsf{Spec}(Z(\mathfrak{g}))$$

be the morphism between affine schemes that corresponds to the Harish-Chandra homomorphism ϕ. The previous discussion can be reformulated as the following result.

Corollary 2.3.9. *The scheme theoretic support of M_λ, viewed as a $Z(\mathfrak{g})$-module, is equal to $\{\varpi(\lambda)\}$.*

As exhibited by the $\mathfrak{sl}_2$-case, the homomorphism $\phi : Z(\mathfrak{g}) \to \mathsf{Sym}(\mathfrak{h})$ is not an isomorphism. Instead, we expect it to be an injection with image equal to a certain invariant subring of $\mathsf{Sym}(\mathfrak{h})$.

Definition 2.3.10. The *dot action* of the Weyl group W on $\mathfrak{h}^*$ is defined to be

$$w \cdot \lambda := w(\lambda + \rho) - \rho,\ \ w \in W,\ \lambda \in \mathfrak{h}^*$$

The *dot action* of W on $\mathsf{Sym}(\mathfrak{h})$ is defined to be the action induced from the isomorphism $\mathsf{Sym}(\mathfrak{h}) \simeq \mathcal{O}(\mathfrak{h}^*)$.

Remark 2.3.11. Note that $-\rho$ is the unique fixed point for the dot W-action on $\mathfrak{h}^*$.

Remark 2.3.12. By definition, for $w \in W$, $\lambda \in \mathfrak{h}^*$ and $r \in \mathsf{Sym}(\mathfrak{h})$, we have

$$(w \cdot r)(\lambda) = r(w^{-1} \cdot \lambda) = r(w^{-1}(\lambda + \rho) - \rho).$$

Example 2.3.13. For $\mathfrak{g} = \mathfrak{sl}_2$, the dot action of the unique non-trivial element $s \in W$ on $\mathfrak{h}^*$ is given by $l \mapsto -l - 2$. The dot action of s on $\mathsf{Sym}(\mathfrak{h})$ is given by $h \mapsto -h - 2$.

Theorem 2.3.14 (Harish-Chandra)**.** *The homomorphism $\phi : Z(\mathfrak{g}) \to \mathsf{Sym}(\mathfrak{h})$ induces an isomorphism*

$$\phi_{\mathsf{HC}} : Z(\mathfrak{g}) \xrightarrow{\simeq} \mathsf{Sym}(\mathfrak{h})^{W_\bullet}.$$

The rest of this subsection is devoted to the proof of the theorem.

Lemma 2.3.15. *Let $\alpha \in \Delta$ be a simple root and $\lambda \in \mathfrak{h}^*$. Suppose that $\langle \lambda + \rho, \check{\alpha} \rangle \in \mathbb{Z}^+$. Then M_λ contains $M_{s_\alpha \cdot \lambda}$ as a submodule.*

Proof. Recall that
$$s_\alpha(\lambda) = \lambda - 2\frac{(\lambda,\alpha)}{(\alpha,\alpha)}\alpha = \lambda - \langle\lambda,\check{\alpha}\rangle\alpha.$$
It follows that
$$s_\alpha\cdot\lambda = \lambda - \langle\lambda+\rho,\check{\alpha}\rangle\alpha = \lambda - m\alpha$$
for $m\in\mathbb{Z}^{\geq 0}$. The lemma is obvious for $m=0$. We assume $m>0$. Then we have $\langle\lambda,\check{\alpha}\rangle = m-1$ because $\langle\rho,\check{\alpha}\rangle = 1$.

Let $f_\alpha\in\mathfrak{g}_{-\alpha}$ be a generator. Note that $f_\alpha^m\cdot v_\lambda$ is a weight vector whose weight is equal to $\lambda - m\alpha = s_\alpha\cdot\lambda$. We claim that
$$\mathfrak{n}\cdot(f_\alpha^m\cdot v_\lambda) = 0. \tag{2.5}$$
Assuming the claim, the map $\mathbb{C}_{s_\alpha\cdot\lambda}\to M_\lambda,\ c\mapsto c(f_\alpha^m\cdot v_\lambda)$ is $\mathfrak{b}$-linear, and thereby induces a $\mathfrak{g}$-linear map $M_{s_\alpha\cdot\lambda}\to M_\lambda$. This map is injective because as a morphism between $U(\mathfrak{n}^-)$-modules, it is given by $-\cdot f_\alpha^m : U(\mathfrak{n}^-)\to U(\mathfrak{n}^-)$, which is injective by the PBW theorem.

It remains to prove (2.5). For each simple root $\beta\in\Delta$, let $e_\beta\in\mathfrak{n}$ be a nonzero vector of weight β. Note that the vectors $(e_\beta)_{\beta\in\Delta}$ generate $\mathfrak{n}$ under Lie brackets. Hence we only need to show $e_\beta\cdot f_\alpha^m\cdot v_\lambda = 0$.

If $\alpha\neq\beta$, we have $[e_\beta,f_\alpha]=0$ because $\beta-\alpha$ is neither a root nor zero. It follows that
$$e_\beta\cdot f_\alpha^m\cdot v_\lambda = f_\alpha^m\cdot e_\beta\cdot v_\lambda = 0$$
as desired.

If $\alpha=\beta$, we can resacle e_α such that $[e_\alpha,f_\alpha]=\check{\alpha}$. Then
$$[\check{\alpha},f_\alpha] = \langle-\alpha,\check{\alpha}\rangle f_\alpha = -2f_\alpha.$$
We have
$$\begin{aligned} e_\alpha\cdot f_\alpha^m\cdot v_\lambda &= \sum_{1\leq i\leq m} f_\alpha^{m-i}\cdot[e_\alpha,f_\alpha]\cdot f_\alpha^{i-1}\cdot v_\lambda + f_\alpha^m\cdot e_\alpha\cdot v_\lambda \\ &= \sum_{1\leq i\leq m} f_\alpha^{m-i}\cdot\check{\alpha}\cdot f_\alpha^{i-1}\cdot v_\lambda \end{aligned}$$
Note that
$$\check{\alpha}\cdot f_\alpha^j = \sum_{1\leq i\leq j} f_\alpha^{j-i}\cdot[\check{\alpha},f_\alpha]\cdot f_\alpha^{i-1} + f_\alpha^j\cdot\check{\alpha} = -2jf_\alpha^j + f_\alpha^j\cdot\check{\alpha}.$$
Hence
$$e_\alpha\cdot f_\alpha^m\cdot v_\lambda = \sum_{1\leq i\leq m}(-2(i-1)f_\alpha^{m-1} + f_\alpha^{m-1}\cdot\check{\alpha})v_\lambda = \big(-m(m-1)+m\langle\lambda,\check{\alpha}\rangle\big)v_\lambda = 0$$
as desired.

□

Lemma 2.3.16. *The image of* $\phi: Z(\mathfrak{g})\to\mathsf{Sym}(\mathfrak{h})$ *is contained in* $\mathsf{Sym}(\mathfrak{h})^{W\bullet}$.

Proof. We only need to show that for any $w\in W$, $z\in Z(\mathfrak{g})$ and $\lambda\in\mathfrak{h}^*$,
$$\phi(z)(\lambda) = \phi(z)(w\cdot\lambda). \tag{2.6}$$
Since W is generated by by simple reflections, we only need to check (2.6) for $w=s_\alpha$, $\alpha\in\Delta$.

Let $\alpha \in \Delta$ be a fixed simple root. It is easy to see that

$$\{\lambda \in \mathfrak{h}^* \mid \langle \lambda + \rho, \check{\alpha} \rangle \in \mathbb{Z}^+\}$$

is a Zariski dense subset of $\mathfrak{h}^*$. Hence we only need to check (2.6) for λ belonging to this set. By Lemma 2.3.15, we have $M_{s_\alpha \cdot \lambda} \subseteq M_\lambda$. This implies (2.6) because z acts on M_μ via the scaler $\phi(z)(\mu)$.

□

The above lemma implies we have a well-defined homomorphism

$$\phi_{\mathsf{HC}} : Z(\mathfrak{g}) \to \mathsf{Sym}(\mathfrak{h})^{W_\bullet}$$

between filtered commutative rings, where the filtrations on the LHS and the RHS are induced by the PBW filtrations on $U(\mathfrak{g})$ and $\mathsf{Sym}(\mathfrak{h})$ respectively. As a consequence, it induces a homomorphism

$$\mathsf{gr}^\bullet \phi_{\mathsf{HC}} : \mathsf{gr}^\bullet Z(\mathfrak{g}) \to \mathsf{gr}^\bullet(\mathsf{Sym}(\mathfrak{h})^{W_\bullet}). \tag{2.7}$$

To show that ϕ_{HC} is an isomorphism, we only need to show $\mathsf{gr}^\bullet \phi_{\mathsf{HC}}$ is so.

Lemma 2.3.17. *There is a unique dotted graded isomorphism making the following diagram commute*

$$\begin{tikzcd}
\mathsf{gr}^\bullet Z(\mathfrak{g}) \arrow[r, dotted, "\simeq"] \arrow[d, "\subseteq"] & \mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}} \arrow[d, "\subseteq"] \\
\mathsf{gr}^\bullet U(\mathfrak{g}) \arrow[r, "\simeq", "\mathsf{PBW}"'] & \mathsf{Sym}(\mathfrak{g}),
\end{tikzcd}$$

where $\mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}}$ *is the invariance of* $\mathsf{Sym}(\mathfrak{g})$ *with respect to the* adjoint action[1].

Proof. Consider the adjoint action of $\mathfrak{g}$ on $U(\mathfrak{g})$ given by

$$\mathfrak{g} \times U(\mathfrak{g}) \to U(\mathfrak{g}), \ \ (x, u) \mapsto [x, u] := xu - ux.$$

One can check that this action preserves the PBW filtration on $U(\mathfrak{g})$, and the induced action on $\mathsf{gr}^\bullet(U(\mathfrak{g})) \simeq \mathsf{Sym}(\mathfrak{g})$ is given by the adjoint $\mathfrak{g}$-action on $\mathsf{Sym}(\mathfrak{g})$.

Note that for any $n \in \mathbb{Z}^+$, the $\mathfrak{g}$-module $\mathsf{F}^{\leq n} U(\mathfrak{g})$ is finite-dimensional. This implies the short exact sequence

$$0 \to \mathsf{F}^{\leq n-1} U(\mathfrak{g}) \to \mathsf{F}^{\leq n} U(\mathfrak{g}) \to \mathsf{Sym}^n(\mathfrak{g}) \to 0$$

splits. Taking $\mathfrak{g}$-invariance, we obtain a split short exact sequence

$$0 \to \mathsf{F}^{\leq n-1} Z(\mathfrak{g}) \to \mathsf{F}^{\leq n} Z(\mathfrak{g}) \to \mathsf{Sym}^n(\mathfrak{g})^{\mathfrak{g}} \to 0.$$

This gives the desired isomorphism $\mathsf{gr}^n Z(\mathfrak{g}) \simeq \mathsf{Sym}^n(\mathfrak{g})^{\mathfrak{g}}$.

□

A similar proof gives the following result.

Lemma 2.3.18. *There is a unique dotted graded isomorphism making the following diagram commute*

$$\begin{tikzcd}
\mathsf{gr}^\bullet(\mathsf{Sym}(\mathfrak{h})^{W_\bullet}) \arrow[r, dotted, "\simeq"] \arrow[d, "\subset"] & \mathsf{Sym}(\mathfrak{h})^{W} \arrow[d, "\subset"] \\
\mathsf{gr}^\bullet \mathsf{Sym}(\mathfrak{h}) \arrow[r, "\simeq"] & \mathsf{Sym}(\mathfrak{h}),
\end{tikzcd}$$

[1]This is the unique action that restricts to the adjoint action on $\mathfrak{g} \subseteq \mathsf{Sym}(\mathfrak{g})$ and satisfies the Lebniz rule.

where the right-top corner is the invariance for the (usual) linear *W-action on* $\mathsf{Sym}(\mathfrak{h})$*.*

By Lemma 2.3.17 and Lemma 2.3.18, the homomorphism (2.7) can be identified with a certain homomorphism

$$\phi_{\mathsf{cl}} : \mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}} \to \mathsf{Sym}(\mathfrak{h})^{W},$$

and we only need to show ϕ_{cl} is an isomorphism.

We are going to give a direct description for the homomorphism ϕ_{cl}. For this purpose, we give the following alternative construction of ϕ.

Exercise 2.3.19 (Problem 4, Homework 2)**.** Consider the adjoint $\mathfrak{h}$-action on $U(\mathfrak{g})$. Show that it preserves the kernel of the surjection

$$U(\mathfrak{g}) \to U(\mathfrak{g}) \underset{U(\mathfrak{n})}{\otimes} \mathbb{C}$$

and induces an $\mathfrak{h}$-action on $U(\mathfrak{g}) \otimes_{U(\mathfrak{n})} \mathbb{C}$. Moreover, we have

$$(U(\mathfrak{g}) \underset{U(\mathfrak{n})}{\otimes} \mathbb{C})^{\mathfrak{h}} \simeq U(\mathfrak{b}) \underset{U(\mathfrak{n})}{\otimes} \mathbb{C} \simeq \mathsf{Sym}(\mathfrak{h}).$$

Exercise 2.3.20 (Problem 5, Homework 2)**.** Show that $\phi : Z(\mathfrak{g}) \to \mathsf{Sym}(\mathfrak{h})$ can be identified with the composition

$$Z(\mathfrak{g}) \xrightarrow{\subseteq} U(\mathfrak{g})^{\mathfrak{h}} \to (U(\mathfrak{g}) \underset{U(\mathfrak{n})}{\otimes} \mathbb{C})^{\mathfrak{h}} \simeq \mathsf{Sym}(\mathfrak{h}),$$

where the last isomorphism is due to Exercise 2.3.19.

It follows that ϕ_{HC} is the unique map that fits into the following commutative diagram

$$\begin{tikzcd} Z(\mathfrak{g}) \arrow[r, dotted, "\phi_{\mathsf{HC}}"] \arrow[d, "\subseteq"] & \mathsf{Sym}(\mathfrak{h})^{W\bullet} \arrow[d, "\subseteq"] \\ U(\mathfrak{g}) \arrow[r] & U(\mathfrak{g}) \otimes_{U(\mathfrak{n})} \mathbb{C}. \end{tikzcd}$$

Taking associated graded pieces, we deduce that ϕ_{cl} is the unique map that fits into the following diagram

$$\begin{tikzcd} \mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}} \arrow[r, dotted, "\phi_{\mathsf{cl}}"] \arrow[d, "\subseteq"] & \mathsf{Sym}(\mathfrak{h})^{W} \arrow[d, "\subseteq"] \\ \mathsf{Sym}(\mathfrak{g}) \arrow[r] & \mathsf{Sym}(\mathfrak{g}/\mathfrak{n}). \end{tikzcd}$$

Remark 2.3.21. Let G be a connected algebraic group with Lie algebra $\mathfrak{g}$. Note that we have $\mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}} = \mathsf{Sym}(\mathfrak{g})^{G}$. The previous discussion implies there is a unique dotted morphism between schemes that fits into the following diagram

$$\begin{tikzcd} \mathfrak{g}^*/\!/G & \mathfrak{h}^*/\!/W \arrow[l, dotted] \\ \mathfrak{g}^* \arrow[u] & (\mathfrak{g}/\mathfrak{n})^*. \arrow[l] \arrow[u] \end{tikzcd}$$

Here $\mathfrak{g}^*/\!/G$ and $\mathfrak{h}^*/\!/W$ are the (GIT) quotient schemes.

Recall that the Killing form induces a G-linear isomorphism $\mathfrak{g} \simeq \mathfrak{g}^*$ and a W-linear isomorphism $\mathfrak{h} \simeq \mathfrak{h}^*$. Via the first isomorphism, $\mathfrak{b} \subseteq \mathfrak{g}$ corresponds to $(\mathfrak{g}/\mathfrak{n})^*$. It follows that there is a unique dotted morphsim between schemes that fits into the following diagram

$$\begin{tikzcd} \mathfrak{g}/\!/G & \mathfrak{h}/\!/W \arrow[l, dotted] \\ \mathfrak{g} \arrow[u] & \mathfrak{b}, \arrow[l] \arrow[u] \end{tikzcd}$$

where the right vertical morphism is the composition $\mathfrak{b} \to \mathfrak{h} \to \mathfrak{h}/\!/W$.

Note that the similar claim where $\mathfrak{b}$ is replaced with $\mathfrak{h}$ is true but weaker than the above claim.

As a byproduct of the above observation, one can prove the following result.

Exercise 2.3.22 (Problem 6, Homework 2)**.** Let $r \in \mathcal{O}(\mathfrak{g})^G$ be a G-invariant regular function on $\mathfrak{g}$. Show that $r(x) = r(0)$ for any nilpotent element $x \in \mathfrak{g}$.

Recall that we only need to show that

$$\phi_{\mathsf{cl}} : \mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}} \to \mathsf{Sym}(\mathfrak{h})^W$$

is an isomorphism. By Remark 2.3.21, this is equivalent to the following result.

Theorem 2.3.23 (Chevalley)**.** *The restriction map $\mathcal{O}(\mathfrak{g}) \to \mathcal{O}(\mathfrak{h})$ induces an isomorphism*

$$\phi_{\mathsf{Chev}} : \mathcal{O}(\mathfrak{g})^{\mathfrak{g}} \xrightarrow{\simeq} \mathcal{O}(\mathfrak{h})^W.$$

Proof. By the previous discussions, the restriction map induces a homomorphism $\phi_{\mathsf{Chev}} : \mathcal{O}(\mathfrak{g})^{\mathfrak{g}} \to \mathcal{O}(\mathfrak{h})^W$. This homomorphism is injective because semisimple elements form a Zariski dense subset of $\mathfrak{g}$, and any such element is conjugate to an element in $\mathfrak{h}$.

It remains to show ϕ_{Chev} is surjective. Recall that the subset Λ^+ of dominant integral weights spans $\mathfrak{h}^*$ as a vector space. This implies that for any $n \geq 0$, the subset $\{\lambda^n \,|\, \lambda \in \Lambda^+\}$ spans $\mathsf{Sym}^n(\mathfrak{h}^*)$. Therefore the elements

$$b_{\lambda,n} \coloneqq \sum_{w \in W} w(\lambda^n), \quad \lambda \in \Lambda^+, n \geq 0$$

span $\mathsf{Sym}(\mathfrak{h}^*)^W \simeq \mathcal{O}(\mathfrak{h})^W$.

It remains to show that each element $b_{\lambda,n}$ defined as above is contained in the image of ϕ_{Chev}. Consider the elements $a_{\lambda,n} \in \mathcal{O}(\mathfrak{g})$ defined by the formula

$$a_{\lambda,n}(x) \coloneqq \mathsf{Tr}(x^n, L_\lambda), \quad \forall x \in \mathfrak{g}.$$

Using the fact that L_λ is G-integrable for simply connected G, we see that $a_{\lambda,n} \in \mathcal{O}(\mathfrak{g})^G = \mathcal{O}(\mathfrak{g})^{\mathfrak{g}}$.

Exercise 2.3.24 (Problem 1, Homework 3)**.** Prove that

$$\phi_{\mathsf{Chev}}(a_{\lambda,n}) = \frac{1}{\#\mathsf{Stab}_W(\lambda)} b_{\lambda,n} \mod \mathsf{span}(b_{\lambda',n},\ \lambda' < \lambda),$$

where $\mathsf{Stab}_W(\lambda) \subseteq W$ is the stabilizer of the W-action at $\lambda \in \mathfrak{h}^*$.

Now one can finish the proof by using the above exercise and induction in λ.

□

□[Theorem 2.3.14]

Definition 2.3.25. The *Chevalley base* of $\mathfrak{g}$ is defined to be

$$\mathfrak{c}_{\mathfrak{g}} := \mathfrak{g}/\!/G.$$

2.4. **Blocks.**

Definition 2.4.1. For a closed point $\chi \in \mathsf{Spec}(Z(\mathfrak{g}))$, let $\mathcal{O}_\chi \subseteq \mathcal{O}$ to be the full subcategory containing objects M such that $\mathsf{supp}_{Z(\mathfrak{g})}(M) = \{\chi\}$. We call it the *block of* $\mathcal{O}$ that corresponds to χ.

We often abuse notations and identify a closed point $\chi \in \mathsf{Spec}(Z(\mathfrak{g}))$ with the corresponding *central character* $Z(\mathfrak{g}) \to \mathbb{C}$.

Example 2.4.2. The Harish-Chandra isomorphism provides a morphism

$$\varpi : \mathfrak{h}^* \to \mathfrak{h}^*/\!/W_\bullet \simeq \mathsf{Spec}(Z(\mathfrak{g}))$$

such that for a closed point $\lambda \in \mathfrak{h}^*$, we have

$$\mathsf{supp}_{Z(\mathfrak{g})}(M_\lambda) = \{\varpi(\lambda)\}.$$

In particular, we have

$$M_\lambda \in \mathcal{O}_\chi,\ \ \forall \lambda \in \varpi^{-1}(\chi).$$

Note that this also implies

$$L_\lambda \in \mathcal{O}_\chi,\ \ \forall \lambda \in \varpi^{-1}(\chi)$$

and therefore

$$\mathsf{Irr}(\mathcal{O}_\chi) \simeq \{L_\lambda \,|\, \lambda \in \varpi^{-1}(\chi)\}.$$

Corollary 2.4.3. *Let $M \in \mathcal{O}$ be an object and $\chi \in \mathsf{Ch}(Z(\mathfrak{g}))$ be a central character. Then $M \in \mathcal{O}_\chi$ iff it is isomorphic to a quotient of a successive extension of the Verma modules M_λ such that $\varpi(\lambda) = \chi$.*

The goal of this subsection is to prove the following result.

Theorem 2.4.4. *We have*

$$\mathcal{O} \simeq \bigoplus_{\chi \in \mathsf{Ch}(Z(\mathfrak{g}))} \mathcal{O}_\chi.$$

Warning 2.4.5. A block $\mathcal{O}_\chi$ can fail to be indecomposable. In some literatures (such as [H2]), blocks are defined to be indecomposable summands of $\mathcal{O}_\chi$.

Let us first explain several geometric properties about the morphism ϖ.

Note that we can canonically identify $\mathfrak{h}^*/\!/W_\bullet$ with $\mathfrak{h}^*/\!/W$ by sending λ to $\lambda + \rho$. Hence ϖ shares the same properties as the morphism $\mathfrak{h}^* \to \mathfrak{h}^*/\!/W$.

Proposition 2.4.6 (Chevalley–Shephard–Todd)**.** *The graded commutative algebra $\mathsf{Sym}(\mathfrak{h})^W$ is* non-canonically *isomorphic to a graded polynomial algebra in* $\dim(\mathfrak{h})$-*variables. The graded $\mathsf{Sym}(\mathfrak{h})^W$-module $\mathsf{Sym}(\mathfrak{h})$ is* non-canonically *isomorphic to a graded free module of rank $|W|$.*

Exercise 2.4.7 (Problem 2, Homework 3)**.** Verify Proposition 2.4.6 for $\mathfrak{g} = \mathfrak{sl}_n$.

Corollary 2.4.8. *The scheme $\mathfrak{h}^*/\!/W$ is* non-canonically *isomorphic to $\mathbb{A}_{\mathbb{C}}^{\dim(\mathfrak{h})}$. The morphism $\mathfrak{h}^* \to \mathfrak{h}^*/\!/W$ is finite and flat.*

Corollary 2.4.9. *The scheme* $\mathsf{Spec}(Z(\mathfrak{g}))$ *is* non-canonically *isomorphic to* $\mathbb{A}_{\mathbb{C}}^{\dim(\mathfrak{h})}$. *The morphism* $\varpi : \mathfrak{h}^* \to \mathsf{Spec}(Z(\mathfrak{g}))$ *is finite and flat.*

Remark 2.4.10. In particular, the morphism ϖ is surjective. Hence each block $\mathcal{O}_\chi$ is nonempty by Example 2.4.2.

We also need the following general result about finite group actions on affine schemes.

Proposition 2.4.11 ([SGA1, Exp. V, Prop. 1.1])**.** *Let* K *be a finite group that acts on an affine scheme* X*. Then* K *acts transitively on each fiber of the morphism* $X \to X/\!/K$.

Corollary 2.4.12. *The Weyl group* W *acts transitively on each fiber of* $\varpi : \mathfrak{h}^* \to \mathsf{Spec}(Z(\mathfrak{g}))$.

Remark 2.4.13. In particular, the Verma modules M_λ and $M_{\lambda'}$ belong to a same block $\mathcal{O}_\chi$ iff $\lambda' = w \cdot \lambda$ for some $w \in W$.

The following result says that the *scheme theoretic* support of an object $M \in \mathcal{O}$ in $\mathsf{Spec}(Z(\mathfrak{g}))$ is 0-dimensional.

Lemma 2.4.14. *Let* $M \in \mathcal{O}$ *be an object and* $\mathsf{Ann}_{Z(\mathfrak{g})}(M) \subseteq Z(\mathfrak{g})$ *be the annihilator of* M *viewed as a* $Z(\mathfrak{g})$*-module. Then* $Z(\mathfrak{g})/\mathsf{Ann}_{Z(\mathfrak{g})}(M)$ *is a finite dimensional vector space.*

Proof. Note that the claim is true for the Verma modules because $\mathsf{Ann}_{Z(\mathfrak{g})}(M_\lambda)$ is the maximal ideal (Lemma 2.3.5). Now the desired claim follows from Proposition 2.1.9 and the following general fact: for a short exact sequence $0 \to M' \to M \to M'' \to 0$ of modules for a commutative algebra R, we have

$$\mathsf{Ann}_R(M')\mathsf{Ann}_R(M'') \subseteq \mathsf{Ann}_R(M).$$

□

Corollary 2.4.15. *Let* $M \in \mathcal{O}$ *be an object. Then* $M \in \mathcal{O}_\chi$ *iff* $\chi : Z(\mathfrak{g}) \to \mathbb{C}$ *is the unique* generalized *central character of* M*, i.e.,* $\mathsf{ker}(\chi)^n \cdot M = 0$ *for some* $n > 0$.

Proof of Theorem 2.4.4. Let $M \in \mathcal{O}$ be an object and χ be a central character. By Lemma 2.4.14, for $n >> 0$, the canonical morphism

$$M/\mathsf{ker}(\chi)^{n+1} \to M/\mathsf{ker}(\chi)^n$$

is an isomorphism. Hence $M_\chi := M/\mathsf{ker}(\chi)^n$, $n >> 0$ is well-defined.

Exercise 2.4.16 (Problem 3, Homework 3)**.** There is a unique $\mathfrak{g}$-module structure on M_χ such that $M \to M_\chi$ is $\mathfrak{g}$-linear. Moreover, M_χ is an object in $\mathcal{O}_\chi$.

One can check that the construction $M \mapsto M_\chi$ is functorial. In other words, we have a functor

$$\mathcal{O} \to \mathcal{O}_\chi, \;\; M \mapsto M_\chi.$$

Moreover, for a fixed object $M \in \mathcal{O}$, by Lemma 2.4.14, there are finitely many central characters χ such that M_χ is nonzero. This implies that we have a functor

$$\mathcal{O} \to \bigoplus_{\chi \in \mathsf{Ch}(Z(\mathfrak{g}))} \mathcal{O}_\chi, \;\; M \mapsto (M_\chi)_{\chi \in \mathsf{Ch}(Z(\mathfrak{g}))}.$$

It remains to check that this functor is inverse to the obvious functor

$$\bigoplus_{\chi \in \mathsf{Ch}(Z(\mathfrak{g}))} \mathcal{O}_\chi \to \mathcal{O}.$$

Exercise 2.4.17 (Problem 4, Homework 3)**.** Verify the above two functors are inverse to each other.

□[Theorem 2.4.4]

2.5. **Linkage principle.** Recall the following formal result.

Proposition-Definition 2.5.1 (Jordan–Hölder)**.** *Let $\mathcal{A}$ be an abelian category and $M \in \mathcal{A}$ be an object. We say M is* of finite length *if the following conditions are equivalent:*

(i) *The object M is both Artinian and Noetherian.*
(ii) *There exists a finite filtration*

$$0 = M_0 \subseteq M_1 \cdots \subseteq M_n = M \tag{2.8}$$

such that each graded piece M_{i+1}/M_i is an irreducible object in $\mathcal{A}$.
(iii) *The length of any non-degenerate finite filtration of M has an upper bound.*

Moreover, if M satisfies the above conditions, for any filtration (2.8) *in (ii) and $L \in \mathsf{Irr}(\mathcal{A})$, the number*

$$[M : L] := |\{i \,|\, M_{i+1}/M_i \simeq L\}|$$

does not depend on the choice of the filtration (2.8)*. We call $[M : L]$ the* multiplicity *of L in M, and call*

$$\mathsf{Length}(M) := \sum_{L \in \mathsf{Irr}(\mathcal{A})} [M : L]$$

the length *of M.*

Proposition 2.5.2. *Each object in the category $\mathcal{O}$ is of finite length.*

Proof. By Theorem 2.4.4, we only need to show each object in a block $\mathcal{O}_\chi$ is of finite length. We will show that for any non-degenerate finite filtration

$$0 = N_0 \subseteq N_1 \cdots \subseteq N_n = N$$

in $\mathcal{O}_\chi$, we have

$$n \leq \sum_{\lambda \in \varpi^{-1}(\chi)} \dim(M^{\mathsf{wt}=\lambda}). \tag{2.9}$$

Note that the RHS is indeed a finite number by Corollary 2.1.11(3) and Corollary 2.4.9.

To prove (2.9), we only need to show each quotient N_{i+1}/N_i has a weight that belongs to $\varpi^{-1}(\chi)$. By Proposition 2.1.9, there exists a nonzero morphism $M_\lambda \to N_{i+1}/N_i$ for some Verma module M_λ. Since $N_{i+1}/N_i \in \mathcal{O}_\chi$, we see that $M_\lambda \in \mathcal{O}_\chi$, and therefore $\lambda \in \varpi^{-1}(\chi)$. Now the image of the highest line of M_λ under the map $N_{i+1}/N_i \in \mathcal{O}_\chi$ is a nonzero λ-weight subspace as desired.

□

We are ready to ask the core question in the study of the category $\mathcal{O}$.

Question 2.5.3. For weights $\lambda, \mu \in \mathfrak{h}^*$, can we find a formula for $[M_\lambda : L_\mu]$?

Example 2.5.4. For any weight λ, we have $[M_\lambda : L_\lambda] = 1$ because the λ-weight subspace of M_λ is 1-dimensional.

In this subsection, we answer the following moderate question.

Question 2.5.5. For weights $\lambda, \mu \in \mathfrak{h}^*$, when is $[M_\lambda : L_\mu]$ nonzero?

Theorem 2.5.6 (Verma–BGG, linkage principle). *For $\lambda, \mu \in \mathfrak{h}^*$, the following conditions are equivalent:*

(i) *The multiplicity $[M_\lambda : L_\mu]$ is nonzero.*
(ii) *The Verma module M_λ has a submodule isomorphic to M_μ.*
(iii) *There exists positive roots $\alpha_1, \cdots, \alpha_n \in \Phi^+$ such that*

$$\lambda \geq s_{\alpha_1} \cdot \lambda \geq (s_{\alpha_2} s_{\alpha_1}) \cdot \lambda \geq \cdots \geq (s_{\alpha_n} \cdots s_{\alpha_1}) \cdot \lambda = \mu.$$

Proof. (ii)⇒(i): obvious.

(iii)⇒(ii): see [H2, Sect. 4.5-4.7][2].

(i)⇒(iii): see [H2, Sect. 5].

□

Remark 2.5.7. In future lectures, at least when the stabilizer of the dot W-action at λ is trivial, we will provide a geometric proof of (i)⇒(iii) using the localization theory. In fact, one can also prove (iii)⇒(ii) in this way, but the geometric tools used in such a proof is beyond the scope of this course.

Exercise 2.5.8 (Problem 5, Homework 3). Verify Theorem 2.5.6 for $\mathfrak{g} = \mathfrak{sl}_2$.

Definition 2.5.9. For $\lambda, \mu \in \mathfrak{h}^*$, we say λ *dot-dominates* μ if the conditions in Theorem 2.5.6 are satisfied.

Note that dot-domination gives a partial ordering on $\mathfrak{h}^*$.

Definition 2.5.10. We say $\lambda \in \mathfrak{h}^*$ is *dot-dominant* (resp. *dot-anti-dominant*) if it is maximal (resp. minimal) with respect to the partial ordering given by dot-domination.

We have the following combinatorial result.

Lemma 2.5.11. *Let λ be a weight. The following conditions are equivalent:*

(i) *The weight λ is dot-dominant.*
(ii) *For any positive root $\alpha \in \Phi^+$, $\langle \lambda + \rho, \check{\alpha} \rangle$ is* not *a negative integer.*
(iii) *For any positive root $\alpha \in \Phi^+$, $\lambda < s_\alpha \cdot \lambda$ is* not *true.*
(iv) *For any $w \in W$, $\lambda < w \cdot \lambda$ is* not *true.*

Proof. See [H2, Sect. 3.4-3.5].

□

Example 2.5.12. For $\mathfrak{g} = \mathfrak{sl}_2$, a weight λ is dot-dominant (resp. dot-anti-dominant) iff $l \neq -2, -3, \cdots$ (resp. $l \neq 0, 1, \cdots$).

Corollary 2.5.13. *A weight λ is dot-anti-donimant iff M_λ is irreducible.*

Proof. Follows immediately from the linkage principle.

□

Note that an orbit $W \cdot \lambda$ can be decomposable with respect to the dot-domination ordering. The following combinatorial result says that this can only happen when λ is not integral.

Lemma 2.5.14. *Let λ be a weight. The following conditions are equivalent:*

(i) *The weight λ is integral.*

[2] Note that the claim for simple roots follows from Lemma 2.3.15.

(ii) *The orbit* $W \cdot \lambda$ *contains exactly one dot-dominant weight.*
(iii) *The orbit* $W \cdot \lambda$ *contains exactly one dot-anti-dominant weight.*
(iv) *For any* $w \in W$*,* $w \cdot \lambda - \lambda \in \mathbb{Z}\Phi$*.*

Proof. See [H2, Sect. 3.4-3.5]. □

Exercise 2.5.15 (Problem 6, Homework 3)**.** Show that for integral weight λ, the block $\mathcal{O}_{\varpi(\lambda)}$ is indecomposable.

Definition 2.5.16. We say a block $\mathcal{O}_\chi$ is integral if $\varpi^{-1}(\chi) \subseteq \Lambda$.

Remark 2.5.17. In this remark, we explain the story for non-integral weights. See [H2, Sect. 3.4-3.5] for more details.

For each residue class $[\lambda] \in \mathfrak{h}^*/\Lambda$, the following subgroup of W is well-defined:

$$W_{[\lambda]} \coloneqq \{w \in W \mid w \cdot \lambda - \lambda \in \mathbb{Z}\Phi\} \subseteq W.$$

One can check that each $W_{[\lambda]} \cdot \lambda$ contains exactly one dot-dominant (resp. dot-anti-dominant) weight.

In particular, for a dot-anti-dominant weight λ, all the weights in $W_{[\lambda]} \cdot \lambda$ dot-dominate λ. Using this fact, one can show that we have a decomposition of $\mathcal{O}$ into *indecomposable* subcategories:

$$\mathcal{O} \simeq \bigoplus_{\lambda \text{ is d.a.d.}} \mathcal{O}_\lambda,$$

where λ ranges over all dot-anti-dominant weights, and

$$\mathcal{O}_\lambda \coloneqq \{M \in \mathcal{O} \mid [M : L_\mu] = 0,\ \forall \mu \notin W_{[\lambda]} \cdot \lambda\} \subseteq \mathcal{O}_{\varpi(\lambda)} \subseteq \mathcal{O}.$$

Moreover, the study of $\mathcal{O}_\lambda$ can be reduced to the study of an *integral* block for another semisimple Lie algebra. Namely, consider the following subset of Φ:

$$\Phi_{[\lambda]} \coloneqq \{\alpha \in \Phi \mid \langle \lambda, \check{\alpha} \rangle \in \mathbb{Z}\} \subseteq \Phi.$$

One can show that $(\mathbb{R}\Phi_{[\lambda]}, \Phi_{[\lambda]})$ is a root system with Weyl group $W_{[\lambda]}$[3]. Let $\mathfrak{g}'$ be the semisimple Lie algebra corresponding to this root system. Then a theorem of Soergel (see [S1, Sect. 2.5, Thm. 11]) says that $\mathcal{O}_\lambda$ is equivalent to an integral block for $\mathfrak{g}'$.

From now on, we will focus on the *integral* blocks $\mathcal{O}_{\varpi(\lambda)}$, $\lambda \in \Lambda$. Note that the number of irreducible objects in $\mathcal{O}_{\varpi(\lambda)}$ is equal to

$$\frac{|W|}{|\mathsf{Stab}_{W_\bullet}(\lambda)|}.$$

Definition 2.5.18. We say a weight λ is *dot-regular* (resp. *dot-singular*) if the stabilizer of the dot W-action at λ is trivial (resp. *non-trivial*).

We say a block $\mathcal{O}_\chi$ is regular (resp. singular) if $\varpi^{-1}(\chi)$ contains dot-regular (resp. dot-singular) weights.

Exercise 2.5.19. A weight λ is dot-regular iff the morphism $\varpi : \mathfrak{h}^* \to \mathsf{Spec}(Z(\mathfrak{g}))$ is smooth at the point $\lambda \in \mathfrak{h}^*$.

[3]One can view $(\mathbb{R}\Phi_{[\lambda]}, \Phi_{[\lambda]})$ as the largest subsystem such that λ is integral with respect to it.

Example 2.5.20. The *principle block* $\mathcal{O}_{\varpi(0)}$ is integral and regular. It contains $|W|$ irreducible objects.

Example 2.5.21. The *most singular* block is $\mathcal{O}_{-\rho}$, which contains exactly one irreducible object, namely $L_{-\rho}$.

In future lectures, we will see that all the integral regular blocks are equivalent to each other. Therefore the study of such blocks is reduced to that of $\mathcal{O}_{\varpi(0)}$.

In particular, for integral regular central characters χ, the partially ordered sets $\varpi^{-1}(\chi)$ (given by the dot-domination relation) should not depend on χ. Note that $|\varpi^{-1}(\chi)| = |W|$. Therefore we should be able to construct a *canonical* partial ordering on W. The following definition serves this purpose.

Proposition-Definition 2.5.22. *Let w and w' be elements in W. The following conditions are equivalent:*

(i) *There exists roots $\alpha_1, \cdots, \alpha_n \in \Phi$ such that*
$$s_{\alpha_n} \cdots s_{\alpha_1} w = w'$$
and
$$\ell(w) < \ell(s_{\alpha_1} w) < \cdot < \ell(s_{\alpha_n} \cdots s_{\alpha_1} w) = \ell(w').$$

(ii) *There exists roots $\alpha_1, \cdots, \alpha_n \in \Phi$ such that*
$$w s_{\alpha_1} \cdots s_{\alpha_n} = w'$$
and
$$\ell(w) < \ell(w s_{\alpha_1}) < \cdot < \ell(w s_{\alpha_1} \cdots s_{\alpha_n}) = \ell(w').$$

(iii) *There exists a reduced presentation $w' = s_1 \cdots s_{\ell(w')}$ and a sequence $1 \le i_1 < i_2 < \cdots < i_{\ell(w)} \le \ell(w')$ such that*
$$w = s_{i_1} \cdots s_{i_{\ell(w)}}$$
is a reduced presentation of w.

If the above conditions are satisfied, we write $w \le w'$. This defines a partial ordering on W, which we call the Bruhat ordering.

Proof. See [H2, Sect. 0.3-0.4] □

We have the following combinatorial result.

Lemma 2.5.23. *Let λ be an integral dot-regular dot-anti-dominant weight. Let w and w' be elements in W. Then $w' \cdot \lambda$ dot-dominates $w \cdot \lambda$ iff $w' \ge w$.*

Proof. See [H2, Sect. 5.2] □

Exercise 2.5.24 (Problem 7, Homework 3)**.** Verify the above lemma for $\mathfrak{g} = \mathfrak{sl}_3$.

2.6. **Projective objects.** Recall the following definition.

Definition 2.6.1. Let $\mathcal{A}$ be an abelian category. We say an object $P \in \mathcal{A}$ is *projective* if the functor $\mathsf{Hom}_{\mathcal{A}}(P, -) : \mathcal{A} \to \mathsf{Ab}$ is exact.

We say $\mathcal{A}$ *has enough projective objects* if any object in $\mathcal{A}$ is a quotient object of a projective object.

Proposition 2.6.2. *The category $\mathcal{O}$ has enough projective objects.*

To prove the proposition, we need the following lemma.

Lemma 2.6.3. *For $\chi = \varpi(\lambda)$, the functor*

$$\mathcal{O}_\chi \to \mathsf{Vect}, \ \ M \mapsto M^{\mathsf{wt}=\lambda}. \tag{2.10}$$

is representable.

Proof. For a positive integer $n > 0$, let $I_{\lambda,n} \subset U(\mathfrak{g})$ be the left ideal genetated by the following elements:

- The element $t - \lambda(t)$, for any $t \in \mathfrak{h}$;
- The element $x_1 x_2 \cdots x_n$, for $x_i \in \mathfrak{n}^+$.

Let $M_{\lambda,n} := U(\mathfrak{g})/I_{\lambda,n}$ be the quotient $U(\mathfrak{g})$-module. Note that $M_{\lambda,1}$ is just the Verma module M_λ. Now the lemma follows from the following two exercises.

Exercise 2.6.4 (Problem 1, Homework 4)**.** Show that $M_{\lambda,n}$ is an object in $\mathcal{O}$ and is a successive extension of Verma modules.

Exercise 2.6.5 (Problem 2, Homework 4)**.** Let $M_{\lambda,n,\chi}$ be the direct summand of $M_{\lambda,n}$ in the block $\mathcal{O}_\chi$. Show that for $n >> 0$, the functor (2.10) is represented by the object $M_{\lambda,n,\chi}$.

□[Lemma 2.6.3]

Remark 2.6.6. The module $M_{\lambda,n}$ is a *generalized Verma module.*

Proof of Proposition 2.6.2. By the block decomposition, we only need to prove that each block $\mathcal{O}_\chi$ has enough projectives. Using dévissage, we only need to show that each irreducible object $L_\lambda \in \mathcal{O}_\chi$ is a quotient object of a projective object.

Recall that we have $\lambda \in \varpi^{-1}(\chi)$. Let P be the object that represents the functor (2.10). Note that P is projective because it represents an exact functor. On the other hand, any nonzero highest weight vector of L_λ corresponds to a nonzero morphism $P \to L_\lambda$. Since L_λ is irreducible, this morphism is surjective. Hence L_λ is isomorphic to a quotient of the projective object P as desired.

□[Proposition 2.6.2]

Definition 2.6.7. Let $\mathcal{A}$ be an abelian category. We say a morphism $P \to M$ in $\mathcal{A}$ exhibits P as a *projective cover* of M if it satisfies the following conditions:

- The object P is projective;
- The morphism $P \to M$ is an *essential surjection*, i.e., for any proper subobject $Q \subset P$, the composition $Q \to P \to M$ is not surjective.

We have the following formal results about projective covers.

Lemma 2.6.8. *Let $\mathcal{A}$ be an abelian category. If $p_1 : P_1 \to M$ and $p_2 : P_2 \to M$ are projective covers in $\mathcal{A}$, there exists an isomorphism $f : P_1 \xrightarrow{\simeq} P_2$ such that $p_1 = p_2 \circ f$.*

Proof. By the lifting property of P_1, the morphism $p_1 : P_1 \to M$ factors as $P_1 \xrightarrow{f} P_2 \xrightarrow{p_2} M$. The morphism $f : P_1 \to P_2$ must be surjective because otherwise $p_2 : P_2 \to M$ is not essential.

By the lifting property of P_2, the identity morphism $P_2 = P_2$ factors as $P_2 \xrightarrow{g} P_1 \xrightarrow{f} P_2$. The morphism $g : P_2 \to P_1$ must be surjective because otherwise $p_1 : P_1 \to M$ is not essential. But $g : P_2 \to P_1$ is also injective because f is a left inverse of it. It follows that both f and g are isomorphisms.

□

Lemma 2.6.9. *Let $\mathcal{A}$ be an abelian category. Suppose that:*

(i) *The category $\mathcal{A}$ has enough projective objects;*
(ii) *The category $\mathcal{A}$ is both Artinian and Noetherian.*

Then any object $M \in \mathcal{A}$ admits a projective cover.

Proof. Let $p : P \to M$ be a surjection such that P is projective and $\mathsf{Length}(P)$ is minimal among all such surjections. We claim that p is an essential surjection.

Suppose that $Q \xrightarrow{i} P$ is a subobject such that $q := p \circ i$ is surjective. We only need to show that i is an isomorphism. We can assume that $\mathsf{Length}(Q)$ is minimal among all such subobjects. By the lifting property of P, the morphism $p : P \to M$ factors as $P \xrightarrow{r} Q \xrightarrow{q} M$. Consider the composition $Q \xrightarrow{i} P \xrightarrow{r} Q$. It must be surjective because otherwise $\mathsf{Im}(r \circ i) \subsetneq Q$ would contradict the minimal assumption about $\mathsf{Length}(Q)$. But then $r \circ i$ must be an isomorphism because $\mathsf{Length}(\mathsf{ker}(r \circ i)) = \mathsf{Length}(Q) - \mathsf{Length}(Q) = 0$. It follows that Q is a direct summand of P, and therefore a projective object. Then we must have $Q \simeq P$ because of the minimal assumption about $\mathsf{Length}(P)$. □

Corollary 2.6.10. *Any object in $\mathcal{O}$ admits a projective cover.*

Notation 2.6.11. For any $\lambda \in \mathfrak{t}^*$, we denote a projective cover of L_λ by P_λ.

Remark 2.6.12. Note that P_λ is well-defined up to (non-unique) isomorphisms.

Exercise 2.6.13 (Problem 3, Homework 4)**.** For dot-dominant weight λ, show that $P_\lambda \simeq M_\lambda$.

Exercise 2.6.14 (Problem 4, Homework 4)**.** Show that $P_\lambda \to L_\lambda$ factors as $P_\lambda \to M_\lambda \to L_\lambda$, and the obtained morphism $P_\lambda \to M_\lambda$ is a projective cover.

Lemma 2.6.15. *Let $\mathcal{A}$ be an abelian category and $L \in \mathcal{A}$ be an irreducible object. Then any projective cover P of L is indecomposable.*

Proof. Suppose that $P \simeq P_1 \bigoplus P_2$ is a nontrivial decomposition. Then both P_1 and P_2 are projective objects. Since $P \to L$ is essential, both $P_1 \to P \to L$ and $P_2 \to P \to L$ are not surjective. Since L is irreducible, both compositions are zero. This implies $P \to L$ is zero, which contradicts the assumption on P. □

Corollary 2.6.16. *For any $\lambda \in \mathfrak{t}^*$, the object P_λ is indecomposable.*

We have the following formal results about indecomposble projective objects.

Lemma 2.6.17. *Let $\mathcal{A}$ be an abelian category and P be an indecomposable projective object with finite length. Then P admits a unique irreducible quotient object.*

Proof. The claim about existence is obvious. For uniqueness, suppose that $P \to L_1$ and $P \to L_2$ are two non-isomorphic irreducible quotients. Let $K_1 \subset P$ and $K_2 \subset P$ be the kernels. Then the morphism $K_1 \oplus K_2 \to P$ must be surjective. By the lifting property of P, the identity morphism $P = P$ factors as $P \to K_1 \oplus K_2 \to P$. Let ϕ_i be the composition $P \to K_i \to P$. Then $\phi_1 + \phi_2 = \mathsf{Id}$. Note that neither ϕ_1 nor ϕ_2 are isomorphisms. This contradicts the following well-known result.

Lemma 2.6.18. *Let $\mathcal{A}$ be any abelian category and $M \in \mathcal{A}$ be an indecomposable object of finite length. Show that*

(i) *Any endomorphism* $\phi : M \to M$ *is either nilpotent or invertible.*
(ii) *If* $\phi_1, \phi_2 : M \to M$ *are endomorphisms such that* $\phi_1 + \phi_2$ *is invertible, then one of them is invertible.*

Proof. For (i), suppose ϕ is not nilpotent. The descending chain $M \supseteq \mathsf{Im}(\phi) \supseteq \mathsf{Im}(\phi^2) \supseteq \cdots$ must stabilize at a nonzero subobject $N \subseteq M$. Then ϕ stabilizes N and $\phi|_N : N \to N$ is an isomorphism. Also, for $n >> 0$, ϕ^n induces a surjection $\varphi_n : M \to N$. By definition $\varphi_n|_N = (\phi|_N)^n$. Hence $\varphi_n|_N$ is also an isomorphism. Then N is a direct summand of M. Since M is indecomposable, we must have $N \simeq M$. This implies that ϕ is an isomorphism.

For (ii), we can replace ϕ_i by $(\phi_1 + \phi_2)^{-1} \circ \phi_i$, and therefore assume $\phi_1 + \phi_2 = \mathsf{id}$. Suppose that ϕ_1 is not invertible. Then ϕ_1 is nilpotent by (i). Then $\phi_2 = \mathsf{id} - \phi_1$ is inverse to $\mathsf{id} + \phi_1 + \phi_1^2 + \cdots$. In particular, ϕ_2 is invertible.

□

□[Lemma 2.6.17]

We leave the proof of the following result to the readers.

Proposition 2.6.19. *Let* $\mathcal{A}$ *be an abelian category satisfying the assumptions in Lemma 2.6.9. Then the map*

$$\begin{array}{rcl} \{\textit{Irreducible objects in } \mathcal{A}\}/\simeq & \to & \{\textit{Indecomposable projective objects in } \mathcal{A}\}/\simeq \\ L & \mapsto & P \end{array}$$

that sends an irreducible object L *to its projective cover* P*, and the map*

$$\begin{array}{rcl} \{\textit{Indecomposable projective objects in } \mathcal{A}\}/\simeq & \to & \{\textit{Irreducible objects in } \mathcal{A}\}/\simeq \\ P & \mapsto & L \end{array}$$

that sends an indecomposable projective object P *to its irreducible quotient object* L*, are inverse to each other.*

Corollary 2.6.20. *The map*

$$\begin{array}{rcl} \mathfrak{h}^* & \to & \{\textit{Indecomposable projective objects in } \mathcal{A}\}/\simeq \\ \lambda & \mapsto & P_\lambda \end{array}$$

is a bijection.

Proposition 2.6.21. *Let* $\mathcal{A}$ *be a* $\mathbb{C}$*-linear abelian category satisfying the assumptions in Lemma 2.6.9. Let* P *and* L *be objects corresponding to each other via the bijection in Proposition 2.6.19. Then for any* $M \in \mathcal{A}$*, we have*

$$\dim_{\mathbb{C}} \mathsf{Hom}_{\mathcal{A}}(P, M) = [M : L].$$

Proof. Using dévissage, we can reduce to the case when M is irreducible. By Lemma 2.6.17 and Schur's lemma, both sides are 0 if M is not isomorphic to L, and are 1 if otherwise.

□

Corollary 2.6.22. *Let* λ *and* μ *be weights. We have*

$$\dim_{\mathbb{C}} \mathsf{Hom}_{\mathcal{O}}(P_\mu, M_\lambda) = [M_\lambda : L_\mu].$$

2.7. Duality.

Definition 2.7.1. Let M be a weight $\mathfrak{h}$-module. We define its *dual weight module* as

$$M^{*,\mathsf{wt}} := \bigoplus_{\lambda\in\mathfrak{h}^*} (M^{\mathsf{wt}=\lambda})^*$$

where $(M^{\mathsf{wt}=\lambda})^*$ is an $\mathfrak{h}$-module of weight $-\lambda$.

Remark 2.7.2. Let M be a weight $\mathfrak{h}$-module. The embedding

$$M^{*,\mathsf{wt}} := \bigoplus_{\lambda\in\mathfrak{h}^*} (M^{\mathsf{wt}=\lambda})^* \to \prod_{\lambda\in\mathfrak{h}^*} (M^{\mathsf{wt}=\lambda})^* =: M^*$$

identifies $M^{*,\mathsf{wt}}$ with the maximal submodule of M^* such that the $\mathfrak{h}$-action is locally finite.

Lemma 2.7.3. *Let M be a weight $\mathfrak{g}$-module, then $M^{*,\mathsf{wt}}$ is a sub-$\mathfrak{g}$-module of M^*.*

Proof. Follows from the root decomposition of $\mathfrak{g}$.

□

For a Verma module M_λ, the above construction would define a *lowest* weight module, which does not belong to $\mathcal{O}$. Hence we need to correct the signs of the weights using the Cartan involution $\tau : \mathfrak{g} \to \mathfrak{g}$. Note that τ induces an involution

$$\tau^* : \mathfrak{g}\mathsf{-mod} \to \mathfrak{g}\mathsf{-mod}$$

on the cateogry $\mathfrak{g}\mathsf{-mod}$.

Definition 2.7.4. Let M be a weight $\mathfrak{g}$-module such that each weight subspace of M is finite-dimensional. We define

$$M^\vee := \tau^*(M^{*,\mathsf{wt}}),$$

and call it the *contragredient dual* of M.

Remark 2.7.5. Note that

$$(M^\vee)^{\mathsf{wt}=\lambda} \simeq (M^{\mathsf{wt}=\lambda})^*.$$

In particular, the dimensions of the λ-weight subspaces of $M^\vee$ and M are equal.

Note that the construction $M \mapsto M^\vee$ is contravariantly functorial. Moreover, we have $(M^\vee)^\vee \simeq M$.

Proposition 2.7.6. *The category $\mathcal{O}$ is stable under the contragredient duality. In particular, we have an involution functor*

$$\mathcal{O}^{\mathsf{op}} \to \mathcal{O},\ \ M \mapsto M^\vee.$$

Proof. Recall that the subcategory $\mathcal{O} \subseteq \mathfrak{g}\mathsf{-mod}^{\mathsf{wt}}$ is stable under taking extensions. Hence we only need to show that $(L_\lambda)^\vee$ belongs to $\mathcal{O}$ for any weight λ. This follows from the following result.

Exercise 2.7.7 (Problem 5, Homework 4)**.** Show that $L_\lambda^\vee \simeq L_\lambda$ for any weight λ

□[Proposition 2.7.6]

Note that Exercise 2.7.7 also implies the following result.

Corollary 2.7.8. *Each block $\mathcal{O}_\chi$ of $\mathcal{O}$ is stable under the contragredient duality.*

Definition 2.7.9. For any weight λ, we call $M_\lambda^\vee$ the *dual Verma module* corresponding to λ.

Remark 2.7.10. The Verma modules are also called the *standard objects* in $\mathcal{O}$. Consequently, the dual Verma modules are called the *costandard objects*.

Corollary 2.7.11. *For any weight λ, the dual Verma module $M_\lambda^\vee$ has a unique irreducible submodule, and this submodule is isomorphic to L_λ.*

Remark 2.7.12. Note that the isomorphism $L_\lambda \to L_\lambda^\vee$ is unique up to a scaler. It follows that the injection $L_\lambda \to M_\lambda^\vee$ is unique up to a scaler.

The following result is often referred as the (non-derived) *orthogonal property* between Verma and dual Verma modules.

Lemma 2.7.13. *For weights λ and μ, we have*

$$\dim \mathsf{Hom}_{\mathcal{O}}(M_\lambda, M_\mu^\vee) = \delta_{\lambda,\mu}.$$

Proof. Knowing a $\mathfrak{g}$-linear map $M_\lambda \to M_\mu^\vee$ is equivalent to knowing a λ-weight vector v in $M_\mu^\vee$ such that $\mathfrak{n} \cdot v = 0$. By definition, this is equivalent to knowing a linear map $f : M_\mu \to \mathbb{C}$ such that

- The map f factors as $M_\mu \to M_\mu^{\mathsf{wt}=\lambda} \to \mathbb{C}$;
- The map f sends $\mathfrak{n}^- \cdot M_\mu$ to zero.

Here the second condition is due to the fact $\tau(\mathfrak{n}) = \mathfrak{n}^-$. Since M_μ is free over $U(\mathfrak{n}^-)$, we have $M_\mu \simeq (\mathfrak{n}^- \cdot M_\mu) \oplus M_\mu^{\mathsf{wt}=\mu}$. Hence $\lambda \neq \mu$ implies $f = 0$. In the case $\lambda = \mu$, f is determined by a linear map $M_\lambda^{\mathsf{wt}=\lambda} \to \mathbb{C}$, and the space of such maps is 1-dimensional.

□

Lemma 2.7.14. *For weights λ and μ, we have*

$$\mathsf{Ext}^1_{\mathcal{O}}(M_\lambda, M_\mu^\vee) = 0.$$

Proof. We need to show any fshort exact sequence in $\mathcal{O}$ of the form

$$0 \to M_\mu^\vee \to N \to M_\lambda \to 0 \tag{2.11}$$

splits. Consider the $\mathfrak{b}$-linear map $\mathbb{C}_\lambda \to M_\lambda$. By pullback along this map, we obtain a short exact sequence of $\mathfrak{b}$-modules:

$$0 \to M_\mu^\vee \to N' \to \mathbb{C}_\lambda \to 0.$$

By the universal property of M_λ, we only need to show this sequence splits.

In the case when $\lambda < \mu$ does not hold, any λ-weight vector of N' would provide a desired splitting.

In the case when $\lambda < \mu$ holds, (2.11) induces a short exact sequence

$$0 \to M_\lambda^\vee \to N^\vee \to M_\mu \to 0,$$

which allows us to reduce to the previous case.

□

2.8. **Standard filtrations.** One of the goals of this subsection is to prove the following result.

Theorem 2.8.1. *For weights λ and μ, we have*

$$\mathsf{Ext}^i_{\mathcal{O}}(M_\lambda, M_\mu^\vee) = 0, \quad \forall i > 0.$$

Definition 2.8.2. We say an object $M \in \mathcal{O}$ *admits a standard filtration* if there exists a finite filtration of M such that each graded piece is a Verma module.

Let $\mathcal{O}^{\Delta} \subset \mathcal{O}$ be the full subcategory of objects admitting standard filtrations.

Example 2.8.3. A generalized Verma modules $M_{\lambda,n}$ admits a standard filtration.

We will prove the following stronger result.

Theorem 2.8.4. *For an object $M \in \mathcal{O}$, the following conditions are equivalent:*

(a) *The object M admits a finite filtration such that each graded piece is a Verma module.*
(b) *For any weight μ and $i > 0$, $\mathsf{Ext}^i_{\mathcal{O}}(M, M_\mu^\vee) = 0$.*
(c) *For any weight μ, $\mathsf{Ext}^1_{\mathcal{O}}(M, M_\mu^\vee) = 0$.*

Lemma 2.8.5. *Let $M \in \mathcal{O}^{\Delta}$ be an object admitting a standard filtration $\mathsf{F}^{\le \bullet}M$ of length m. Let $v \in M$ be a nonzero highest weight vector with weight λ, and $M_\lambda \to M$ be the unique $\mathfrak{g}$-linear map sending v_λ to v. Let i be the smallest index such that the image $\mathsf{Im}(f)$ is contained in $\mathsf{F}^{\le i}M$. Then:*

(1) *The composition $M_\lambda \to M \to \mathsf{gr}^i M$ is an isomorphism. In particular, $M_\lambda \to M$ is injective.*
(2) *The quotient M/M_λ admits a standard filtration of length $m-1$.*

Proof. By assumption $\mathsf{gr}^i M$ is a Verma module M_μ and the composition $M_\lambda \to M \to \mathsf{gr}^i M$ is nonzero. This implies $\lambda \le \mu$. Since λ is a highest weight of M, we have $\lambda = \mu$. This implies $M_\lambda \to M \to \mathsf{gr}^i M$ is an isomorphism. This proves (1).

For (2), we have a short exact sequence $0 \to \mathsf{F}^{\le i-1}M \to M/M_\lambda \to M/\mathsf{F}^{\le i}M \to 0$. By assumption, $\mathsf{F}^{\le i-1}M$ admits a standard filtration of length $i-1$, and $M/\mathsf{F}^{\le i}M$ admits a standard filtration of length $m-i$. It follows that M/M_λ admits a standard filtration of length $(i-1)+(m-i) = m-1$ as desired.

□

Lemma 2.8.6. *The subcategory $\mathcal{O}^{\Delta} \subseteq \mathcal{O}$ is stable under taking direct summands.*

Proof. Let $M \in \mathcal{O}^{\Delta}$ be an object and $M \simeq M' \bigoplus M''$ be a decomposition in $\mathcal{O}$. We will show that M' and M'' belong to $\mathcal{O}^{\Delta}$.

We prove the above claim by induction on the length m of the shortest possible standard filtration of M. The claim is trivial if $m = 0$. For $m > 0$, without lose of generality, we can assume M' contains a nonzero highest weight vector v of M. Let λ be the weight of v. By Lemma 2.8.5, we have injections $M_\lambda \to M' \to M$ such that M/M_λ admits a standard filtration of length $m-1$. Since $M/M_\lambda \simeq M'/M_\lambda \oplus M''$, by induction hypothesis, M'/M_λ and $M'' \in \mathcal{O}^{\Delta}$ belong to $\mathcal{O}^{\Delta}$. Note that M' also belong to $\mathcal{O}^{\Delta}$ because it is an extension of M'/M_λ by M_λ.

□

Lemma 2.8.7. *Let $0 \to K \to M \to N \to 0$ be a short exact sequence in $\mathcal{O}$ such that $N \in \mathcal{O}^{\Delta}$. Then $M \in \mathcal{O}^{\Delta}$ iff $K \in \mathcal{O}^{\Delta}$.*

Proof. The "if" claim is obvious. For the "only if" claim, using induction, we can reduce to the case when N is a Verma module M_μ. Let $M_\lambda \to M$ be as in Lemma 2.8.5. Then $M/M_\lambda \in \mathcal{O}^{\Delta}$. Note that the composition $M_\lambda \to M \to M_\mu$ is either 0 or an isomorphism by considering the weights. If this is the zero map, then

$M/M_\lambda \to M_\mu$ is still a surjection and we can finish the proof by using induction. Otherwise $M \simeq K \oplus M_\mu$ and the claim follows from Lemma 2.8.6.

□

Proposition 2.8.8. *Every projective object of $\mathcal{O}$ admits a standard filtration.*

Proof. In the proof of Proposition 2.6.2, we have seen that any object M in $\mathcal{O}$ is a quotient of a direct sum of generalized Verma modules. In particular, M is a quotient of an object S that belongs to $\mathcal{O}^\Delta$. Note that M is a direct summand of S if M is projective. Hence the claim follows from Lemma 2.8.6.

□

Proof of Theorem 2.8.4. We first prove $(a) \Rightarrow (b)$. We prove by induction on i. The $i = 1$ case follows from Lemma 2.7.14. For $i > 1$, let

$$0 \to N \to P \to M \to 0$$

be a short exact sequence such that P is projective. We have

$$\mathsf{Ext}^{i-1}_{\mathcal{O}}(N, M_\mu^\vee) \simeq \mathsf{Ext}^{i}_{\mathcal{O}}(M, M_\mu^\vee).$$

By Theorem 2.8.8 and Lemma 2.8.7, $N \in \mathcal{O}^\Delta$. Hence $\mathsf{Ext}^{i-1}_{\mathcal{O}}(N, M_\mu^\vee) \simeq 0$ by the induction hypothesis.

$(b) \Rightarrow (c)$ is obvious.

We now prove $(c) \Rightarrow (a)$ by induction on $\mathsf{Length}(M)$. The case $\mathsf{Length}(M) = 0$ is obvious. If $\mathsf{Length}(M) > 0$, let λ be a highest weight vector of M. There is a nonzero $\mathfrak{g}$-linear map $M_\lambda^{\oplus n} \to M$ that restricts to an bijection between λ-weight subspaces. Let N_1 and N_2 be the kernel and cokernel of this map, and $M' \neq 0$ be the image of it. We have short exact sequences

$$0 \to N_1 \to M_\lambda^{\oplus n} \to M' \to 0,$$
$$0 \to M' \to M \to N_2 \to 0.$$

They induce long exact sequences

$$0 \to \mathsf{Hom}_{\mathcal{O}}(M', M_\mu^\vee) \to \mathsf{Hom}_{\mathcal{O}}(M_\lambda^{\oplus n}, M_\mu^\vee) \to \mathsf{Hom}_{\mathcal{O}}(N_1, M_\mu^\vee) \to \mathsf{Ext}^1_{\mathcal{O}}(M', M_\mu^\vee) \to \cdots$$

$$\cdots \to \mathsf{Hom}_{\mathcal{O}}(M, M_\mu^\vee) \to \mathsf{Hom}_{\mathcal{O}}(M', M_\mu^\vee) \to \mathsf{Ext}^1_{\mathcal{O}}(N_2, M_\mu^\vee) \to \mathsf{Ext}^1_{\mathcal{O}}(M, M_\mu^\vee) \to$$
$$\to \mathsf{Ext}^1_{\mathcal{O}}(M', M_\mu^\vee) \to \mathsf{Ext}^2_{\mathcal{O}}(N_2, M_\mu^\vee) \to \cdots$$

By assumption, $\mathsf{Ext}^1_{\mathcal{O}}(M, M_\mu^\vee) = 0$. We claim $\mathsf{Ext}^1_{\mathcal{O}}(N_2, M_\mu^\vee) = 0$. There are two cases.

- In the case when $\lambda \neq \mu$, Lemma 2.7.13 implies $\mathsf{Hom}_{\mathcal{O}}(M_\lambda^{\oplus n}, M_\mu^\vee) = 0$. Then the first long exact sequence implies $\mathsf{Hom}_{\mathcal{O}}(M', M_\mu^\vee) = 0$. Then the second long exact sequence implies $\mathsf{Ext}^1_{\mathcal{O}}(N_2, M_\mu^\vee) = 0$.
- In the case when $\lambda = \mu$, note that μ is a highest weight of both M' and M, and we have $(M')^{\mathsf{wt}=\mu} \simeq M^{\mathsf{wt}=\mu}$. It follows that

  $$\mathsf{Hom}_{\mathcal{O}}(M', M_\mu^\vee) \simeq ((M')^{\mathsf{wt}=\mu})^* \simeq (M^{\mathsf{wt}=\mu})^* \simeq \mathsf{Hom}_{\mathcal{O}}(M, M_\mu^\vee).$$

  By the second long exact sequence sequence, $\mathsf{Ext}^1_{\mathcal{O}}(N_2, M_\mu^\vee) = 0$ as desired.

Note that $\mathsf{Length}(N_2) < \mathsf{Length}(M)$. By the induction hypothesis, $N_2 \in \mathcal{O}^\Delta$. Using $(a) \Rightarrow (b)$, we get $\mathsf{Ext}^2_{\mathcal{O}}(N_2, M_\mu^\vee) = 0$. By the second long exact sequence, $\mathsf{Ext}^1_{\mathcal{O}}(M', M_\mu^\vee) = 0$. Now we have two cases.

- In the case when $N_2 \neq 0$, we have $\mathsf{Length}(M') < \mathsf{Length}(M)$. By the induction hypothesis, $M' \in \mathcal{O}^{\Delta}$. Then $M \in \mathcal{O}^{\Delta}$ because it is an extension of N_2 by M'.
- In the case when $N_2 = 0$, we have $M \simeq M'$. An argument similar to that in the last paragraph implies $\mathsf{Hom}_{\mathcal{O}}(N_1, M_\mu^\vee) = 0$. Note that μ can be any weight. This implies $N_1 = 0$ and therefore $M \simeq M_\lambda^{\oplus n} \in \mathcal{O}^{\Delta}$ as desired.

□[Theorem 2.8.4]
□[Theorem 2.8.1]

We now turn to the study of general standard filtrations.

Proposition-Definition 2.8.9. *Let M be an object in $\mathcal{O}^{\Delta}$ and $\mathsf{F}^{\leq\bullet}M$ be a standard filtration of M. For any weight λ, we have*

$$|\{i \,|\, \mathsf{gr}^i M \simeq M_\lambda\}| = \dim_{\mathbb{C}}(\mathsf{Hom}_{\mathcal{O}}(M, M_\lambda^\vee)).$$

In particular, the LHS does not depend on the choice of the standard filtration. We call this number the multiplicity of M_λ in M, *and denote it by* $(M : M_\lambda)$.

Proof. Follows from Lemma 2.7.13 and Theorem 2.8.1 by a dévissage argument. □

Theorem 2.8.10 (BGG reciprocity)**.** *For weights $\lambda, \mu \in \mathfrak{t}^*$, we have*

$$(P_\mu : M_\lambda) = [M_\lambda^\vee : L_\mu] = [M_\lambda : L_\mu].$$

Proof. The last identity follows from $L_\mu^\vee \simeq L_\mu$. The first one follows from Proposition-Definition 2.8.9 and Corollary 2.6.22. □

Remark 2.8.11. The previous discussions on standard filtrations and BGG reciprocity can be axiomized using the language of *highest weight categories*. See [CPS].

Remark 2.8.12. In the category $\mathcal{O}$, or any highest weight category, a *tilting object* is an object that admits both a standard filtration and a costandard filtration. One can show that the set of indecomposable tilting objects is bijective to the set of irreducible objects. Tilting objects play an important role in representation theory. See [H2, Chapter 11] and [BBM].

2.9. **Translation functors.** In this subsection, among other things, we show that all integral regular blocks of $\mathcal{O}$ are equivalent.

Construction 2.9.1. Let V be a finite-dimensional $\mathfrak{g}$-module. Consider the functor

$$\mathfrak{g}\text{−}\mathsf{mod} \to \mathfrak{g}\text{−}\mathsf{mod},\ M \mapsto V \otimes M,$$

where $\mathfrak{g}$ acts on $V \otimes M$ according to the Lebniz rule. It is easy to see that this functor preserves $\mathcal{O}$. We denote the obtained functor by $T_V : \mathcal{O} \to \mathcal{O}$.

Lemma 2.9.2. *The functor $T_V : \mathcal{O} \to \mathcal{O}$ commutes with the contragredient duality.*

Proof. For any finite dimensional $\mathfrak{g}$-module, we have $V^\vee \simeq V$ because $L_\lambda^\vee \simeq L_\lambda$. This implies T_V commutes with the contragredient duality. □

Exercise 2.9.3 (Problem 6, Homework 4)**.** Show that T_{V^*} is both left and right adjoint to T_V.

Remark 2.9.4. The same argument shows that T_{V^*} and T_V are adjoint in the derived sense, i.e.

$$\mathsf{Ext}^i(T_V(M), N) \simeq \mathsf{Ext}^i(M, T_{V^*}(N)).$$

Lemma 2.9.5. *For any weight λ, the module $T_V(M_\lambda)$ admits a standard filtration $\mathsf{F}^{\leq\bullet}(T_V(M_\lambda))$ such that the highest weights of $\mathsf{gr}^k(T_V(M_\lambda))$ form a (non-strictly) decreasing sequence. Moreover,*

$$(T_V(M_\lambda) : M_\mu) = \dim V^{\mathsf{wt}=\mu-\lambda}.$$

Proof. Follows from the *projection formula*

$$\mathsf{Ind}_{\mathfrak{b}}^{\mathfrak{g}}(V_1) \otimes V_2 \simeq \mathsf{Ind}_{\mathfrak{b}}^{\mathfrak{g}}(V_1 \otimes V_2),\ V_1 \in \mathfrak{b}\text{-}\mathsf{mod}, V_2 \in \mathfrak{g}\text{-}\mathsf{mod}.$$

Namely, any finite-dimensional weight $\mathfrak{b}$-module, such as $V \otimes \mathbb{C}_\lambda$, admits a finite filtration whose graded pieces are 1-dimensional modules with decreasing weights. □

Definition 2.9.6. Let χ_1, χ_2 be two central characters. For any finite-dimensional $\mathfrak{g}$-module V, consider the composition

$$T_{\chi_1,V,\chi_2} : \mathcal{O}_{\chi_1} \to \mathcal{O} \xrightarrow{T_V} \mathcal{O} \to \mathcal{O}_{\chi_2},$$

where the first and third functors are respectively the embedding and projection functors for the blocks. We call T_{χ_1,V,χ_2} the *translation functor* from $\mathcal{O}_{\chi_1}$ to $\mathcal{O}_{\chi_2}$ given by V.

Corollary 2.9.7. *The functor T_{χ_1,V,χ_2} is both left and right adjoint to T_{χ_2,V^*,χ_1}.*

Construction 2.9.8. Let λ and μ be weights such that $\mu - \lambda$ is integral. Then the W-orbit $W(\mu - \lambda)$ (for the linear W-action on $\mathfrak{h}^*$) contains a unique dominant integral weight ν. Consider the finite-dimensional $\mathfrak{g}$-module L_ν. We write

$$T_\lambda^\mu := T_{\varpi(\lambda), L_\nu, \varpi(\mu)}$$

for the translation functor from $\mathcal{O}_{\varpi(\lambda)}$ to $\mathcal{O}_{\varpi(\mu)}$ given by L_ν.

Exercise 2.9.9 (Problem 7, Homework 4)**.** Let λ, μ and ν be as in Construction 2.9.8. Show that

$$T_\mu^\lambda = T_{\varpi(\mu), L_{-w_0(\nu)}, \varpi(\lambda)}.$$

Moreover, T_μ^λ is both left and right adjoint to T_λ^μ.

To proceed, we need the following definition.

Definition 2.9.10. Let λ and μ be real weights, i.e., elements in $\mathfrak{h}_{\mathbb{R}}^* := \mathbb{R}\Phi$. We say λ and μ belongs to the same *dot-Weyl facet* if the signs of $\langle \lambda + \rho, \check{\alpha} \rangle$ and $\langle \mu + \rho, \check{\alpha} \rangle$ are the same for any $\alpha \in \Phi$.

For $\lambda \in \mathfrak{h}_{\mathbb{R}}^*$, we write $F_\lambda \subseteq \mathfrak{h}_{\mathbb{R}}^*$ for the dot-Weyl facet containing λ.

Definition 2.9.11. For a dot-Weyl facet F_λ, its *upper closure* F_λ^+ consists of real weights μ such that

$$\langle \mu + \rho, \check{\alpha} \rangle \begin{cases} > 0 & \text{if } \langle \lambda + \rho, \check{\alpha} \rangle > 0, \\ = 0 & \text{if } \langle \lambda + \rho, \check{\alpha} \rangle = 0, \\ \leq 0 & \text{if } \langle \lambda + \rho, \check{\alpha} \rangle < 0 \end{cases}$$

Exercise 2.9.12. Draw the dot-Weyl facets for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$. Find the upper closure of each facet.

Remark 2.9.13. When equipped with the standard topology, F_λ is a locally closed subset of $\mathfrak{h}_R^*$, and both the closure $\overline{F}_\lambda$ and the upper closure F_λ^+ of F_λ are unions of dot-Weyl facets.

Definition 2.9.14. We say a dot-Weyl facet F_λ is a *dot-Weyl chamber* if $\langle \mu+\rho, \check{\alpha}\rangle$ is nonzero for any $\alpha \in \Phi$.

Remark 2.9.15. Note that each dot-Weyl facet is contained in the upper closure of a unique *dot-Weyl chamber.* Also, λ is dot-regular iff F_λ is a dot-Weyl chamber.

We have the following theorem. For complete proofs and its generalization to non-integral weights, see [H2, Sect. 7].

Theorem 2.9.16. *Let λ and μ be dot-anti-dominant integral weights such that $F_\mu \subseteq \overline{F_\lambda}$. Then for any $w \in W$, we have*

$$T_\lambda^\mu(M_{w\cdot\lambda}) \simeq M_{w\cdot\mu},\;\; T_\lambda^\mu(M_{w\cdot\lambda}^\vee) \simeq M_{w\cdot\mu}^\vee$$

and

$$T_\lambda^\mu(L_{w\cdot\lambda}) \simeq \begin{cases} L_{w\cdot\mu} & \textit{if } F_{w\cdot\mu} \subseteq F_{w\cdot\lambda}^+, \\ 0 & \textit{otherwise.} \end{cases}$$

Sketch. Let $\nu \in W(\mu-\lambda)$ be the unique dominant integral weight in this orbit. By Lemma 2.9.5, $T_\lambda^\mu(M_{w\cdot\lambda})$ admits a standard filtration and we have

$$(T_\lambda^\mu(M_{w\cdot\lambda}) : M_{w'\cdot\mu}) = \dim L_\nu^{\mathsf{wt}=w'\cdot\mu-w\cdot\lambda}.$$

The RHS is nonzero unless $w'\cdot\mu - w\cdot\lambda \leq \nu$. Now a combinatorial argument (see [H2, Lemma 7.5]) shows that the latter can happen only if $w'\cdot\mu = w\cdot\mu$. Note that $w\cdot\mu - w\cdot\lambda = w(\mu-\lambda) \in W(\nu)$. This implies $L_\nu^{\mathsf{wt}=w'\cdot\mu-w\cdot\lambda}$ is 1-dimensional, and therfore $T_\lambda^\mu(M_{w\cdot\lambda}) \simeq M_{w\cdot\mu}$ as desired.

The claim for the dual Verma modules follows from the contragredient duality.

Now consider the chain $M_{w\cdot\lambda} \to L_{w\cdot\lambda} \to M_{w\cdot\lambda}^\vee$. Since T_λ^μ is exact, we obtain a chain $M_{w\cdot\mu} \to T_\lambda^\mu(L_{w\cdot\lambda}) \to M_{w\cdot\mu}^\vee$. By Lemma 2.7.13, $T_\lambda^\mu(L_{w\cdot\lambda})$ is either zero or isomorphic to $L_{w\cdot\mu}$.

It remains to determine when these two cases happen. Since the exact functor T_λ^μ sends $M_{w\cdot\lambda}$ to $M_{w\cdot\mu}$, there exists a unique composition factor $L_{w'\cdot\lambda}$ of $M_{w\cdot\lambda}$ such that $T_\lambda^\mu(L_{w'\cdot\lambda}) \simeq L_{w\cdot\mu}$. By the same argument as in the first paragraph of this proof, we have $w'\cdot\mu = w\cdot\mu$. There are two cases.

In the case when $F_{w\cdot\mu}$ is contained in the upper closure $F_{w\cdot\lambda}^+$, a combinatorial argument shows that $w'\cdot\lambda = w\cdot\lambda$ and therefore $T_\lambda^\mu(L_{w\cdot\lambda}) \simeq L_{w\cdot\mu}$ as desired.

In the case when $F_{w\cdot\mu}$ is not contained in the upper closure $F_{w\cdot\lambda}^+$, there exists a reflection s_α such that $s_\alpha w\cdot\lambda \leq w\cdot\lambda$ while $s_\alpha w\cdot\mu = w\cdot\mu$. By the Verma–BGG theorem, there exists a proper embedding $M_{s_\alpha w\cdot\lambda} \to M_{w\cdot\lambda}$. Note that T_λ^μ sends $M_{s_\alpha w\cdot\lambda}$ and $M_{w\cdot\lambda}$ to isomorphic objects. This implies T_λ^μ sends $M_{s_\alpha w\cdot\lambda} \to M_{w\cdot\lambda}$ to an isomorphism. It follows that the quotient object $M_{w\cdot\lambda}/M_{s_\alpha w\cdot\lambda} \simeq L_{w\cdot\lambda}$ is sent to zero by T_λ^μ.

□

The following result is a formal consequence of the above theorem:

Theorem 2.9.17. *Let λ and μ be dot-anti-dominant integral weights such that $F_\lambda = F_\mu$. Then $T_\lambda^\mu : \mathcal{O}_{\varpi(\lambda)} \to \mathcal{O}_{\varpi(\mu)}$ is an equivalence*

Proof. By the previous theorem, the exact functor $F := T_\lambda^\mu$ is left adjoint to the exact functor $G := T_\mu^\lambda$, and they induce a bijection between the sets of irreducible objects. In general, such adjoint functors are inverse to each other.

Indeed, we only to show the adjunctions $\mathsf{Id} \to G \circ F$ and $F \circ G \to \mathsf{Id}$ are equivalences. We will prove the first equivalence and leave the second to the readers. Note that for any object M, the objects M and $G \circ F(M)$ have the same composition factors and multiplicities. Hence we only need to show $M \to G \circ F(M)$ is injective. Since F sends nonzero objects to nonzero objects, we only need to show $F(M) \to F \circ G \circ F(M)$ is injective. By the axiom of adjoint functors, this morphism has a left inverse, hence is injective as desired.

□

Corollary 2.9.18. *Any regular integral block of $\mathcal{O}$ is equivalent to the principle block $\mathcal{O}_{\varpi(0)}$ via a translation functor.*

Construction 2.9.19. Let μ be a dot-anti-dominant integral weight. The functor $T_{-\rho}^\mu : \mathcal{O}_{\varpi(-\rho)} \to \mathcal{O}_{\varpi(\mu)}$ is *not* covered by the above theorems (although its adjoint $T_\mu^{-\rho}$ is). Recall that the most singular block $\mathcal{O}_{\varpi(-\rho)}$ is semi-simple and contains a unique irreducible object $L_{-\rho}$, and we have $L_{-\rho} \simeq M_{-\rho} \simeq M_{-\rho}^\vee \simeq P_{-\rho}$. It follows that

$$\Xi_\mu := T_{-\rho}^\mu(L_{-\rho})$$

is both projective and injective.

Exercise 2.9.20 (Problem 8, Homework 4)**.** Let μ be any dot-anti-dominant integral weight. Prove that:

(1) For any $w \in W$, $(\Xi_\mu : M_{w\cdot\mu}) = 1$ and there is a surjection $\Xi_\mu \to M_\mu$.
(2) For any $w \in W$, $(P_\mu : M_{w\cdot\mu}) \geq 1$ and there is a surjection $P_\mu \to M_\mu$.
(3) There exists an isomorphism $\Xi_\mu \simeq P_\mu$ compatible with the surjections to M_μ.
(4) For any $w \in W$, $[M_{w\cdot\mu} : L_\mu] = 1$.

Remark 2.9.21. For dot-anti-dominant weight μ, the object P_μ is called the *big projective module*, which plays an important role in representation theory.

3. Flag variety

Throughout this section, let G be a *simply connected* semisimple algebraic group over $\mathbb{C}$, and $\mathfrak{g}$ be the Lie algebra of G. Recall that

$$G\text{–}\mathsf{mod}_{\mathsf{fd}} \simeq \mathfrak{g}\text{–}\mathsf{mod}_{\mathsf{fd}}.$$

In other words, any *finite-dimensional* $\mathfrak{g}$-module is G-integrable.

3.1. Definition of flag variety.

Recall that a *Borel subgroup* of G is a maximal connected solvable subgroup of G. We have the following fundamental result.

Theorem 3.1.1. *Let B be a Borel subgroup of G. Then the quotient scheme G/B exists and is smooth and projective over $\mathbb{C}$.*

Remark 3.1.2. For any algebraic group H over a field k, and for any subgroup K of H, the quotient H/K exists (see [M, Section 5.c]). This means that the fppf sheafification of the functor $\mathsf{CAlg}_k \to \mathsf{Set}$, $R \mapsto H(R)/K(R)$ is represented by a k-scheme H/K, and the map $H \to H/K$ exhibits H as a fppf K-torsor on H/K. When $\mathsf{char}(k) = 0$, the fppf topology can be replaced by the étale topology.

The quotient H/K is always quasi-projective over k (see [M, Section 8.k]).

The quotient G/B is a smooth $\mathbb{C}$-scheme because G and B are smooth.

The proof of the completeness of the quotient G/B requires several representation theoretic ingredients (see [M, Section 17]), including Chevalley's theorem[4] and the Lie–Kolchin theorem[5].

Example 3.1.3. Let $G = \mathsf{SL}_n$ and B be its standard Borel subgroup. Then G/B classifes *complete flags* in $\mathbb{C}^{\oplus n}$, i.e., subspaces $0 = V_0 \subset V_1 \subset \cdots \subset V_n = \mathbb{C}^{\oplus n}$ such that $\dim(V_i) = i$. One can use this description to construct homogenous coordinates for G/B. See [M, Section 7.g].

Theorem 3.1.1, together with the Borel fixed point theorem[6], imply the following results. See [M, Section 17].

Proposition 3.1.4. *Any two Borel pairs of G are conjugate by an element of $G(\mathbb{C})$.*

Proposition 3.1.5. *The normalizer subgroup of a Borel subgroup B in G is equal to B:*

$$N_G(B) = B.$$

Exercise 3.1.6 (Problem 1, Homework 5)**.** For two Borel subgroups B and B' of G, there is a unique G-equivariant isomorphism $\mathfrak{c}_{B,B'} : G/B \xrightarrow{\simeq} G/B'$.

Definition 3.1.7. The *(abstract) flag variety* for G is a $\mathbb{C}$-scheme Fl_G equipped with a system of isomorphisms $\{\mathfrak{r}_B : \mathsf{Fl}_G \xrightarrow{\simeq} G/B\}$ indexed by Borel subgroups B of G, such that for two Borel subgroups B and B', we have

$$\mathfrak{c}_{B,B'} \circ \mathfrak{r}_B = \mathfrak{r}_{B'}.$$

[4] Every subgroup K of an affine algebraic group H can be realized as the stabilizer of a 1-dimensional subspace in a finite-dimensional representation of H. See [M, Section 4.h].

[5] Irreducible representations of a smooth connected solvable algebraic group are 1-dimensional. See [M, Section 16.d].

[6] Any action of a smooth connected solvable algebraic group B on a finite type separated k-scheme X has a fixed k-point.

Remark 3.1.8. The data $(\mathsf{Fl}_G, \{\mathfrak{r}_B\})$ is unique up to unique isomorphism.

Remark 3.1.9. There is a unique G-action on Fl_G such that the isomorphisms $\mathfrak{r}_B$ are G-linear.

Note that we have $(G/B)(\mathbb{C}) \simeq G(\mathbb{C})/B(\mathbb{C})$ because any fppf cover of $\mathsf{Spec}(\mathbb{C})$ admits a section. Hence Proposition 3.1.4 and Proposition 3.1.5 imply the following result.

Corollary 3.1.10. *Let B be a Borel subgroup of G. We have a bijection*

$$(G/B)(\mathbb{C}) \xrightarrow{\sim} \{\textit{Borel subgroups of } G\}, \ gB \mapsto \mathsf{Ad}_g(B). \tag{3.1}$$

Exercise 3.1.11 (Problem 2, Homework 5)**.** Show that the bijection

$$\mathsf{Fl}_G(\mathbb{C}) \xrightarrow{\mathfrak{r}_B} (G/B)(\mathbb{C}) \xrightarrow{(3.1)} \{\text{Borel subgroups of } G\}$$

does not depend on the choice of B.

In other words, we have a *canonical* bijection

$$\mathsf{Fl}_G(\mathbb{C}) \simeq \{\text{Borel subgroups of } G\} \simeq \{\text{Borel subalgebras of } \mathfrak{g}\}.$$

For a $\mathbb{C}$-point x of Fl_G, we often denote the corresponding Borel subgroup (resp. Borel subalgebra) by B_x (resp. $\mathfrak{b}_x$).

Remark 3.1.12. In fact, one can *define* a moduli problem classifying Borel subalgebras of $\mathfrak{g}$ and show that it is represented by the $\mathbb{C}$-scheme Fl_G. Namely, let $n = \dim(G)$ and d be the common dimension of the Borel subgroups. Then the above moduli problem is a closed subfunctor of the Grassmannian $\mathsf{Gr}(d, \mathfrak{g})$ that classifies d-dimensional subspaces of $\mathfrak{g}$.

It is a well-known fact that the obvious map $\mathsf{Gr}(d, V) \to \mathbb{P}(\wedge^d V)$ is a closed immersion, known as the *Plücker embedding*. It follows that the composition $\mathsf{Fl}_G \to \mathsf{Gr}(d, \mathfrak{g}) \to \mathbb{P}(\wedge^d \mathfrak{g})$ gives a closed immersion of Fl_G into a projective space.

3.2. **Bruhat decomposition.** Let H be a Cartan subgroup contained in the chosen Borel subgroup B. Let $N_G(H)$ be the nomalizer subgroup for H inside G. Note that $N_G(H)$ acts on the Lie algebra $\mathfrak{h}$ of H.

We have the following basic result (see [M, Section 21.d]).

Proposition 3.2.1. *There is a unique isomorphism*

$$W \simeq N_G(H)/H$$

compatible with the actions of both sides on $\mathfrak{h}$.

In particular, for any $w \in W$, we can choose a lifting $\widetilde{w} \in N_G(H)$ of it. Note that the double coset $B(\mathbb{C})\widetilde{w}B(\mathbb{C}) \subseteq G(\mathbb{C})$ does not depend on the choice of the lifting. Therefore we denote this double coset by $B(\mathbb{C})wB(\mathbb{C})$.

We have the following fundamental results (see [M, Section 21.h]).

Theorem 3.2.2 (Bruhat decomposition, set version)**.** *We have*

$$G(\mathbb{C}) = \bigsqcup_{w \in W} B(\mathbb{C})wB(\mathbb{C}).$$

Consider the obvious (B, B)-action on G. For any $w \in W$, it is easy to see that the (B, B)-orbit of G that contains $\widetilde{w} \in G(\mathbb{C})$ does not depend on the lifting $\widetilde{w}$. Therefore we denote this orbit by BwB. Note that we have $(BwB)(\mathbb{C}) \simeq B(\mathbb{C})wB(\mathbb{C})$.

Theorem 3.2.3 (Bruhat decomposition, scheme version)**.** *The orbit BwB is a smooth locally closed subscheme of G, and the underlying topological space of G is a union of these orbits.*

Example 3.2.4. For $G = \mathsf{SL}_n$, the Bruhat decomposition follows from Gaussian elimination.

Remark 3.2.5. Using $B = NH$, we have $B(\mathbb{C})wB(\mathbb{C}) = N(\mathbb{C})wB(\mathbb{C}) = B(\mathbb{C})wN(\mathbb{C})$ and similarly $BwB = NwB = BwN$.

Taking quotient for the right B-action, we obtain B-orbits

$$\mathsf{Fl}_G^{=w} := \mathfrak{r}_B^{-1}(BwB/B) \subseteq \mathsf{Fl}_G$$

of the abstract flag variety, where $\mathfrak{r}_B : \mathsf{Fl}_G \xrightarrow{\simeq} G/B$ is the canonical identification.

Definition 3.2.6. The orbit $\mathsf{Fl}_G^{=w}$ is called the *Bruhat cell* of Fl_G labelled by w.

Corollary 3.2.7 (Bruhat decomposition for Fl_G)**.** *The orbit $\mathsf{Fl}_G^{=w}$ is a smooth locally closed subscheme of Fl_G, and the underlying topological space of Fl_G is a union of these orbits.*

Example 3.2.8. For $G = \mathsf{SL}_2$, via the isomorphism $\mathsf{Fl}_G \simeq \mathbb{P}^1$ in Example 1.5.1. We have $\mathsf{Fl}_G^{=\mathsf{id}} \simeq \{\infty\}$ and $\mathsf{Fl}_G^{=s} \simeq \mathbb{A}^1$.

The following lemma is easy on the level of $\mathbb{C}$-points. A careful argument shows it is also true on the level of schemes.

Lemma 3.2.9. *We have*

$$\mathsf{Fl}_G^{=w} \simeq N/(\mathsf{Ad}_w(N) \cap N) \simeq \mathsf{Ad}_w(N) \cap N^-.$$

Remark 3.2.10. Note that $\mathsf{Ad}_w(N) \cap N^-$ is an uniportent algebraic group.

In general, for any unipotent algebraic group U over $\mathbb{C}$. We can view its Lie algebra $\mathfrak{u}$ as an affine space. There is an algebraic version $\exp : \mathfrak{u} \to U$ of the usual exponential map, which is well-defined because U is unipotent. Moreover, exp is an isomorphism between schemes (see [M, Section 14.d]). It follows that $\mathsf{Fl}_G^{=w}$ is isomorphic to an affine space.

Exercise 3.2.11 (Problem 3, Homework 5)**.** Show that $\dim(\mathsf{Fl}_G^{=w}) = \ell(w)$. Deduce that $\dim(\mathsf{Fl}_G) = \ell(w_0)$.

Definition 3.2.12. The reduced closure

$$\mathsf{Fl}_G^{\leq w} := \overline{\mathsf{Fl}_G^{=w}}$$

of $\mathsf{Fl}_G^{=w}$ inside Fl_G is called the *Schubert variety* labelled by w.

Remark 3.2.13. In general, the Schubert varieties are not smooth. For example, for $G = \mathsf{SL}_4$, the Schubert variety $\mathsf{Fl}_G^{\leq(3142)}$ is singular, where we identify W with Σ_4 in the standard way.

Theorem 3.2.14. *The underlying topological space of $\mathsf{Fl}_G^{\leq w}$ is the union of the Bruhat cells $\mathsf{Fl}_G^{=w'}$ for $w' \leq w$.*

3.3. **Digression: equivariant sheaves.** In this subsection, we explain how to construct quasi-coherent modules on a quotient scheme.

Definition 3.3.1. Let K be an algebraic group and $\mathsf{mult}: K \to K$ be the multiplication morphism. Let Y be a k-scheme equipped with an action of K and $\mathsf{act}: K \times Y \to Y$ be the action morphism. A *K-equivariant quasi-coherent $\mathcal{O}_Y$-module* is a quasi-coherent $\mathcal{O}_Y$-module $\mathcal{F}$ equipped with an isomorphism $\alpha : \mathsf{act}^*\mathcal{F} \to \mathsf{pr}_2^*\mathcal{F}$ that satisfies the *cocycle condition*.

Remark 3.3.2. The cocycle condition says that the following diagram commutes

$$\begin{tikzcd}
(\mathsf{id}\times\mathsf{act})^*\circ\mathsf{act}^*(\mathcal{F}) \arrow[r, "(\mathsf{id}\times\mathsf{act})^*(\alpha)", "\simeq"'] \arrow[d, "\simeq"] & (\mathsf{id}\times\mathsf{act})^*\circ\mathsf{pr}_2^*(\mathcal{F}) \arrow[r, "\simeq"] & \mathsf{pr}_{23}^*\circ\mathsf{act}^*(\mathcal{F}) \arrow[d, "\simeq"', "\mathsf{pr}_{23}^*(\alpha)"] \\
(\mathsf{mult}\times\mathsf{id})^*\circ\mathsf{act}^*(\mathcal{F}) \arrow[r, "\simeq", "(\mathsf{mult}\times\mathsf{id})^*(\alpha)"'] & (\mathsf{mult}\times\mathsf{id})^*\circ\mathsf{pr}_2^*(\mathcal{F}) \arrow[r, "\simeq"] & \mathsf{pr}_{23}^*\circ\mathsf{pr}_2^*(\mathcal{F}),
\end{tikzcd}$$

where the unexplained isomorphisms are induced by the following identities in $\mathsf{Hom}(K\times K\times Y, Y)$:

$$\begin{aligned}
\mathsf{act}\circ(\mathsf{id}\times\mathsf{act}) &= \mathsf{act}\circ(\mathsf{mult}\times\mathsf{id}),\\
\mathsf{pr}_2\circ(\mathsf{id}\times\mathsf{act}) &= \mathsf{act}\circ\mathsf{pr}_{23},\\
\mathsf{pr}_2\circ(\mathsf{mult}\times\mathsf{id}) &= \mathsf{pr}_2\circ\mathsf{pr}_{23}.
\end{aligned}$$

Remark 3.3.3. Let $\mathcal{F}$ be a K-equivariant quasi-coherent $\mathcal{O}_Y$-module. For any k-point $g \in K(k)$, the fiber of α at g gives an isomorphism $\alpha_g : g^*\mathcal{F} \to \mathcal{F}$, where we abuse notation by writing $g : Y \to Y$ for the morphism $\mathsf{act}(g,-)$. Then the cocycle condition implies that

$$\alpha_{g_1 g_2} = \alpha_{g_2}\circ(g_2^*(\alpha_{g_1})). \tag{3.2}$$

Remark 3.3.4. One can define the category $\mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}}^K$ of K-equivariant quasi-coherent $\mathcal{O}_Y$-modules. It is an abelian category and the forgetful functor

$$\mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}}^K \to \mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}}$$

is faithful and exact. However, this functor is *not* fully faithful. In other words, *being equivariant is a structure rather than a property.*

Example 3.3.5. The structure sheaf $\mathcal{O}_Y$ has an obvious K-equivariant structure.

Example 3.3.6. The structure sheaf $\mathcal{O}_K$ has an obvious $(K\times K)$-equivariant structure, where the two factors of $K\times K$ act on K via the left and right multiplications respectively. Alternatively, we say $\mathcal{O}_K$ is equipped with a *left* K-equivariant structure and a *right* K-equivariant structure that are compatible to each other.

Construction 3.3.7. Suppose that the fppf quotient Y/K is represented by a k-scheme. Let $\pi : Y \to Y/K$ be the projection morphism. Note that $\pi\circ\mathsf{act} = \pi\circ\mathsf{pr}_2$ in $\mathsf{Hom}(K\times Y, Y/K)$. Hence we have a functor

$$\mathcal{O}_{Y/K}\text{-}\mathsf{mod}_{\mathsf{qc}} \to \mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}}^K,\ \ \mathcal{M}\mapsto\pi^*\mathcal{M}, \tag{3.3}$$

where $\pi^*\mathcal{M}$ is equipped with the K-equivariant structure given by

$$\mathsf{act}^*\circ\pi^*(\mathcal{M}) \simeq (\pi\circ\mathsf{act})^*(\mathcal{M}) = (\pi\circ\mathsf{pr}_2)^*(\mathcal{M}) \simeq \mathsf{pr}_2^*\circ\pi^*(\mathcal{M}).$$

The following result follows from the flat descent of quasi-coherent sheaves.

Proposition 3.3.8. *The functor* (3.3) *is an equivalence.*

When K is affine, the inverse of (3.3) can be constructed as follows.

Construction 3.3.9. Let $\mathcal{F} \in \mathcal{O}_Y\text{–}\mathsf{mod}_{\mathsf{qc}}^K$. The K-equivariant structure

$$\alpha : \mathsf{act}^*\mathcal{F} \xrightarrow{\simeq} \mathsf{pr}_2^*\mathcal{F}$$

induces a morphism

$$\mathcal{F} \to \mathsf{act}_* \circ \mathsf{pr}_2^*\mathcal{F}$$

Taking global sections, we obtain a map

$$\mathcal{F}(Y) \to (\mathsf{pr}_2^*\mathcal{F})(K \times Y) \simeq \mathcal{O}(K) \otimes \mathcal{F}(Y). \tag{3.4}$$

One can check that the cocycle condition implies that (3.4) defines a left coaction of the coalgebra $\mathcal{O}(K)$ on $\mathcal{F}(Y)$. By definition, we obtain a *right* K-module structure on $\mathcal{F}(Y)$.

One can check that the above construction is functorial in $\mathcal{F}$. In other words, we have upgraded the taking global sections functor $\Gamma(Y, -)$ to a functor

$$\Gamma(Y, -) : \mathcal{O}_Y\text{–}\mathsf{mod}_{\mathsf{qc}}^K \to K\text{–}\mathsf{mod}^r. \tag{3.5}$$

Remark 3.3.10. It follows formally that we can upgrade the taking cohomology functor $\mathsf{H}^i(Y, -)$ to a functor

$$\mathsf{H}^i(Y, -) : \mathcal{O}_Y\text{–}\mathsf{mod}_{\mathsf{qc}}^K \to K\text{–}\mathsf{mod}^r.$$

Remark 3.3.11. The functor (3.5) can also be defined as the direct image functor along the morphism $Y/K \to \mathsf{pt}/K$, where we use the identification

$$K\text{–}\mathsf{mod}^r \simeq \mathcal{O}_{\mathsf{pt}/K}\text{–}\mathsf{mod}.$$

Exercise 3.3.12. Show that the multiplication map $\mathcal{O}(Y) \otimes \mathcal{O}(Y) \to \mathcal{O}(Y)$ and the action map $\mathcal{O}(Y) \otimes \mathcal{F}(Y) \to \mathcal{F}(Y)$ are K-linear with respect to the right K-module structures in Construction 3.3.9, where K acts diagonally on $\mathcal{O}(Y) \otimes \mathcal{O}(Y)$ and $\mathcal{O}(Y) \otimes \mathcal{F}(Y)$.

Remark 3.3.13. For any k-point $g \in K(k)$, the isomorphism $\alpha_g : g^*\mathcal{F} \to \mathcal{F}$ induces an isomorphism $\beta_g : \mathcal{F} \to g_*\mathcal{F}$, which induces an isomorphism

$$\gamma_g : \mathcal{F}(Y) \xrightarrow{\beta_g(Y)} (g_*\mathcal{F})(Y) \simeq \mathcal{F}(g^{-1}(Y)) = \mathcal{F}(Y).$$

Then (3.2) implies that

$$\gamma_{g_1 g_2} = \gamma_{g_2} \circ \gamma_{g_1}.$$

In other words, $\mathcal{F}(Y)$ is indeed equipped with a *right* $K(k)$-action.

Construction 3.3.14. Let us continue the discussion in Construction 3.3.9.

For any open subset $U \subseteq Y/K$, the inverse image $\pi^{-1}(U) \subseteq Y$ is stabilized by the K-action. Replacing Y with $\pi^{-1}(U)$, we see that

$$(\pi_*\mathcal{F})(U) \simeq \mathcal{F}(\pi^{-1}(U))$$

is naturally a right K-module.

One can check that $\pi_*\mathcal{F}$ is naturally a sheaf of K-modules on Y. In other words, the restriction maps $(\pi_*\mathcal{F})(U) \to (\pi_*\mathcal{F})(V)$ for $V \subseteq U$ are K-linear.

Also, Exercise 3.3.12 implies that the multiplication map $\pi_*\mathcal{O}_Y \otimes \pi_*\mathcal{O}_Y \to \pi_*\mathcal{O}_Y$ and the action map $\pi_*\mathcal{O}_Y \otimes \pi_*\mathcal{F} \to \pi_*\mathcal{F}$ are K-linear. Taking K-invariance, we see that $(\pi_*\mathcal{O}_Y)^K$ is a sheaf of commutative rings on $\mathcal{O}_{Y/K}$, and $(\pi_*\mathcal{F})^K$ is a sheaf of $(\pi_*\mathcal{O}_Y)^K$-modules.

Exercise 3.3.15. We have $(\pi_*\mathcal{O}_Y)^K \simeq \mathcal{O}_{Y/K}$. Moreover, $(\pi_*\mathcal{F})^K$ is a quasi-coherent $\mathcal{O}_{Y/K}$-modules.

One can check that the above construction is functorial in $\mathcal{F}$. In other words, we obtain a functor

$$\mathcal{O}_Y\text{−}\mathsf{mod}_{\mathsf{qc}}^K \to \mathcal{O}_{Y/K}\text{−}\mathsf{mod}_{\mathsf{qc}},\ \ \mathcal{F} \mapsto (\pi_*\mathcal{F})^K. \tag{3.6}$$

Exercise 3.3.16. The functors (3.3) and (3.6) are inverse to each other.

3.4. **Line bundles.** In this subsection, we classify line bundles on the flag variety Fl_G. We fix a Borel subgroup $B \subseteq G$. Recall that we have a canonical isomorphism $\mathsf{Fl}_G \simeq G/B$. For simplicity, we choose a Cartan subgroup H insider B.

We start with the following general construction.

Construction 3.4.1. Let K be an affine algebraic group over $\mathbb{C}$ and Y be a $\mathbb{C}$-scheme equipped with a K-action. Suppose that the fppf quotient Y/K is represented by a $\mathbb{C}$-scheme. Let $\pi : Y \to Y/K$ be the canonical morphism.

Let V be a finite-dimensional K-module, viewed as an affine space equipped with a K-action. Consider the diagonal K-action on $Y\times V$. One can show that $(Y\times V)/K$ is represented by a $\mathbb{C}$-scheme and the obvious morphism $(Y\times V)/K \to Y/K$ exhibits it as a vector bundle over Y/K. Let $V_{Y/K}$ be the corresponding locally free $\mathcal{O}_{Y/K}$-module. The construction $V \mapsto V_{Y/K}$ is functorial. Hence we obtain a functor

$$K\text{−}\mathsf{mod}_{\mathsf{fd}} \to \mathcal{O}_{Y/K}\text{−}\mathsf{mod}_{\mathsf{coh}},\ \ V \mapsto V_{Y/K}.$$

Using ind-extension, we obtain a functor

$$K\text{−}\mathsf{mod} \to \mathcal{O}_{Y/K}\text{−}\mathsf{mod}_{\mathsf{qc}},\ \ V \mapsto V_{Y/K}. \tag{3.7}$$

Remark 3.4.2. Let $U \subseteq Y/K$ be an open subset. Unwinding the definition, we have

$$\Gamma(U, V_{Y/K}) \simeq (\mathcal{O}(\pi^{-1}(U)) \otimes V)^K,$$

where the K-module structure on $\mathcal{O}(\pi^{-1}(U))$ is induced by the K-action on $\pi^{-1}(U)$.

Remark 3.4.3. Via the equivalence

$$\mathcal{O}_{Y/K}\text{−}\mathsf{mod}_{\mathsf{qc}} \simeq \mathcal{O}_Y\text{−}\mathsf{mod}_{\mathsf{qc}}^K,$$

the $\mathcal{O}_{Y/K}$-module $V_{Y/K}$ corresponds to the free $\mathcal{O}_Y$-module $\mathcal{O}_Y \otimes V$ equipped with the K-equivariant structure that combines the K-equivariant structure on $\mathcal{O}_Y$ with the K-module structure on V.

Remark 3.4.4. The functor (3.7) can also be defined as the pullback functor along the morphism $Y/K \to \mathsf{pt}/K$, where we use the identification

$$K\text{−}\mathsf{mod} \simeq \mathcal{O}_{\mathsf{pt}/K}\text{−}\mathsf{mod}.$$

We now apply the above construction to the flag variety G/B.

Notation 3.4.5. Let $\lambda \in \mathbb{X}(B) := \mathsf{Hom}_{\mathsf{Grp}}(B, \mathbb{G}_{\mathsf{m}})$ be a character of B. Consider the 1-dimensional representation $\mathbb{C}_{-\lambda}$ of B corresponding to $-\lambda$. Write

$$\mathcal{O}(\lambda) := (\mathbb{C}_{-\lambda})_{G/B}$$

for the line bundle on G/B obtained by Construction 3.4.1.

Remark 3.4.6. Via the equivalence

$$\mathcal{O}_{G/B}\text{–}\mathsf{mod}_{\mathsf{qc}} \simeq \mathcal{O}_G\text{–}\mathsf{mod}_{\mathsf{qc}}^B,$$

the line bundle $\mathcal{O}(\lambda)$ corresponds to the trivial line bundle $\mathcal{O}_G$ equipped with the B-equivariant structure that combines the *right* B-equivariant structure on $\mathcal{O}_G$ (see Example 3.3.6) with the K-module structure on V. As a consequence, the *left* G-equivariant structure on $\mathcal{O}_G$, which is compatible with the above B-equivariant structure, induces a G-equivariant structure on $\mathcal{O}(\lambda)$. In other words, we can upgrade $\mathcal{O}(\lambda)$ to an object

$$\mathcal{O}(\lambda) \in \mathcal{O}_{G/B}\text{–}\mathsf{mod}_{\mathsf{qc}}^G. \tag{3.8}$$

In fact, we have

$$\mathcal{O}_{G/B}\text{–}\mathsf{mod}_{\mathsf{qc}}^G \simeq \mathcal{O}_G\text{–}\mathsf{mod}_{\mathsf{qc}}^{G\times B} \simeq \mathcal{O}_{\mathsf{pt}/B}\text{–}\mathsf{mod}_{\mathsf{qc}} \simeq B\text{–}\mathsf{mod}$$

and the object (3.8) corresponds to the representation $\mathbb{C}_{-\lambda}$.

Exercise 3.4.7 (Problem 4, Homework 5)**.** The map

$$\mathbb{X}(B) \to \mathsf{Pic}(G/B),\ \ \lambda \mapsto \mathcal{O}(\lambda)$$

is a group homomorphism.

Exercise 3.4.8 (Problem 5, Homework 5)**.** Let $G = \mathsf{SL}_2$ and B be its standard Borel subgroup. Let λ_n, $n \in \mathbb{Z}$ be the character of B given by $\left(\begin{smallmatrix} t & * \\ 0 & t^{-1}\end{smallmatrix}\right) \mapsto t^n$. Show that the isomorphism $G/B \simeq \mathbb{P}^1$ in Example 1.5.1 sends $\mathcal{O}(\lambda_n) \in \mathsf{Pic}(G/B)$ to $\mathcal{O}(n) \in \mathsf{Pic}(\mathbb{P}^1)$.

Remark 3.4.9. Note that the restriction map

$$\mathbb{X}(H) \to \mathbb{X}(B)$$

along the projection $B \to B/[B,B] \simeq H$ is an isomorphism. On the other hand, it is easy to see the map

$$\mathbb{X}(H) \coloneqq \mathsf{Hom}_{\mathsf{Grp}}(H, \mathbb{G}_{\mathsf{m}}) \to \mathsf{Hom}_{\mathsf{Lie}}(\mathfrak{h}, \mathbb{C}) \simeq \mathfrak{h}^*$$

is injective and its image is the lattice Λ of integral weights. It follows that we have a canonical identification $\mathbb{X}(B) \simeq \Lambda$.

See [M, Section 18] for a complete proof of the following result.

Theorem 3.4.10. *The map*

$$\mathbb{X}(B) \to \mathsf{Pic}(G/B),\ \ \lambda \mapsto \mathcal{O}(\lambda)$$

is an isomorphism.

Exercise 3.4.11 (Problem 6, Homework 5)**.** Show that any line bundle on G/B has a unique G-equivariant structure.

3.5. **Borel–Weil–Bott theorem.** Recall that the line bundle $\mathcal{O}(\lambda) \in \mathsf{Pic}(G/B)$ has a canonical (in fact unique) G-equivariant structure. By Construction 3.3.9 and Remark 3.3.10, the cohomology groups

$$\mathsf{H}^i(G/B, \mathcal{O}(\lambda))$$

are naturally right G-modules. We view them as left G-modules via the inversion map of G.

Theorem 3.5.1 (Borel–Weil–Bott)**.** *Let λ be an integral weight.*

(1) *If λ is not dot-regular, i.e., the dot W-action at λ has nontrivial stabilizer, then*
$$\mathsf{H}^i(G/B, \mathcal{O}(\lambda)) \simeq 0 \ \ \forall i \geq 0.$$
(2) *If λ is dot-regular, let $w \in W$ be the unique element such that $\mu := w \cdot \lambda$ is dot-dominant, then $\mathsf{H}^i(G/B, \mathcal{O}(\lambda)) \simeq 0$ unless $i = \ell(w)$, and*
$$\mathsf{H}^{\ell(w)}(G/B, \mathcal{O}(\lambda)) \simeq L_\mu^* \simeq L_{-w_0(\mu)}.$$

We will only prove the $i = 0$ part of the above theorem. The general case can be reduced to this case by studying the $\mathbb{P}^1$-bundle $G/P_\alpha \to P/B$, where P_α is a minimal parabolic subgroup of G. See [D1], [D2] or [L].

We start with the following well-known result.

Theorem 3.5.2 (Peter–Weyl)**.** *Consider the $(G \times G)$-action on G and the corresponding $(G \times G)$-module $\mathcal{O}(G)$. We have*
$$\mathcal{O}(G) \simeq \bigoplus_{\lambda \in \Lambda^+} L_\lambda^* \otimes L_\lambda \in (G \times G)\text{−}\mathsf{mod}$$

We provide a proof of the Peter–Weyl theorem in the following exercise.

Exercise 3.5.3 (Problem 7, Homework 5)**.** Let V be a G-module. Consider the *second* G-module structure on $\mathcal{O}(G)$, and the corresponding diagonal G-module structure on $\mathcal{O}(G) \otimes V$. Let
$$\mathcal{O}(G) \overset{G}{\otimes} V$$
be the invariance for this G-action. Consider the G-action on $\mathcal{O}(G) \otimes^G V$ induced from the *first* G-module structure on $\mathcal{O}(G)$. Show that
$$\mathcal{O}(G) \overset{G}{\otimes} V \simeq V \in G\text{−}\mathsf{mod}.$$
Deduce the Peter–Weyl theorem from the above calculation.

We are ready to prove the $i = 0$ case of Theorem 3.5.1[7].

Theorem 3.5.4 (Borel–Weil)**.** *Let λ be an integral weight. We have*
$$\Gamma(G/B, \mathcal{O}(\lambda)) \simeq \begin{cases} L_\lambda^* & \text{if } \lambda \in \Lambda^+, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. By Remark 3.4.2, we have
$$\Gamma(G/U, \mathcal{O}(\lambda)) \simeq (\mathcal{O}(G) \otimes \mathbb{C}_{-\lambda})^B,$$
where B acts on $\mathcal{O}(G)$ via the second G-module structure of it, and B acts diagonally on $\mathcal{O}(G) \otimes \mathbb{C}_{-\lambda}$. By the Peter–Weyl theorem, we obtain
$$\Gamma(G/U, \mathcal{O}(\lambda)) \simeq \bigoplus_{\mu \in \Lambda^+} L_\mu^* \otimes (L_\mu \otimes \mathbb{C}_{-\lambda})^B.$$
Now the claim follows from the following calculation:
$$(L_\mu \otimes \mathbb{C}_{-\lambda})^B \simeq (L_\mu^N \otimes \mathbb{C}_{-\lambda})^H \simeq (\mathbb{C}_\mu \otimes \mathbb{C}_{-\lambda})^H \simeq \begin{cases} \mathbb{C} & \text{if } \lambda = \mu, \\ 0 & \text{otherwise.} \end{cases}$$
□

[7] Note that for an integral weight λ, being dominant is equivalent to being dot-dominant and dot-regular.

3.6. **Global vector fields.** The goal of this subsection is to prove a canonical isomorphism between Lie algebras

$$\mathfrak{g} \simeq \mathcal{T}(\mathsf{Fl}_G).$$

We start with the following general construction.

Construction 3.6.1. Let X be a finite type scheme equipped with a *right* action of an affine algebraic group K. Let $\mathfrak{k}$ be the Lie algebra of K. Consider the action map $\mathsf{act}: X \times K \to X$ and the $\mathcal{O}_{X\times K}$-linear morphisms

$$\mathcal{O}_X \boxtimes \mathcal{T}_K \to (\mathcal{O}_X \boxtimes \mathcal{T}_K) \oplus (\mathcal{T}_X \boxtimes \mathcal{O}_K) \simeq \mathcal{T}_{X\times G} \xrightarrow{d\mathsf{act}} \mathsf{act}^* \mathcal{T}_X.$$

By restricting along $X \xrightarrow{\mathsf{Id}\times e} X \times G$, we otain an $\mathcal{O}_X$-linear morphism

$$a : \mathcal{O}_X \otimes \mathfrak{k} \to \mathcal{T}_X.$$

This induces a $\mathbb{C}$-linear map

$$a : \mathfrak{k} \to \mathcal{T}(X),$$

which we denote by the same notation.

Remark 3.6.2. Using the language of differential geometry, for $x \in \mathfrak{g}$, the vector field $a(x)$ corresponds to the flow $\mathsf{act}(-, \exp(tx)) : X \to X$.

Lemma 3.6.3. *The above map $a : \mathfrak{k} \to \mathcal{T}(X)$ is a Lie algebra homomorphism.*

Variant 3.6.4. *Let X be a finite type scheme equipped with a* left *action of an algebraic group K. Consider the right K-action on X given by the inversion map of K. We obtain a canonical Lie algebra homomorphism $a : \mathfrak{k} \to \mathcal{T}(X)$.*

To prove the lemma, we need to recall some facts about $\mathfrak{k}$. By definition, we have canonical identifications

$$\mathfrak{k} \simeq \mathcal{T}_K|_e \simeq \mathsf{Der}(\mathcal{O}(K), \mathbb{C}_e), \tag{3.9}$$

where the RHS is the space of derivations of $\mathcal{O}(K)$ into $\mathbb{C}_e$.

Exercise 3.6.5. Via the identification (3.9), the Lie bracket of $\mathfrak{k}$ corresponds to the Lie bracket on $\mathsf{Der}(\mathcal{O}(K), \mathbb{C}_e)$ defined by the following formula

$$[\partial_1, \partial_2] := (\partial_1 \otimes \partial_2 - \partial_2 \otimes \partial_1) \circ \Delta, \ \ \forall \partial_1, \partial_2 \in \mathsf{Der}(\mathcal{O}(K), \mathbb{C}_e) \tag{3.10}$$

where $\Delta : \mathcal{O}(K) \to \mathcal{O}(K) \otimes \mathcal{O}(K)$ is the comultiplication map.

Remark 3.6.6. The formula (3.10) is the algebraic analogue of the following formula in differential geometry:

$$\exp(tu) \cdot \exp(tv) \cdot \exp(-tu) \cdot \exp(-tv) = \exp(t^2[u, v])(1 + O(t^3)), \ \ \forall u, v \in \mathfrak{k}.$$

Proof of Lemma 3.6.3. Unwinding the definitions, the map $a : \mathfrak{k} \to \mathcal{T}(X)$ can be identified with the map

$$\mathsf{Der}(\mathcal{O}(K), \mathbb{C}_e) \to \mathsf{Der}(\mathcal{O}(K), \mathcal{O}(K))$$

that sends $\partial : \mathcal{O}(K) \to \mathbb{C}_e$ to the composition

$$a(\partial) : \mathcal{O}(X) \xrightarrow{\mathsf{coact}} \mathcal{O}(X) \otimes \mathcal{O}(K) \xrightarrow{\mathsf{id}\otimes\partial} \mathcal{O}(X),$$

where coact is the composition

$$\mathcal{O}(X) \xrightarrow{-\circ\mathsf{act}} \mathcal{O}(X \times K) \simeq \mathcal{O}(X) \otimes \mathcal{O}(K).$$

Then $a(\partial_1)\circ a(\partial_2)$ is equal to

$$\mathcal{O}(X)\xrightarrow{\text{coact}}\mathcal{O}(X)\otimes\mathcal{O}(K)\xrightarrow{\text{coact}\otimes\text{id}}\mathcal{O}(X)\otimes\mathcal{O}(K)\otimes\mathcal{O}(K)\xrightarrow{\text{id}\otimes\partial_1\otimes\partial_2}\mathcal{O}(X).$$

By the axiom of group actions, the above composition is equal to

$$\mathcal{O}(X)\xrightarrow{\text{coact}}\mathcal{O}(X)\otimes\mathcal{O}(K)\xrightarrow{\text{id}\otimes\Delta}\mathcal{O}(X)\otimes\mathcal{O}(K)\otimes\mathcal{O}(K)\xrightarrow{\text{id}\otimes\partial_1\otimes\partial_2}\mathcal{O}(X).$$

It follows that $[a(\partial_1),a(\partial_2)]$ is equal to

$$\mathcal{O}(X)\xrightarrow{\text{coact}}\mathcal{O}(X)\otimes\mathcal{O}(K)\xrightarrow{\text{id}\otimes\Delta}\mathcal{O}(X)\otimes\mathcal{O}(K)\otimes\mathcal{O}(K)\xrightarrow{\text{id}\otimes(\partial_1\otimes\partial_2-\partial_2\otimes\partial_1)}\mathcal{O}(X).$$

By Exercise 3.6.5, this is equal to $a([\partial_1,\partial_2])$. In other words, we obtain the desired equation

$$[a(\partial_1),a(\partial_2)]=a([\partial_1,\partial_2]).$$

□[Lemma 3.6.3]

Example 3.6.7. Consider the left and right multplication actions of K on itself. We obtain Lie algebra homomorphisms

$$\mathfrak{k}\xrightarrow{a_l}\mathcal{T}(K)\xleftarrow{a_r}\mathfrak{k}. \tag{3.11}$$

By construction, the image of a_r consists of *left* invariant vector fields on K, while the image of a_l consists of *right* invariant ones. Note that the corresponding maps

$$\mathcal{O}_K\otimes\mathfrak{k}\xrightarrow{a_l}\mathcal{T}_K\xleftarrow{a_r}\mathcal{O}_K\otimes\mathfrak{k} \tag{3.12}$$

are isomorphisms between sheaves of Lie algebras on K, because both maps induce isomorphisms between stalks[8].

Note that the images of (3.11) commute with each other because the left and right multiplication actions commute with each other.

Theorem 3.6.8 (Bott)**.** *The action of G on Fl_G induces an isomorphism between Lie algebras*

$$a:\mathfrak{g}\to\mathcal{T}(\mathsf{Fl}_G).$$

The rest of this subsection is devoted to the proof of the above theorem.

Exercise 3.6.9 (Problem 1, Homework 6)**.** The canonical morphism

$$a:\mathcal{O}_{\mathsf{Fl}_G}\otimes\mathfrak{g}\to\mathcal{T}_{\mathsf{Fl}_G}$$

is surjective. Moreover, the fiber of it at a point $x\in\mathsf{Fl}_G(\mathbb{C})$ can be identified with the obvious map

$$\mathfrak{g}\to\mathfrak{g}/\mathfrak{b}_x,$$

where $\mathfrak{b}_x$ is the Borel algebra corresponding to x.

Exercise 3.6.10 (Problem 2, Homework 6)**.** Let B be a Borel subgroup of G. Show that

$$(G\times\mathfrak{g})/B\to G/B\times\mathfrak{g},\ \ \overline{(g,u)}\mapsto(\overline{g},gug^{-1})$$

is a well-defined isomorphism between vector bundles on G/B. Deduce that there is a canonical isomorphism

$$\mathfrak{g}_{G/B}\simeq\mathcal{O}_{\mathsf{Fl}_G}\otimes\mathfrak{g}. \tag{3.13}$$

[8]Unwinding the definitions, the stalk of (3.12) at $e\in K$ is the diagram

$$\mathfrak{k}\xrightarrow{-\mathsf{Id}}\mathfrak{k}\xleftarrow{\mathsf{Id}}\mathfrak{k}.$$

Exercise 3.6.11 (Problem 3, Homework 6)**.** Show that the kernel of the composition

$$\mathfrak{g}_{G/B} \xrightarrow{(3.13)} \mathcal{O}_{\mathsf{Fl}_G} \otimes \mathfrak{g} \xrightarrow{a} \mathcal{T}_{\mathsf{Fl}_G}$$

is equal to $\mathfrak{b}_{G/B} \subseteq \mathfrak{g}_{G/B}$. Deduce that

$$\mathcal{T}_{\mathsf{Fl}_G} \simeq (\mathfrak{g}/\mathfrak{b})_{G/B}.$$

and

$$\mathcal{K}_{\mathsf{Fl}_G} \simeq \mathcal{O}(-2\rho).$$

where $\mathcal{K}_{\mathsf{Fl}_G}$ is the canonical (a.k.a. dualizing) line bundle on Fl_G.

Example 3.6.12. We have $\mathcal{K}_{\mathbb{P}^1} \simeq \mathcal{O}(-2)$.

Proof of Theorem 3.6.8. We only need to show the map $a : \mathfrak{g} \to \mathcal{T}(\mathsf{Fl}_G)$ is a bijection.

Let B be a Borel subgroup of G. By Exercise 3.6.11, we have a short exact sequence

$$0 \to \mathfrak{b}_{G/B} \to \mathcal{O}_{\mathsf{Fl}_G} \otimes \mathfrak{g} \xrightarrow{a} \mathcal{T}_{\mathsf{Fl}_G} \to 0.$$

Taking global sections, we obtain a long exact sequence

$$0 \to \Gamma(G/B, \mathfrak{b}_{G/B}) \to \mathfrak{g} \xrightarrow{a} \mathcal{T}(\mathsf{Fl}_G) \to \mathsf{H}^1(G/B, \mathfrak{b}_{G/B}).$$

Hence we only need to show

$$\mathsf{H}^i(G/B, \mathfrak{b}_{G/B}) \simeq 0, \quad i = 0, 1$$

By Exercise 3.6.9, we have

$$\Gamma(G/B, \mathfrak{b}_{G/B}) \simeq \ker(a : \mathfrak{g} \to \mathcal{T}(\mathsf{Fl}_G)) \simeq \bigcap_{x \in \mathsf{Fl}_G(\mathbb{C})} \ker(\mathfrak{g} \to \mathfrak{g}/\mathfrak{b}_x) \simeq \bigcap_{x \in \mathsf{Fl}_G(\mathbb{C})} \mathfrak{b}_x \simeq 0,$$

where the last isomorphism follows from semisimpleness of $\mathfrak{g}$.

To prove $\mathsf{H}^1(G/B, \mathfrak{b}_{G/B}) \simeq 0$, consider the short exact sequence

$$0 \to \mathfrak{n}_{G/B} \to \mathfrak{b}_{G/B} \to \mathfrak{h}_{G/B} \to 0,$$

which induces a long exact sequence

$$0 \to \Gamma(G/B, \mathfrak{h}_{G/B}) \to \mathsf{H}^1(G/B, \mathfrak{n}_{G/B}) \to \mathsf{H}^1(G/B, \mathfrak{b}_{G/B}) \to \mathsf{H}^1(G/B, \mathfrak{h}_{G/B}).$$

Since the adjoint B-action on $\mathfrak{h}$ is trivial, we have $\mathfrak{h}_{G/B} \simeq \mathcal{O}_{G/B} \otimes \mathfrak{h}$. It follows that

$$\Gamma(G/B, \mathfrak{h}_{G/B}) \simeq \mathfrak{h} \text{ and } \mathsf{H}^1(G/B, \mathfrak{h}_{G/B}) \simeq 0.$$

Hence to finish the proof, we only need to show

$$\dim \mathsf{H}^1(G/B, \mathfrak{n}_{G/B}) \leq \dim \mathfrak{h}. \tag{3.14}$$

Consider the short exact sequence

$$0 \to [\mathfrak{n}, \mathfrak{n}]_{G/B} \to \mathfrak{n}_{G/B} \to (\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])_{G/B} \to 0,$$

which induces an exact sequence

$$\mathsf{H}^1(G/B, [\mathfrak{n}, \mathfrak{n}]_{G/B}) \to \mathsf{H}^1(G/B, \mathfrak{n}_{G/B}) \to \mathsf{H}^1(G/B, (\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}])_{G/B}).$$

We claim that

$$\mathsf{H}^1(G/B, [\mathfrak{n}, \mathfrak{n}]_{G/B}) \simeq 0 \tag{3.15}$$

while

$$\dim \mathsf{H}^1(G/B, (\mathfrak{n}/[\mathfrak{n},\mathfrak{n}])_{G/B}) = \dim \mathfrak{h}. \tag{3.16}$$

Note that (3.14) immediately follows from the claim.

Note that $[\mathfrak{n},\mathfrak{n}]$ has a filtration such that the graded pieces are of the form $\mathbb{C}_\alpha$ for $\alpha \in \Phi^+ \smallsetminus \Delta$, while

$$\mathfrak{n}/[\mathfrak{n},\mathfrak{n}] \simeq \bigoplus_{\alpha\in\Delta} \mathbb{C}_\alpha.$$

Hence (3.15) and (3.16) follow respectively from the two claims in the following exercise.

Exercise 3.6.13. Let α be a positive root. Deduce the following claims from the Borel–Weil–Bott theorem.

(1) If α is not simple, then $\mathsf{H}^1(G/B, \mathcal{O}(-\alpha)) \simeq 0$.
(2) If α is simple, then $\mathsf{H}^1(G/B, \mathcal{O}(-\alpha)) \simeq \mathbb{C}$.

□[Theorem 3.6.8]

Remark 3.6.14. See [D3] for an alternative proof of Theorem 3.6.8.

4. Localization theory - differential operators

4.1. Roadmap.

Definition 4.1.1. Let X be a smooth scheme. An $(\mathcal{O}\oplus\mathcal{T})_X$*-module* is an abelian sheaf $\mathcal{M}$ on X equipped with an action

$$\mathsf{act}_{\mathcal{O}} : \mathcal{O}_X \otimes \mathcal{M} \to \mathcal{M}$$

by the sheaf $\mathcal{O}_X$ of commutative rings, and an action

$$\mathsf{act}_{\mathcal{T}} : \mathcal{T}_X \otimes \mathcal{M} \to \mathcal{M}$$

by the sheaf $\mathcal{T}_X$ of Lie algebras, such that they satisfy the following conditions.

(i) The map $\mathsf{act}_{\mathcal{T}}$ is $\mathcal{O}_X$-linear, where $\mathcal{O}_X$ acts on $\mathcal{T}_X \otimes \mathcal{M}$ via the first factor $\mathcal{T}_X$;

(ii) The map $\mathsf{act}_{\mathcal{O}}$ is $\mathcal{T}_X$-linear, where $\mathcal{T}_X$ acts on $\mathcal{O}_X \otimes \mathcal{M}$ via the Lebniz rule.

We say an $(\mathcal{O}\oplus\mathcal{T})_X$-module $\mathcal{M}$ is *quasi-coherent* if it is quasi-coherent as an $\mathcal{O}_X$-module.

Remark 4.1.2. Let $\partial \in \mathcal{T}(U)$, $f \in \mathcal{O}(U)$ and $m \in \mathcal{M}(U)$ be local sections. We often write

$$\partial(m) := \mathsf{act}_{\mathcal{T}}(\partial, m) \text{ and } f \cdot m := \mathsf{act}_{\mathcal{O}}(f, m).$$

Then (i) and (ii) become

$$(f\partial)(m) = f \cdot (\partial(m)) \text{ and } \partial(f \cdot m) = \partial(f) \cdot m + f \cdot \partial(m).$$

Example 4.1.3. The structure sheaf $\mathcal{O}_X$, equipped with the standard $\mathcal{O}_X$-action and $\mathcal{T}_X$-action, is a quasi-coherent $(\mathcal{O}\oplus\mathcal{T})_X$-module.

Construction 4.1.4. Let

$$(\mathcal{O}\oplus\mathcal{T})_X\text{–}\mathsf{mod}_{\mathsf{qc}}$$

be the abelian category of quasi-coherent $(\mathcal{O}\oplus\mathcal{T})_X$-modules. We have a left exact functor

$$(\mathcal{O}\oplus\mathcal{T})_X\text{–}\mathsf{mod}_{\mathsf{qc}} \to \mathcal{T}(X)\text{–}\mathsf{mod},\ \ \mathcal{M} \mapsto \mathcal{M}(X).$$

In particular, for $X = \mathsf{Fl}_G$, we obtain a left exact functor

$$(\mathcal{O}\oplus\mathcal{T})_{\mathsf{Fl}_G}\text{–}\mathsf{mod}_{\mathsf{qc}} \to \mathcal{T}(\mathsf{Fl}_G)\text{–}\mathsf{mod} \to \mathfrak{g}\text{–}\mathsf{mod},\ \ \mathcal{M} \mapsto \mathcal{M}(X) \tag{4.1}$$

where the last functor in the composition is induced by the Lie algebra homomorphism $a : \mathfrak{g} \to \mathcal{T}(\mathsf{Fl}_G)$.

Remark 4.1.5. Recall that Theorem 3.6.8 says $a : \mathfrak{g} \to \mathcal{T}(\mathsf{Fl}_G)$ is an isomorphism. However, we will not use this fact in this section. Instead, we will provide another proof for it.

Motivated by the isomorphism $\mathfrak{g} \simeq \mathcal{T}(\mathsf{Fl}_G)$, one may conjecture that (4.1) is an equivalence between categories. However, this is not true even for SL_2.

Exercise 4.1.6 (Problem 4, Homework 6)**.** For $G := \mathsf{SL}_2$ and a quasi-coherent $(\mathcal{O}\oplus\mathcal{T})_{\mathsf{Fl}_G}$-module $\mathcal{M}$, show that the Casimir element $\Omega \in U(\mathfrak{sl}_2)$ acts on $\mathcal{M}(\mathsf{Fl}_G)$ by 0.

Let $\chi_0 : Z(\mathfrak{g}) \to \mathbb{C}$ be the central character for the trivial $\mathfrak{g}$-module. Let

$$\mathfrak{g}\text{–}\mathsf{mod}_{\chi_0} \subseteq \mathfrak{g}\text{–}\mathsf{mod}$$

be the full subcategory consisting of $\mathfrak{g}$-modules such that $Z(\mathfrak{g})$ acts via χ_0 on them. In future lectures, we will prove the following theorem.

Theorem 4.1.7. *The functor* (4.1) *induces an equivalence*

$$(\mathcal{O} \oplus \mathcal{T})_{\mathsf{Fl}_G}\text{–}\mathsf{mod}_{\mathsf{qc}} \xrightarrow{\simeq} \mathfrak{g}\text{–}\mathsf{mod}_{\chi_0}.$$

To prove the theorem, we need a geometric incarnation of the associative ring

$$U(\mathfrak{g})_{\chi_0} := U(\mathfrak{g}) \underset{Z(\mathfrak{g})}{\otimes} \mathbb{C}_{\chi_0}.$$

More precisely, we want to construct a *sheaf* $\mathcal{D}_{\mathsf{Fl}_G}$ of associative rings on Fl_G, such that

(i) The sheaf $\mathcal{D}_{\mathsf{Fl}_G}$ is a universal enveloping algebra of $\mathcal{O} \oplus \mathcal{T}$ in the sense that we have a canonical equivalence

$$\mathcal{D}_{\mathsf{Fl}_G}\text{–}\mathsf{mod}_{\mathsf{qc}} \simeq (\mathcal{O} \oplus \mathcal{T})_{\mathsf{Fl}_G}\text{–}\mathsf{mod}_{\mathsf{qc}}.$$

(ii) The Lie algebra homomorphism $a : \mathfrak{g} \to \mathcal{T}(\mathsf{Fl}_G)$ can be extended to an algebra homomorphism

$$a : U(\mathfrak{g})_{\chi_0} \to \mathcal{D}(\mathsf{Fl}_G) \tag{4.2}$$

such that the composition

$$\mathcal{D}_{\mathsf{Fl}_G}\text{–}\mathsf{mod}_{\mathsf{qc}} \xrightarrow{\Gamma} \mathcal{D}(\mathsf{Fl}_G)\text{–}\mathsf{mod} \to U(\mathfrak{g})_{\chi_0}\text{–}\mathsf{mod} \tag{4.3}$$

is an equivalence.

Note that these conditions actually imply (4.2) is an isomorphism. Namely, one can show that (4.3) admits a left adjoint functor that sends $U(\mathfrak{g})_{\chi_0}$ to $\mathcal{D}_{\mathsf{Fl}_G}$. Then (4.2) can be identified with the isomorphism

$$\mathsf{End}_{U(\mathfrak{g})_{\chi_0}\text{–}\mathsf{mod}}(U(\mathfrak{g})_{\chi_0}) \to \mathsf{End}_{\mathcal{D}_{\mathsf{Fl}_G}\text{–}\mathsf{mod}_{\mathsf{qc}}}(\mathcal{D}_{\mathsf{Fl}_G}, \mathcal{D}_{\mathsf{Fl}_G}).$$

In this section, we will introduce the desired sheaf $\mathcal{D}_{\mathsf{Fl}_G}$, known as the sheaf of *differential operators* on Fl_G, and prove the desired isomorphism (4.2). In the next section, we will prove the desired equivalence (4.3), known as the *Beilinson–Bernstein localization theorem*.

4.2. **Sheaf of differential operators.**

Throughout this subsection, X is a smooth scheme over $\mathbb{C}$.

The following definition is due to Grothendieck (see [EGA4, Section 16.8]). See [HTT, Sect. 1.1] for a more direct definition.

Definition 4.2.1. A differential operator of order -1 on X is the zero morphism $\mathcal{O}_X \xrightarrow{0} \mathcal{O}_X$.

For $n \geq 0$, a *differential operator of order* n on X is a $\mathbb{C}$-linear morphism $\phi : \mathcal{O}_X \to \mathcal{O}_X$ of sheaves such that for any open subscheme $U \subseteq X$ and any section $f \in \mathcal{O}(U)$, the $\mathbb{C}$-linear morphism

$$[\phi|_U, f] := \phi|_U \circ f - f \circ \phi|_U \in \mathsf{Hom}_{\mathbb{C}}(\mathcal{O}_U, \mathcal{O}_U)$$

is a differential operator of order $n-1$ on U, where we view f as an element in $\mathsf{End}_{\mathbb{C}}(\mathcal{O}(X))$ via the multiplication map.

Let $\mathcal{D}^{\leq n}(X)$ be the space of differential operators of order n on X, and

$$\mathcal{D}(X) := \bigcup_n \mathcal{D}^{\leq n}(X)$$

be the space of all differential operators on X.

Example 4.2.2. Multplication by any function $f \in \mathcal{O}(X)$ gives a differential operator of order 0 on X

Example 4.2.3. A vector field $\partial \in \mathcal{T}(X) \simeq \mathsf{Der}(\mathcal{O}_X, \mathcal{O}_X)$ is a differential operator of order 1 on X.

Exercise 4.2.4 (Problem 5, Homework 6)**.** Show that the obtained maps

$$\mathcal{O}(X) \to \mathcal{D}^{\leq 0}(X) \text{ and } \mathcal{O}(X) \oplus \mathcal{T}(X) \to \mathcal{D}^{\leq 1}(X)$$

are isomorphisms.

Exercise 4.2.5. Let $\phi_1 \in \mathcal{D}^{\leq m}(X)$ and $\phi_2 \in \mathcal{D}^{\leq n}(X)$ be differential operators. Show that:

(1) The composition $\phi_1 \circ \phi_2$ is a differential operator of order $m+n$.
(2) The commutator $[\phi_1, \phi_2] \coloneqq \phi_1 \circ \phi_2 - \phi_2 \circ \phi_1$ is a differential operator of order $m+n-1$.

The above exercise can be rephrased as the following lemma.

Lemma 4.2.6. *Differential operators on X form an almost commutative filtered associative algebra $\mathcal{D}^{\leq \bullet}(X)$.*

Warning 4.2.7. Definition 4.2.1 makes sense for singular schemes, but the obtained algebra $\mathcal{D}(X)$ is ill-behaved. In the modern point of view, the "correct" $\mathcal{D}(X)$ for a singular scheme X should be a DG algebra.

Exercise 4.2.8. Let $U \subseteq V$ be open subschemes of X. The restriction map

$$\mathsf{Hom}_{\mathbb{C}}(\mathcal{O}_V, \mathcal{O}_V) \to \mathsf{Hom}_{\mathbb{C}}(\mathcal{O}_U, \mathcal{O}_U)$$

sends $\mathcal{D}^{\leq n}(V)$ into $\mathcal{D}^{\leq n}(U)$.

In other words, the assignment $U \mapsto \mathcal{D}^{\leq n}(U)$ is contravariantly functorial. This defines a presheaf $\mathcal{D}_X^{\leq n}$ on X, where

$$\mathcal{D}_X^{\leq n}(U) \coloneqq \mathcal{D}^{\leq n}(U).$$

One can check that $\mathcal{D}_X^{\leq n}$ satisfies the sheaf condition.

Definition 4.2.9. We call $\mathcal{D}_X^{\leq n}$ the sheaf of differential operators of order n on X, and call

$$\mathcal{D}_X \coloneqq \bigcup_n \mathcal{D}_X^{\leq n}$$

the sheaf of differential operators on X.

Example 4.2.10. Exercise 4.2.4 implies

$$\mathcal{O}_X \simeq \mathcal{D}^{\leq 0}\mathcal{D}_X \text{ and } \mathcal{O}_X \oplus \mathcal{T}_X \simeq \mathcal{D}^{\leq 1}\mathcal{D}_X.$$

Lemma 4.2.6 implies the following result.

Lemma 4.2.11. *The filtered sheaf $\mathcal{D}_X^{\leq \bullet}$ is naturally an almost commutative filtered associative algebra object in $\mathsf{Shv}(X, \mathsf{Ab})$.*

Note that the embedding morphism $\mathcal{O}_X \to \mathcal{D}_X$ is a ring homomorphism. We obtain an $\mathcal{O}_X$-bimodule structure on $\mathcal{D}_X$ given by left and right multiplications.

The following result is often referred as the PBW theorem for $\mathcal{D}_X$.

Theorem 4.2.12 ([EGA4, Theorem 16.11.2])**.** *The isomorphism $\mathcal{T}_X \simeq \mathsf{gr}^1\mathcal{D}_X$ induces an isomorphism between commutative $\mathcal{O}_X$-algebras:*

$$\mathsf{Sym}_{\mathcal{O}_X}\mathcal{T}_X \xrightarrow{\simeq} \mathsf{gr}^\bullet\mathcal{D}_X.$$

In particular, each $\mathsf{gr}^n\mathcal{D}_X$ is a locally free $\mathcal{O}_X$-module. This implies the following result.

Corollary 4.2.13. *For $n \geq 0$, the sheaf $\mathcal{D}_X^{\leq n}$ is locally free for both $\mathcal{O}_X$-module structures.*

We also have the following corollary of the PBW theorem for $\mathcal{D}_X$.

Corollary 4.2.14. *The sheaf $\mathcal{D}_X$ of associative algebras is generated by its subsheaves $\mathcal{O}_X$ and $\mathcal{T}_X$ subject to the following relations:*

$$f_1 \star f_2 = f_1 f_2,\ \partial_1 \star \partial_2 - \partial_2 \star \partial_1 = [\partial_1, \partial_2],\ f \star \partial = f\partial,\ \partial \star f - f \star \partial = \partial(f),$$

where $f, f_1, f_2 \in \mathcal{O}(U)$ and $\partial, \partial_1, \partial_2 \in \mathcal{T}(U)$ are local sections. Here we temporarily use $\star$ to denote the multplication map of $\mathcal{D}_X$.

Example 4.2.15. For $X = \mathbb{A}^d$, we have $\mathcal{D}(X) = k[x_1, \cdots, x_d, \partial_1, \cdots, \partial_d]$ such that $[x_i, x_j] = [\partial_i, \partial_j] = 0$ and $[\partial_i, x_j] = \delta_{ij}$. This is known as the n-th *Weyl algebra*.

Remark 4.2.16. For any $\mathbb{C}$-point $x \in X$, there exists an open neighborhood $U \subseteq X$ of x and an étale morphism $f : U \to \mathbb{A}^d$, where $d = \dim_x(X)$. Let $f_1, \cdots, f_d \in \mathcal{O}(U)$ be the functions corresponding to the morphism f. It follows that:

- The $\mathcal{O}(U)$-module $\Omega^1(U)$ is free, and $\{df_1, \cdots, df_n\}$ is a basis for it.
- Let $\{\partial_1, \cdots, \partial_2\}$ be the basis for the free $\mathcal{O}(U)$-module $\mathcal{T}(U)$ that is dual to $\{df_1, \cdots, df_n\}$. We have

  $$\mathcal{D}(U) \simeq \mathcal{O}(U)[\partial_1, \cdots, \partial_d],$$

  where

  $$[\partial_i, \partial_j] = 0 \text{ and } \partial_i(f_j) = \delta_{ij}.$$

4.3. **Definition of D-modules.** Throughout this subsection, X is a smooth scheme over $\mathbb{C}$.

Definition 4.3.1. A *(quasi-coherent) left (resp. right) $\mathcal{D}$-module* on X is a left (resp. right) $\mathcal{D}_X$-module $\mathcal{M}$ such that the underlying $\mathcal{O}_X$-module is quasi-coherent.

Let

$$\mathcal{D}_X\text{–}\mathsf{mod}^l_{\mathsf{qc}} \text{ and } \mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}}$$

be the category of left (resp. right) $\mathcal{D}$-modules on X.

The following result follows immediately from Corollary 4.2.14.

Corollary 4.3.2. *Restriction along $\mathcal{D}_X^{\leq 1} \to \mathcal{D}_X$ induces an equivalence of categories:*

$$\mathcal{D}_X\text{–}\mathsf{mod}_{\mathsf{qc}} \simeq (\mathcal{O} \oplus \mathcal{T})_X\text{–}\mathsf{mod}_{\mathsf{qc}}.$$

Our next goal is to identify left $\mathcal{D}$-modules with $\mathcal{O}$-modules equipped with a flat connection. We first recall the latter notion.

Definition 4.3.3. Let $\mathcal{M}$ be a quasi-coherent $\mathcal{O}_X$-module. A *connection* on $\mathcal{M}$ is a $\mathbb{C}$-linear morphism
$$\nabla : \mathcal{M} \to \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \Omega^1_X$$
that satisfies the Lebniz rule:
$$\nabla(f \cdot m) = f \cdot \nabla(m) + m \otimes df,$$
where $f \in \mathcal{O}(U)$ and $m \in \mathcal{M}(U)$ are local sections.

Construction 4.3.4. Let ∇ be a connection on $\mathcal{M}$. For any $n \geq 0$, we obtain a $\mathbb{C}$-linear morphism
$$\nabla : \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \Omega^n_X \to \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \Omega^{n+1}_X$$
given by the formula
$$\nabla(m \otimes \omega) := \nabla(m) \wedge \omega + m \otimes d\omega$$

Definition 4.3.5. We say a connection ∇ on $\mathcal{M}$ is *flat* if the composition
$$\nabla^2 : \mathcal{M} \to \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \Omega^1_X \to \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \Omega^2_X$$
is zero.

Let
$$\mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc},\nabla}$$
be the category of quasi-coherent $\mathcal{O}_X$-modules equipped with a flat connection.

Exercise 4.3.6. Let ∇ be a flat connection on $\mathcal{M}$. Show that $(\mathcal{M} \otimes_{\mathcal{O}_X} \Omega^\bullet_X, \nabla)$ is a complex of sheaves. This is known as the *de-Rham complex for* $\mathcal{M}$.

Warning 4.3.7. There is an obvious flat connection on $\mathcal{O}_X$. However, the corresponding de-Rham complex $(\Omega^\bullet_X, \nabla)$ is *not* exact, i.e., the Poincaré Lemma does not hold in the algebraic geometry setting.

Exercise 4.3.8. Let $X = \mathbb{A}^1 \setminus 0$ be the punctured affine line. Show that the de-Rham complex $(\Omega^\bullet_X, \nabla)$ is not exact.

Construction 4.3.9. Let $\mathcal{M}$ be a quasi-coherent $\mathcal{C}_X$-module. Since $\mathcal{T}_X$ and Ω^1_X are dual as $\mathcal{O}_X$-modules, we have an isomorphism
$$\mathsf{Hom}_{\mathbb{C}}(\mathcal{M}, \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \Omega^1_X) \simeq \mathsf{Hom}_{\mathcal{O}_X}(\mathcal{T}_X \otimes \mathcal{M}, \mathcal{M}),$$
where $\mathcal{O}_X$ acts on $\mathcal{T}_X \otimes \mathcal{M}$ via the first factor.

Exercise 4.3.10 (Problem 6, Homework 6)**.** Show that an element
$$\nabla \in \mathsf{Hom}_{\mathbb{C}}(\mathcal{M}, \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \Omega^1_X)$$
is a flat connection iff the corresponding element
$$\mathsf{act}_{\mathcal{T}} \in \mathsf{Hom}_{\mathcal{O}_X}(\mathcal{T}_X \otimes \mathcal{M}, \mathcal{M})$$
makes $\mathcal{M}$ an $(\mathcal{O} \oplus \mathcal{T})_X$-module.

Corollary 4.3.11. *There are canonical equivalence*
$$\mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc},\nabla} \simeq (\mathcal{O} \oplus \mathcal{T})_X\text{-}\mathsf{mod}_{\mathsf{qc}} \simeq \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}.$$

We end this subsection with some examples of $\mathcal{D}$-modules.

Example 4.3.12. The sheaf $\mathcal{D}_X$ itself is both a left and a right $\mathcal{D}$-module.

Example 4.3.13. The sheaf $\mathcal{O}_X$ is a left $\mathcal{D}$-module with the action morphism given by $D \cdot f := D(f)$, where $D \in \mathcal{D}_X(U)$ and $f \in \mathcal{O}_X(U)$ are local sections.

Example 4.3.14. There is a right $\mathcal{D}$-module structure on the canonical line bundle $\mathcal{K}_X$ given by the following formula

$$\omega \cdot f = f\omega, \;\; \omega \cdot \partial = -\mathcal{L}_\partial(\omega),$$

where $f \in \mathcal{O}_X(U)$, $\partial \in \mathcal{T}_X(U)$ and $\omega \in \mathcal{K}_X(U)$ are local sections.

Example 4.3.15. Let $X = \mathbb{A}^1 = \mathsf{Spec}(k[x])$ be the affine line. We define a left $\mathcal{D}$-module $\mathcal{M}_{\exp(x)}$ as follows.

The underlying $\mathcal{O}_X$-module of $\mathcal{M}_{\exp(x)}$ is a rank 1 free module, with a generator denoted by "$\exp(x)$". The $\mathcal{T}_X$-action on $\mathcal{M}_{\exp(x)}$ is determined by the formula

$$\partial_x \cdot \exp(x) = \exp(x).$$

In other words,

$$\mathcal{M}_{\exp(x)} := \mathcal{D}_X / \mathcal{D}_X \cdot (\partial_x - 1),$$

where $\mathcal{D}_X \cdot (\partial_x - 1)$ is the left ideal of $\mathcal{D}_X$ generated by the section $\partial_x - 1$. It is called the *exponential local system* on $\mathbb{A}^1$.

Example 4.3.16. Let $X = \mathbb{A}^1 \smallsetminus 0 = \mathsf{Spec}(k[x^\pm])$ be the punctured affine line. We define a left $\mathcal{D}$-module $\mathcal{M}_{x^\lambda}$ as follows.

The underlying $\mathcal{O}_X$-module of $\mathcal{M}_{x^\lambda}$ is a rank 1 free module, with a generator denoted by "x^λ". The $\mathcal{T}_X$-action on $\mathcal{M}_{x^\lambda}$ is determined by the formula

$$\partial_x \cdot x^\lambda = \lambda x^{-1} \cdot x^\lambda.$$

In other words,

$$\mathcal{M}_{x^\lambda} := \mathcal{D}_X / \mathcal{D}_X \cdot (x\partial_x - \lambda).$$

It is called the *Kummer local system* on $\mathbb{A}^1 \smallsetminus 0$ with monodromy λ.

Exercise 4.3.17 (Problem 7, Homework 6)**.** Let $X = \mathbb{A}^1 \smallsetminus 0$, decide whether $\mathcal{O}_X$, $\mathcal{M}_{x^\lambda}$ and (the restriction of) $\mathcal{M}_{\exp(x)}$ are isomorphic to as left $\mathcal{D}$-modules on X.

4.4. **Global differential operators.** Recall that the G-action on Fl_G induces an isomorphism between Lie algebras (see Theorem 3.6.8)

$$a : \mathfrak{g} \xrightarrow{\simeq} \mathcal{T}(\mathsf{Fl}_G).$$

By the definition of the universal enveloping algebras, the composition

$$\mathfrak{g} \to \mathcal{T}(\mathsf{Fl}_G) \to \mathcal{D}(\mathsf{Fl}_G)$$

induces a homomorphism between filtered rings

$$a : U(\mathfrak{g}) \to \mathcal{D}(\mathsf{Fl}_G). \tag{4.4}$$

The rest of this section is devoted to the proof of the following result.

Theorem 4.4.1. *The homomorphism* (4.4) *induces an isomorphism*

$$U(\mathfrak{g})_{\chi_0} \to \mathcal{D}(\mathsf{Fl}_G).$$

We first show that (4.4) factors through $U(\mathfrak{g})_{\chi_0}$.

Proposition 4.4.2. *The homomorphism* (4.4) *sends* $\mathsf{ker}(\chi_0) \subseteq Z(\mathfrak{g})$ *to zero.*

Proof. Let $z \in \mathsf{ker}(\chi_0)$. We only need to show $a(z) = 0$. We can assume $z \in \mathsf{F}^{\leq n} Z(\mathfrak{g})$ and therefore $a(z) \in \mathcal{D}^{\leq n}(\mathsf{Fl}_G)$. Since $\mathcal{D}^{\leq n}(\mathsf{Fl}_G)$ is a coherent $\mathcal{O}_X$-module, we only need to show the fiber

$$a(z)|_x \in \Gamma(\mathsf{Fl}_G, \mathbb{C}_x \underset{\mathcal{O}_{\mathsf{Fl}_G}}{\otimes} \mathcal{D}_{\mathsf{Fl}_G})$$

of $a(z)$ at any closed point $x \in \mathsf{Fl}_G(\mathbb{C})$ is zero. In other words, we only need to show the composition

$$U(\mathfrak{g}) \to \Gamma(\mathsf{Fl}_G, \mathcal{D}_{\mathsf{Fl}_G}) \to \Gamma(\mathsf{Fl}_G, \delta_x) \tag{4.5}$$

sends $\mathsf{ker}(\chi_0)$ to zero, where

$$\delta_x := \mathbb{C}_x \underset{\mathcal{O}_{\mathsf{Fl}_G}}{\otimes} \mathcal{D}_{\mathsf{Fl}_G}$$

is called the *right δ-D-module* at x. We need the following general results about right δ-D-modules.

Exercise 4.4.3 (Problem 1, Homework 7)**.** Let X be a smooth scheme and $x \in X(\mathbb{C})$ be a closed point. Consider the right D-module $\delta_x := \mathbb{C}_x \otimes_{\mathcal{O}_X} \mathcal{D}_X$. Let

$$\mathsf{Dirac}_x \in \Gamma(X, \delta_x)$$

be the image of $1 \in \mathbb{C}$ under the composition

$$\mathbb{C} \simeq \Gamma(X, \mathbb{C}_x \underset{\mathcal{O}_X}{\otimes} \mathcal{O}_X) \to \Gamma(X, \mathbb{C}_x \underset{\mathcal{O}_X}{\otimes} \mathcal{D}_X).$$

Show that:

(1) The support of the sheaf δ_x is equal to $\{x\}$.
(2) The right D-module δ_x is generated by Dirac_x.
(3) For a section $f \in \mathcal{O}(U)$ over a neighborhood U of x, we have
$$\mathsf{Dirac}_x \cdot f = f(x)\mathsf{Dirac}_x.$$
(4) For a section $\partial \in \mathcal{T}(U)$ over a neighborhood U of x such that $\partial|_x = 0$, we have
$$\mathsf{Dirac}_x \cdot \partial = 0.$$
(5) The canonical filtration on $\mathcal{D}_X$ induces a filtration on δ_x such that
$$\mathsf{gr}^\bullet \delta_x \simeq \mathsf{Sym}_{\mathcal{O}_X}(\mathbb{C}_x \underset{\mathcal{O}_X}{\otimes} \mathcal{T}_X).$$

Note that we can view $\Gamma(\mathsf{Fl}_G, \delta_x)$ as a right $U(\mathfrak{g})$-module via the homomorphism (4.4), which makes (4.5) an $U(\mathfrak{g})$-linear map. Also note that (4.5) sends 1 to Dirac_x.

Exercise 4.4.4 (Problem 2, Homework 7)**.** Use Exercise 4.4.3 to show that (4.5) induces an isomorphism

$$\mathbb{C} \underset{U(\mathfrak{b}_x)}{\otimes} U(\mathfrak{g}) \xrightarrow{\simeq} \Gamma(\mathsf{Fl}_G, \delta_x),$$

where $\mathfrak{b}_x \subseteq \mathfrak{g}$ is the Borel subalgebra corresponding to $x \in \mathsf{Fl}_G(\mathbb{C})$.

By the prvious exercise, it remains to show that the composition

$$Z(\mathfrak{g}) \to U(\mathfrak{g}) \to \mathbb{C} \underset{U(\mathfrak{b}_x)}{\otimes} U(\mathfrak{g})$$

sends $\mathsf{ker}(\chi_0)$ to 0. Write $\mathfrak{b}^- := \mathfrak{b}_x$ and let $\mathfrak{b}$ be an opposite Borel subalgebra. By (the dual version of) Exericse 2.3.19 and 2.3.20, the compoisition

$$Z(\mathfrak{g}) \to U(\mathfrak{g}) \to \mathbb{C} \underset{U(\mathfrak{n}^-)}{\otimes} U(\mathfrak{g})$$

is equal to

$$Z(\mathfrak{g}) \xrightarrow{\phi} \mathsf{Sym}(\mathfrak{h}) \simeq \mathbb{C} \underset{U(\mathfrak{n}^-)}{\otimes} U(\mathfrak{b}) \to \mathbb{C} \underset{U(\mathfrak{n}^-)}{\otimes} U(\mathfrak{g}).$$

Hence we only need to show χ_0 is equal to

$$Z(\mathfrak{g}) \xrightarrow{\phi} \mathsf{Sym}(\mathfrak{h}) \to \mathbb{C},$$

where $\mathsf{Sym}(\mathfrak{h}) \to \mathbb{C}$ is the trivial character. But this follows from the definition of ϕ (see the proof of Proposition 2.3.6).

□[Proposition 4.4.2]

By the previous proposition, we obtain a well-defined homomorphism between filtered rings:

$$U(\mathfrak{g})_{\chi_0} \to \mathcal{D}(\mathsf{Fl}_G),$$

where the filtration on $U(\mathfrak{g})_{\chi_0}$ is induced by the PBW filtration on $U(\mathfrak{g})$. To prove Theorem 4.4.1, we only need to show the induced homomorphism

$$\mathsf{gr}^\bullet(U(\mathfrak{g})_{\chi_0}) \to \mathsf{gr}^\bullet(\mathcal{D}(\mathsf{Fl}_G)) \tag{4.6}$$

is an isomorphism.

The right exact functor $\Gamma(\mathsf{Fl}_G, -)$ induces an *injection*

$$\mathsf{gr}^\bullet(\mathcal{D}(\mathsf{Fl}_G)) \to \Gamma(\mathsf{Fl}_G, \mathsf{gr}^\bullet\mathcal{D}) \simeq \Gamma(\mathsf{Fl}_G, \mathsf{Sym}_{\mathcal{O}}(\mathcal{T})) \simeq \mathcal{O}(\mathsf{T}^*\mathsf{Fl}_G), \tag{4.7}$$

where

$$\mathsf{T}^*\mathsf{Fl}_G \simeq \mathsf{Spec}_{\mathsf{Fl}_G}(\mathsf{Sym}_{\mathcal{O}}(\mathcal{T}))$$

is the cotangent bundle of Fl_G.

On the other hand, the surjection $U(\mathfrak{g}) \to U(\mathfrak{g})_{\chi_0}$ induces a surjection

$$\mathsf{Sym}(\mathfrak{g}) \simeq \mathsf{gr}^\bullet U(\mathfrak{g}) \to \mathsf{gr}^\bullet(U(\mathfrak{g})_{\chi_0}) \tag{4.8}$$

which sends $\mathsf{gr}^\bullet(\mathsf{ker}(\chi_0))$ to 0. Recall that (see Lemma 2.3.17)

$$\mathsf{gr}^\bullet(Z(\mathfrak{g})) \simeq \mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}} \simeq \mathsf{Sym}(\mathfrak{g})^G.$$

One can check that this isomorphism sends $\mathsf{gr}^\bullet(\mathsf{ker}(\chi_0))$ to the augmentation ideal of $\mathsf{Sym}(\mathfrak{g})^G$. Hence (4.8) induces a surjection

$$\mathcal{O}(\mathfrak{g}^* \underset{\mathfrak{g}^*/\!/G}{\times} 0) \simeq \mathsf{Sym}(\mathfrak{g}) \underset{\mathsf{Sym}(\mathfrak{g})^G}{\otimes} \mathbb{C} \to \mathsf{gr}^\bullet(U(\mathfrak{g})_{\chi_0}). \tag{4.9}$$

In summary, we obtain a chain of homomorphisms

$$\mathcal{O}(\mathfrak{g}^* \underset{\mathfrak{g}^*/\!/G}{\times} 0) \xrightarrow{(4.9)} \mathsf{gr}^\bullet(U(\mathfrak{g})_{\chi_0}) \xrightarrow{(4.6)} \mathsf{gr}^\bullet(\mathcal{D}(\mathsf{Fl}_G)) \xrightarrow{(4.7)} \mathcal{O}(\mathsf{T}^*\mathsf{Fl}_G) \tag{4.10}$$

such that the first homomorphism is a surjection, while the last homomorphism is an injection. Hence to prove Theorem 4.4.1, we only need to show the composition (4.10) is an isomorphism.

Consider the morphism

$$\mathsf{T}^*\mathsf{Fl}_G \to \mathfrak{g}^* \underset{\mathfrak{g}^*/\!/G}{\times} 0$$

corresponding to the homomorphism (4.10). Unwinding the definitions, the composition

$$\mathsf{T}^*\mathsf{Fl}_G \to \mathfrak{g}^* \underset{\mathfrak{g}^*/\!/G}{\times} 0 \to \mathfrak{g}^*$$

sends a closed point $(x,\xi) \in \mathsf{T}^*\mathsf{Fl}_G(\mathbb{C})$ to the image of $\xi \in \mathsf{T}^*_x\mathsf{Fl}_G$ under the map

$$\mathsf{T}^*_x\mathsf{Fl}_G \simeq (\mathfrak{g}/\mathfrak{b}_x)^* \to \mathfrak{g}^*.$$

Such a morphism makes sense for any smooth scheme X equipped with a G-action.

Construction 4.4.5. Let X be a smooth scheme equipped with a G-action. The $\mathcal{O}_X$-linear morphism (see Construction 3.6.1)

$$a : \mathcal{O}_X \otimes \mathfrak{g} \to \mathcal{T}_X$$

induces an X-morphism

$$\mathsf{T}^*X \to X \times \mathfrak{g}^*.$$

The *momentum map* is defined to be the composition

$$\mu : \mathsf{T}^*X \to X \times \mathfrak{g}^* \to \mathfrak{g}^*.$$

Remark 4.4.6. Unwinding the definitions, μ is the unique morphism that sends a closed point $(x,\xi) \in \mathsf{T}^*X(\mathbb{C})$ to the image of $\xi \in \mathsf{T}^*_xX$ under the map

$$\mathsf{T}^*_xX \to \mathfrak{g}^*$$

that is dual to the fiber of $a : \mathcal{O}_X \otimes \mathfrak{g} \to \mathcal{T}_X$ at x.

By the previous discussions, the momentum map $\mu : \mathsf{T}^*\mathsf{Fl}_G \to \mathfrak{g}^*$ factors through $\mathfrak{g}^* \times_{\mathfrak{g}^*/\!/G} 0$, and Theorem 4.4.1 has been reduced to the following result.

Theorem 4.4.7 (Kostant)**.** *The canonical morphism*

$$\mathsf{T}^*\mathsf{Fl}_G \to \mathfrak{g}^* \underset{\mathfrak{g}^*/\!/G}{\times} 0$$

induces an isomorphism

$$\mathcal{O}(\mathfrak{g}^* \underset{\mathfrak{g}^*/\!/G}{\times} 0) \xrightarrow{\simeq} \mathcal{O}(\mathsf{T}^*\mathsf{Fl}_G).$$

Theorem 4.4.7 will be proved in §5.2 using algebro-geometric methods.

5. Geometry of $\mathfrak{g}$

In this section, we study the geometry of the semisimple algebra $\mathfrak{g}$, which will ultimately leads to a proof of Theorem 4.4.7.

5.1. **Nilpotent cone.**

Recall that the Killing form induces a G-linear isomorphism $\mathfrak{g} \simeq \mathfrak{g}^*$. Hence we have

$$\mathfrak{g}^* \underset{\mathfrak{g}^*/\!/G}{\times} 0 \simeq \mathfrak{g} \underset{\mathfrak{c}_\mathfrak{g}}{\times} 0,$$

where $\mathfrak{c}_\mathfrak{g} := \mathfrak{g}/\!/G$ is the Chevalley base of $\mathfrak{g}$.

To prove Theorem 4.4.7, we first study these fiber products.

Definition 5.1.1. The *nilpotent cone* $\mathcal{N}$ of $\mathfrak{g}$ is defined to be the following fiber product

$$\begin{tikzcd} \mathcal{N} \arrow[r] \arrow[d] & \mathfrak{g} \arrow[d] \\ 0 \arrow[r] & \mathfrak{c}_\mathfrak{g}. \end{tikzcd}$$

Remark 5.1.2. Note that $\mathcal{N} \to \mathfrak{g}$ is a closed immersion because so is $0 \to \mathfrak{c}_\mathfrak{g}$.

The name *nilpotent cone* is justified by the following result.

Lemma 5.1.3. *A closed point v of $\mathfrak{g}$ is contained in $\mathcal{N}$ iff it is an nilpotent element.*

Proof. Recall that for any Borel pair $(\mathfrak{b}, \mathfrak{h})$, we have a commutative diagram

$$\begin{tikzcd} \mathfrak{b} \arrow[r] \arrow[d] & \mathfrak{c}_\mathfrak{g} \\ \mathfrak{h} \arrow[r] & \mathfrak{h}/\!/W, \arrow[u, "\simeq"'] \end{tikzcd}$$

where the right vertical isomorphism is due to Chevalley's restriction theorem (see Remark 2.3.21). Hence a closed point v of $\mathfrak{b} \subseteq \mathfrak{g}$ is sent to 0 by the projection $\mathfrak{g} \to \mathfrak{c}_\mathfrak{g}$ iff v is sent to 0 by the composition

$$\mathfrak{b} \to \mathfrak{h} \to \mathfrak{h}/\!/W.$$

Recall that W acts transitively on the fibers of the map $\mathfrak{h} \to \mathfrak{h}/\!/W$ (Proposition 2.4.11). Hence the above condition is equivalent to v being sent to 0 by the morphism $\mathfrak{b} \to \mathfrak{h}$, i.e., v being nilpotent. Now the desired claim follows from the fact that any vector in $\mathfrak{g}$ is contained in a Borel subalgebra of $\mathfrak{g}$.

□

The following result will be proved in the end of this subsection.

Proposition 5.1.4. *The scheme $\mathcal{N}$ is normal.*

Remark 5.1.5. In particular, $\mathcal{N}$ is reduced. Therefore the closed subscheme $\mathcal{N} \subseteq \mathfrak{g}$ is characterized by Lemma 5.1.3.

Exercise 5.1.6 (Problem 3, Homework 7)**.** Show that the tangent space of $\mathcal{N}$ at 0 is isomorphic to $\mathfrak{g}$. Deduce that $\mathcal{N}$ is singular at 0.

To prove Proposition 5.1.4, we need some preparations.

Recall that the Chevalley–Shephard–Todd theorem (Proposition 2.4.6) says that $\mathsf{Sym}(\mathfrak{h})$ is a graded free $\mathsf{Sym}(\mathfrak{h})^W$-module of rank $|W|$.

Proposition 5.1.7 (Bernstein–Lunts)**.** *Let $\{a_w\}_{w\in W}$ be a homogeneous basis of $\mathsf{Sym}(\mathfrak{h})$ over $\mathsf{Sym}(\mathfrak{h})^W$. Then $\{a_w\}_{w\in W}$ is also a free basis of $\mathsf{Sym}(\mathfrak{g})$ over $\mathsf{Sym}(\mathfrak{n}^-)\otimes \mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}}\otimes\mathsf{Sym}(\mathfrak{n})$.*

Warning 5.1.8. The BL theorem does *not* immediately follow from the CST theorem because the images of the following homomorphisms are not equal:

$$\mathsf{Sym}(\mathfrak{n}^-)\otimes\mathsf{Sym}(\mathfrak{g})^{\mathfrak{g}}\otimes\mathsf{Sym}(\mathfrak{n}) \xrightarrow{\mathsf{mult}} \mathsf{Sym}(\mathfrak{g}), \tag{5.1}$$

$$\mathsf{Sym}(\mathfrak{n}^-)\otimes\mathsf{Sym}(\mathfrak{t})^{W}\otimes\mathsf{Sym}(\mathfrak{n}) \xrightarrow{\mathsf{mult}} \mathsf{Sym}(\mathfrak{g}). \tag{5.2}$$

Namely, the CST theorem implies $\mathsf{Sym}(\mathfrak{g})$ is a free module over the source of (5.2), while the BL thoerem claims the same is true for (5.1)

That said, [BL] constructed a *novel* filtration on both sides of (5.1) such that the induced homomorphism between their graded pieces can be identified with (5.2). Then an elementary linear algebra argument reduces the BL theorem to the CST theorem.

Exercise 5.1.9. Verify the BL theorem for $\mathfrak{g}=\mathfrak{sl}_2$. Challenge: using this example to invent the BL filtration.

We also record the following corollary of Proposition 5.1.7, which can be proved by passing to the graded pieces.

Corollary 5.1.10 (Kostant's freeness)**.** *The ring $U(\mathfrak{g})$ is a free $Z(\mathfrak{g})$-module.*

Let us come back to the proof of Proposition 5.1.4. With the help of the Killing form, Proposition 5.1.7 implies the following result.

Corollary 5.1.11. *The obvious morphism*

$$\mathfrak{g}\to\mathfrak{n}\times\mathfrak{c}_{\mathfrak{g}}\times\mathfrak{n}^-$$

is finite and flat. In particular, $\mathfrak{g}\to\mathfrak{c}_{\mathfrak{g}}$ is flat.

Corollary 5.1.12. *We have*

$$\dim(\mathcal{N})\simeq\dim(\mathfrak{g})-\dim(\mathfrak{c}_{\mathfrak{g}})=\dim(\mathfrak{g})-\dim(\mathfrak{h}).$$

Lemma 5.1.13. *The closed immersion $\mathcal{N}\to\mathfrak{g}$ is regular*[9]*. In particular, the scheme $\mathcal{N}$ is Cohen–Macaulay*[10].

Proof. The closed immersion $0\to\mathfrak{c}_{\mathfrak{g}}$ is regular because $\mathfrak{c}_{\mathfrak{g}}\simeq\mathfrak{h}/\!/W$ is isomorphic to an affine space (CST theorem). Now the claim follows from Corollary 5.1.11 and the fact that regular immersions are closed under flat base-changes (see [EGA4, Proposition 19.1.5]).

□

By Serre's criterion (see [EGA4, Theorem 5.8.6]), Proposition 5.1.4 is reduced to the following result, which will be proved in §5.4.

Lemma 5.1.14. *The scheme $\mathcal{N}$ satisfies Serre's condition (R1), i.e., it is regular in codimension* 1.

[9]See [EGA4, Section 16.9] for what this means.
[10]See [EGA4, Section 5.7] for what this means.

5.2. **Springer resolution.** Exercise 3.6.9 implies the following result.

Corollary 5.2.1. *We have a canonical injection*

$$\mathsf{T}^*\mathsf{Fl}_G \to \mathsf{Fl}_G \times \mathfrak{g}^*$$

between vector bundles on Fl_G*, and a closed point* (x,v) *of* $\mathsf{Fl}_G \times \mathfrak{g}^*$ *is contained in its image iff* $v \in (\mathfrak{g}/\mathfrak{b}_x)^*$.

Via the identification $\mathfrak{g} \simeq \mathfrak{g}^*$ given by the Killing form, we have $\mathfrak{n}_x \simeq (\mathfrak{g}/\mathfrak{b}_x)^*$. Hence we can rewrite the previous result as follows.

Corollary 5.2.2. *We have a canonical injection*

$$\mathsf{T}^*\mathsf{Fl}_G \to \mathsf{Fl}_G \times \mathfrak{g} \tag{5.3}$$

between vector bundles on Fl_G*, and a closed point* (x,v) *of* $\mathsf{Fl}_G \times \mathfrak{g}$ *is contained in the image iff* $v \in \mathfrak{n}_x$.

Definition 5.2.3. Let $\widetilde{\mathcal{N}} \subseteq \mathsf{Fl}_G \times \mathfrak{g}$ be the closed subscheme that is equal to the image of (5.3). We call it the *Springer resolution* of $\mathcal{N}$.

Remark 5.2.4. By definition, $\widetilde{\mathcal{N}}$ is the unique reduced closed subscheme of $\mathsf{Fl}_G \times \mathfrak{g}$ whose closed points are $(x,v) \in (\mathsf{Fl}_G \times \mathfrak{g})(\mathbb{C})$ satisfying $v \in \mathfrak{n}_x$.

We provide an alternative definition of $\widetilde{\mathcal{N}}$ as follows. Let $B \subseteq G$ be a Borel subgroup of G. Recall that we have a canonical identification of vector bundles on $G/B \simeq \mathsf{Fl}_G$ (see Exercise 3.6.10):

$$(G \times \mathfrak{g})/B \simeq \mathsf{Fl}_G \times \mathfrak{g}. \tag{5.4}$$

Exercise 5.2.5 (Problem 4, Homework 7)**.** Via the above identification, the subbundle

$$(G \times \mathfrak{n})/B \subseteq (G \times \mathfrak{g})/B$$

corresponds to the subbundle

$$\widetilde{\mathcal{N}} \subseteq \mathsf{Fl}_G \times \mathfrak{g}.$$

Remark 5.2.6. As a consequence of the above exercise, we can alternatively *define* $\widetilde{\mathcal{N}}$ as $(G \times \mathfrak{n})/B$. Note however that this definition *a priori* depends on the choice of B.

Lemma 5.2.7. *The morphism* $\mathfrak{p} : \widetilde{\mathcal{N}} \to \mathcal{N}$ *is proper and surjective.*

Proof. We have a commutative diagram

$$\begin{tikzcd} \widetilde{\mathcal{N}} \arrow[r] \arrow[d] & \mathsf{Fl}_G \times \mathfrak{g} \arrow[d] \\ \mathcal{N} \arrow[r] & \mathfrak{g}. \end{tikzcd}$$

The top horizontal map is proper because it is a closed embedding. The right vertical map is proper because Fl_G is projective. Hence the composition $\widetilde{\mathcal{N}} \to \mathfrak{g}$ is proper. Since $\mathcal{N} \to \mathfrak{g}$ is separated, the map $\widetilde{\mathcal{N}} \to \mathcal{N}$ is proper.

It remains to show that $\mathfrak{p}$ is surjective on closed points. This follows from the fact that any nilpotent element of $\mathfrak{g}$ is contained in some Borel subalgebras. □

Note that $\widetilde{\mathcal{N}}$ is irreducible because it is a vector bundle over Fl_G. Hence Lemma 5.2.7 implies the following result.

Corollary 5.2.8. *The scheme $\mathcal{N}$ is irreducible.*

The following lemma justifies the name of $\widetilde{\mathcal{N}}$. We will prove it in §5.4.

Lemma 5.2.9. *The morphism $\mathfrak{p} : \widetilde{\mathcal{N}} \to \mathcal{N}$ is a birational equivalence.*

Exercise 5.2.10 (Problem 5, Homework 7)**.** For $G = \mathsf{SL}_2$, show that $\widetilde{\mathcal{N}}$ can be identified with the blow-up of $\mathcal{N}$ at the point $0 \in \mathcal{N}$.

Using the schemes $\mathcal{N}$ and $\widetilde{\mathcal{N}}$, Theorem 4.4.7 can be rephased as follows.

Theorem 5.2.11 (Kostant)**.** *The obvious morphism $\mathfrak{p} : \widetilde{\mathcal{N}} \to \mathcal{N}$, $(x, v) \mapsto v$ induces an isomorphism*

$$\mathcal{O}(\mathcal{N}) \to \mathcal{O}(\widetilde{\mathcal{N}}). \tag{5.5}$$

Proof. Note that $\widetilde{\mathcal{N}}$ and $\mathcal{N}$ are both integral (Proposition 5.1.4 and Corollary 5.2.8). Therefore both $\mathcal{O}(\mathcal{N})$ and $\mathcal{O}(\widetilde{\mathcal{N}})$ are integral domains. By Lemma 5.2.9, the homomorphism (5.5) induces an isomorphism

$$\mathsf{Frac}(\mathcal{O}(\mathcal{N})) \xrightarrow{\simeq} \mathsf{Frac}(\mathcal{O}(\widetilde{\mathcal{N}})).$$

Since $\mathfrak{p}$ is proper, $\mathfrak{p}_*\mathcal{O}_{\widetilde{\mathcal{N}}}$ is a coherent $\mathcal{O}_{\mathcal{N}}$-algebra. Since $\mathcal{N}$ is affine, $\mathcal{O}(\widetilde{\mathcal{N}})$ is a finite $\mathcal{O}(\mathcal{N})$-algebra. This implies $\mathcal{O}(\mathcal{N}) \simeq \mathcal{O}(\widetilde{\mathcal{N}})$ because $\mathcal{O}(\mathcal{N})$ is integrally closed. □

5.3. **Grothendieck alteration.** To proceed, we need even more geometric ingredients.

Exercise 5.3.1 (Problem 6, Homework 7)**.** Let B be a Borel subgroup of G. Consider the following subbundle of $(G \times \mathfrak{g})/B$ over G/B

$$(G \times \mathfrak{b})/B \subseteq (G \times \mathfrak{g})/B. \tag{5.6}$$

Let

$$\widetilde{\mathfrak{g}} \subseteq \mathsf{Fl}_G \times \mathfrak{g}$$

be the subbundle of $\mathsf{Fl}_G \times \mathfrak{g}$ over Fl_G that corresponds to (5.6) via the canonical identification (5.4). Show that a closed point (x, v) of $\mathsf{Fl}_G \times \mathfrak{g}$ is contained in $\widetilde{\mathfrak{g}}$ iff $v \in \mathfrak{b}_x$. Deduce that $\widetilde{\mathfrak{g}}$ does not depend on the choice of B.

Definition 5.3.2. We call $\widetilde{\mathfrak{g}}$ the *Grothendieck alteration* of $\mathfrak{g}$.

Lemma 5.3.3. *The morphism $\mathfrak{q} : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ is proper and surjective.*

Proof. Similar to Lemma 5.3.3. □

The following lemma justifies the name of $\widetilde{\mathfrak{g}}$. We will prove it in §5.4.

Lemma 5.3.4. *The morphism $\mathfrak{q} : \widetilde{\mathfrak{g}} \to \mathfrak{g}$ is generically a finite Galois covering.*

Note that there is an obvious injection

$$\widetilde{\mathcal{N}} \to \widetilde{\mathfrak{g}}$$

between subbundles of $\mathsf{Fl}_G \times \mathfrak{g}$. We have the following key observation.

Lemma 5.3.5. *Let B be a Borel subgroup of G and $\mathfrak{b}$ be the Cartan* quotient *algebra of $\mathfrak{b}$. There is a canonical short exact sequence of vector bundles on Fl_G:*

$$0 \to \widetilde{\mathcal{N}} \to \widetilde{\mathfrak{g}} \to \mathsf{Fl}_G \times \mathfrak{h} \to 0$$

Proof. The claim follows from Exercise 5.2.5 and the short exact sequence

$$0 \to (G \times \mathfrak{n})/B \to (G \times \mathfrak{b})/B \to G/B \times \mathfrak{h} \to 0,$$

where we used the fact that the adjoint B-action on $\mathfrak{h}$ is trivial. □

As a cool byproduct of the above lemma, we see that the Cartan quotient algebra $\mathfrak{h}$ of $\mathfrak{g}$ does not depend on *any* choice! Indeed, it can be identified with the global section of the quotient bundle of $\widetilde{\mathfrak{g}}$ by $\widetilde{\mathcal{N}}$.

Remark 5.3.6. The discussion below is only for aesthetic reasons: one could fix a Borel subgroup and finish the proofs.

Exercise 5.3.7 (Problem 7, Homework 7)**.** Let B and B' be Borel subgroups of G. Write $H := B/[B,B]$ and $H' := B'/[B',B']$ for the corresponding Cartan quotient groups.

(1) Let $g \in G(\mathbb{C})$ be a point such that $B' = \mathsf{Ad}_g(B)$. Show that the isomorphism $\overline{\mathsf{Ad}}_g : H \to H'$ does not depend on the choice of g.
(2) Let $\mathfrak{c}_{B,B'} : H \to H'$ be the isomorphism obtained in (1). Show that

$$\mathfrak{c}_{B',B''} \circ \mathfrak{c}_{B,B'} = \mathfrak{c}_{B,B''}.$$

Definition 5.3.8. The *abstract Cartan group* for G is an algebraic group H equipped with a system of isomorphisms $\{\mathfrak{r}_B : \mathsf{H} \xrightarrow{\simeq} B/[B,B]\}$ indexed by Borel subgroups B of G, such that for two Borel subgroups B and B', we have

$$\mathfrak{c}_{B,B'} \circ \mathfrak{r}_B = \mathfrak{r}_{B'}.$$

Remark 5.3.9. The data $(\mathsf{H}, \{\mathfrak{r}_B\})$ is unique up to unique isomorphism.

Let h be the Lie algebra of H. We call it the *abstract Cartan algebra* for $\mathfrak{g}$.

Unwinding the definitions, we obtain the following corollary of Lemma 5.3.5.

Corollary 5.3.10. *There is a canonical short exact sequence of vector bundles over* Fl_G*:*

$$0 \to \widetilde{\mathcal{N}} \to \widetilde{\mathfrak{g}} \to \mathsf{Fl}_G \times \mathsf{h} \to 0.$$

We can also define an abstract version of the Weyl group.

Exercise 5.3.11. Let (B,H) and (B',H') be Borel pairs.

(1) Show that there exists an element $g \in G(\mathbb{C})$ such that

$$(B',H') = \mathsf{Ad}_g(B,H).$$

(2) Let $g \in G(\mathbb{C})$ be an element as in (1). Show that the induced isomorphism

$$\overline{\mathsf{Ad}}_g : N_G(H)/H \to N_G(H')/H'$$

does not depend on the choice of g.
(3) Let

$$\mathfrak{c}_{(B,H),(B',H')} : N_G(H)/H \to N_G(H')/H'$$

be the isomorphism in (2). Show that

$$\mathfrak{c}_{(B',H'),(B'',H'')} \circ \mathfrak{c}_{(B,H),(B',H')} = \mathfrak{c}_{(B,H),(B'',H'')}.$$

Definition 5.3.12. The *abstract Weyl group* for G is a finite group W equipped with a system of isomorphisms $\{\mathfrak{r}_{(B,H)} : \mathsf{W} \xrightarrow{\simeq} N_G(H)/H\}$ indexed by Borel pairs (B,H) of G, such that for two Borel pairs (B,H) and (B',H'), we have

$$\mathfrak{c}_{(B,H),(B',H')} \circ \mathfrak{r}_{(B,H)} = \mathfrak{r}_{(B',H')}.$$

Remark 5.3.13. The data $(\mathsf{W}, \{\mathfrak{r}_{(B,H)}\})$ is unique up to unique isomorphism.

Construction 5.3.14. Let (B,H) be a Borel pair. Consider the adjoint action of $N_G(H)/H$ on $\mathfrak{h}$. Via the isomorphisms

$$\mathsf{W} \xrightarrow{\mathfrak{r}_{(B,H)}} N_G(H)/H, \quad \mathsf{h} \xrightarrow{\mathfrak{r}_B} B/[B,B] \xleftarrow{\simeq} H, \tag{5.7}$$

we obtain an action of the abstract Weyl group W on the abstract Cartan algebra h. One can check that this action does not depend on the choice of the Borel pair (B,H). In other words, we obtain a *canonical* W-action on h.

Construction 5.3.15. Let (B,H) be a Borel pair and $W \simeq N_G(H)/H$ be the corresponding Weyl group. The isomorphisms (5.7) induces an isomorphism

$$\mathsf{h}/\!/\mathsf{W} \xrightarrow{\mathfrak{r}_{(B,H)}} \mathfrak{h}/\!/W.$$

One can check the composition

$$\mathsf{h}/\!/\mathsf{W} \xrightarrow{\mathfrak{r}_{(B,H)}} \mathfrak{h}/\!/W \xrightarrow{\simeq} \mathfrak{g}/\!/G =: \mathfrak{c}$$

does not depend on the choice of the Borel pair (B,H). We call it the *abstract Chevalley isomorphism.*

As a consequence, we obtain a *canonical* map $\mathsf{h} \to \mathfrak{c}$.

Exercise 5.3.16 (Problem 1, Homework 8)**.** Show that the following diagram commute:

$$\begin{tikzcd} \widetilde{\mathfrak{g}} \arrow[r] \arrow[d] & \mathfrak{g} \arrow[d] \\ \mathsf{h} \arrow[r] & \mathfrak{c}. \end{tikzcd} \tag{5.8}$$

As an application of the Grothendieck alteration, we sketch a new proof of the flatness of the morphism $\mathfrak{g} \to \mathfrak{c}$ in the following exercise.

Exercise 5.3.17 (Problem 2, Homework 8)**.** Consider the commutative diagram (5.8).

(1) Show that the fibers of the morphism $\widetilde{\mathfrak{g}} \to \mathfrak{c}$ all have dimension $\dim(\mathfrak{g}) - \dim(\mathsf{h})$.
(2) Use (1) to deduce that the fibers of the morphism $\mathfrak{g} \to \mathfrak{c}$ all have dimension $\dim(\mathfrak{g}) - \dim(\mathsf{h})$.
(3) Use the Miracle Flatness Theorem (see [EGA4, Proposition 6.1.5]) to deduce that $\mathfrak{g} \to \mathfrak{c}$ is flat.

5.4. **Regular loci I.** The commutative diagram (5.8) fits into the following commutative cube:

(5.9)
$$\begin{tikzcd}
 & \mathcal{N} \arrow[rr] \arrow[dd] & & \mathfrak{g} \arrow[dd] \\
\widetilde{\mathcal{N}} \arrow[ur] \arrow[rr] \arrow[dd] & & \widetilde{\mathfrak{g}} \arrow[ur] \arrow[dd] & \\
 & 0 \arrow[rr] & & \mathfrak{c} \\
0 \arrow[ur] \arrow[rr] & & \mathsf{h} \arrow[ur] &
\end{tikzcd}$$

such that the front and the back squares are Cartesian. In order to show $\widetilde{\mathcal{N}} \to \mathcal{N}$ is a birational equivalence, we only need to show the right square is Cartesian over an open subset of $\mathfrak{g}$ that intersects $\mathcal{N}$. We first introduce this open subset.

Lemma 5.4.1. *Let $v \in \mathfrak{g}$ be an element and $G^v \subseteq G$ be its centralizer. We have*

$$\min_{v \in \mathfrak{g}} \dim(G^v) = \dim(\mathsf{h}).$$

Proof. Consider the universal centralizer defined as the following fiber product

$$\begin{tikzcd}
C_G \arrow[r] \arrow[d] & G \times \mathfrak{g} \arrow[d, "(\mathsf{Ad},\mathsf{id})"] \\
\mathfrak{g} \arrow[r, "\Delta"] & \mathfrak{g} \times \mathfrak{g}.
\end{tikzcd}$$

By definition, the fiber of $C_G \to \mathfrak{g}$ at the $\mathbb{C}$-point v is isomorphic to G^v. Note that $\dim(G^v) \geq \dim(\mathsf{h})$ if v is an semisimple element because v is contained in a Cartan subalgebra. Since semisimple elements are dense in $\mathfrak{g}$, the Semicontinuity Theorem (see [EGA4, Theorem 13.1.3]) implies that $\dim(G^v) \geq \dim(\mathsf{h})$ for any closed point $v \in \mathfrak{g}$.

It remains to find an element $v \in \mathfrak{g}$ such that $\dim(G^v) = \dim(\mathsf{h})$. Let $\mathfrak{h} \subseteq \mathfrak{g}$ be a Cartan subalgebra. Choose a faithful representation V of $\mathfrak{g}$. Let $\{\lambda_i\} \subseteq \mathfrak{h}^*$ be the weights of V. Choose an element v of $\mathfrak{h}$ such that the numbers $\lambda_i(v)$ are all distinct. Since V is a faithful representation, an element $g \in G(\mathbb{C})$ commutes with v iff the action Ad_g on V commutes with Ad_v. The definition of v implies the latter is equivalent to the condition that Ad_g stablizes the weight subspaces of V. This implies that the centralizer of v is equal to the centralizer of $\mathfrak{h}$, and therefore is equal to H. In particular, $\dim(G^v) = \dim(\mathsf{h})$.

□

Definition 5.4.2. We say an element $v \in \mathfrak{g}$ is *regular* if $\dim(G^v) = \dim(\mathsf{h})$.

Exercise 5.4.3 (Problem 3, Homework 8)**.** An element v in $\mathfrak{sl}_n$ is regular iff the Jordan blocks of v have distinct eigenvalues.

The Semicontinuity Theorem implies the following result.

Corollary 5.4.4. *There exists an unique dense open subscheme $\mathfrak{g}^{\mathsf{reg}} \subseteq \mathfrak{g}$ such that $v \in \mathfrak{g}^{\mathsf{reg}}(\mathbb{C})$ iff v is a regular element in $\mathfrak{g}$.*

Definition 5.4.5. Let

$$\mathcal{N}^{\mathsf{reg}} \subseteq \mathcal{N}, \ \ \widetilde{\mathcal{N}}^{\mathsf{reg}} \subseteq \widetilde{\mathcal{N}}, \ \ \widetilde{\mathfrak{g}}^{\mathsf{reg}} \subseteq \widetilde{\mathfrak{g}}$$

be the inverse images of the open subscheme $\mathfrak{g}^{\mathsf{reg}} \subseteq \mathfrak{g}$. We call them the *regular loci* of the corresponding schemes.

The following lemma characterizes regular nilpotent elements.

Lemma 5.4.6. *Let $(\mathfrak{b},\mathfrak{h})$ be a Borel pair of $\mathfrak{g}$ and $v \in \mathfrak{n} := [\mathfrak{b},\mathfrak{b}]$ be a nilpotent element in $\mathfrak{b}$. If v is regular, then for any simple positive root α with respect to $(\mathfrak{b},\mathfrak{h})$, the image v_α of v under the projection*

$$\mathfrak{n} \simeq \bigoplus_{\alpha \in \Phi^+} \mathfrak{g}_\alpha \to \mathfrak{g}_\alpha$$

is nonzero.

Proof. Consider the orbit Bv for the adjoint B-action on $\mathfrak{n}$. We have

$$\dim(\mathfrak{n}) \geq \dim(Bv) = \dim(B) - \dim(\mathsf{Stab}_B(v)) \geq \dim(B) - \dim(G^v). \tag{5.10}$$

If v is regular, this inequality becomes an equality. Hence Bv is dense in $\mathfrak{n}$. This implies Hv_α is dense in $\mathfrak{g}_\alpha$. In particular, $v_\alpha \neq 0$. □

Remark 5.4.7. The converse statement of Lemma 5.4.6 is also true. We will prove it in §5.6.

Lemma 5.4.8. *Let v be a regular nilpotent element in $\mathfrak{g}$. There is a unique Borel subalgebra that contains v.*

Proof. Let $\mathfrak{b}$ and $\mathfrak{b}'$ be two Borel subalgebras both containing v. Let $\mathfrak{h} \subseteq \mathfrak{b} \cap \mathfrak{b}'$ be a common Cartan subalgebra. Consider the corresponding root decomposition

$$\mathfrak{g} \simeq \mathfrak{h} \oplus (\bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha).$$

Let Φ^+ and Φ'^+ be the subsets of positive roots with respect to $\mathfrak{b}$ and $\mathfrak{b}'$. Let Δ and Δ' be the corresponding subsets of simple positive roots. Lemma 5.4.6 implies $\Delta \subseteq \Phi'^+$. It is easy to see this implies $\Phi^+ = \Phi'+$ and therefore $\mathfrak{b} = \mathfrak{b}'$. □

Corollary 5.4.9. *The morphism $\widetilde{\mathcal{N}}^{\mathsf{reg}} \to \mathcal{N}^{\mathsf{reg}}$ induces a bijection between $\mathbb{C}$-points.*

Remark 5.4.10. Note that the above result does not imply Lemma 5.2.9 unless we can prove $\mathcal{N}^{\mathsf{reg}}$ is smooth (and dense in $\mathcal{N}$). However, I know no proof of this fact without using the following theorem.

Theorem 5.4.11. *The following commutative square is Cartesian:*

$$\begin{tikzcd} \widetilde{\mathfrak{g}}^{\mathsf{reg}} \arrow[r] \arrow[d] & \mathfrak{g}^{\mathsf{reg}} \arrow[d] \\ \mathfrak{h} \arrow[r] & \mathfrak{c}. \end{tikzcd} \tag{5.11}$$

We will prove Theorem 5.4.11 in §5.5. Let us first state some corollaries of it.

Corollary 5.4.12. *The morphism $\mathfrak{g}^{\mathsf{reg}} \to \mathfrak{c}$ is smooth.*

Proof. Note that $\widetilde{\mathfrak{g}}^{\mathsf{reg}}$ is smooth over h by Corollary 5.3.10. Then the claim follows from Theorem 5.4.11 and the flat descent for smooth morphisms (see [EGA4, Corollary 17.7.3]).

□

Corollary 5.4.13. *The scheme $\mathcal{N}^{\mathsf{reg}}$ is smooth.*

The following result will be proved in §5.6.

Proposition 5.4.14. *We have*

$$\dim(\mathcal{N} \smallsetminus \mathcal{N}^{\mathsf{reg}}) \leq \dim(\mathcal{N}) - 2.$$

Proof of Lemma 5.1.14. Follows by combining Corollary 5.4.13, Lemma 5.4.14 and Corollary 5.2.8.

□[Lemma 5.1.14]

Proof of Lemma 5.2.9. By Proposition 5.4.14 and Corollary 5.2.8, $\mathcal{N}^{\mathsf{reg}}$ is a dense open subset of $\mathcal{N}$. Since $\widetilde{\mathcal{N}}$ is integral, we only need to show

$$\widetilde{\mathcal{N}}^{\mathsf{reg}} \to \mathcal{N}^{\mathsf{reg}}$$

is an isomorphism.

The commutative cube 5.9 induces a commutative cube

$$\begin{tikzcd}
& \mathcal{N}^{\mathsf{reg}} \arrow[rr] \arrow[dd] & & \mathfrak{g}^{\mathsf{reg}} \arrow[dd] \\
\widetilde{\mathcal{N}}^{\mathsf{reg}} \arrow[ur] \arrow[rr] \arrow[dd] & & \widetilde{\mathfrak{g}}^{\mathsf{reg}} \arrow[ur] \arrow[dd] & \\
& 0 \arrow[rr] & & \mathfrak{c} \\
0 \arrow[ur] \arrow[rr] & & \mathsf{h}. \arrow[ur] &
\end{tikzcd}$$

By definition, the back square is Cartesian. By Corollary 5.3.10, the front square is Cartesian. By Theorem 5.4.11, the right square is Cartesian. These formally imply that the left square is Cartesian. Therefore $\widetilde{\mathcal{N}}^{\mathsf{reg}} \simeq \mathcal{N}^{\mathsf{reg}}$ as desired.

□[Lemma 5.2.9]

To prove Theorem 5.4.11, we need to study a smaller locus of $\mathfrak{g}^{\mathsf{reg}}$ consisting of regular *semisimple* elements. The following exercise characterizes these elements.

Exercise 5.4.15 (Problem 4, Homework 8)**.** Let v be a semisimple elements in $\mathfrak{g}$.

(1) Show that v is regular iff v is contained in a unique Cartan subalgebra of $\mathfrak{g}$.
(2) Let $\mathfrak{h}$ be a Cartan subalgebra. Use (1) to deduce that the following open subschemes of $\mathfrak{h}$ are equal:
   - The intersection $\mathfrak{h}^{\mathsf{reg}} := \mathfrak{g} \cap \mathfrak{g}^{\mathsf{reg}}$;
   - The open locus where the Weyl group acts freely;
   - The open locus where the morphism $\mathfrak{h} \to \mathfrak{c}$ is smooth.

Definition 5.4.16. Let $\mathsf{h}^{\mathsf{reg}} \subseteq \mathsf{h}$ be the open subscheme where $\mathsf{h} \to \mathfrak{c}$ is smooth. Let $\mathfrak{c}^{\mathsf{reg}} \subseteq \mathfrak{c}$ be the open subscheme that is equal to the image of $\mathsf{h}^{\mathsf{reg}}$. We call them the *regular loci* of the corresponding schemes.

Exercise 5.4.15 implies the following result.

Corollary 5.4.17. *The abstract Weyl group* W *acts freely on* $\mathsf{h}^{\mathsf{reg}}$. *Moreover, the morphism* $\mathsf{h}^{\mathsf{reg}} \to \mathfrak{c}^{\mathsf{reg}}$ *induces an isomorphism*

$$\mathsf{h}^{\mathsf{reg}}/\mathsf{W} \xrightarrow{\simeq} \mathfrak{c}^{\mathsf{reg}}.$$

Lemma 5.4.18. *The closed points of the open subscheme*

$$\mathfrak{g} \underset{\mathfrak{c}}{\times} \mathfrak{c}^{\mathsf{reg}} \subseteq \mathfrak{g}$$

are exactly regular semisimple elements in $\mathfrak{g}$.

Proof. Let v be an element in $\mathfrak{g}$. Choose a Borel subalgebra $\mathfrak{b}$ containing v. Let $\mathfrak{h}$ be the Cartan quotient algebra of $\mathfrak{b}$. Recall that we have a commutative square

$$\begin{tikzcd} \mathfrak{b} \arrow[r] \arrow[d] & \mathfrak{g} \arrow[d] \\ \mathsf{h} \arrow[r] & \mathfrak{c}. \end{tikzcd}$$

We first show that regular semisimple elements are contained in $(\mathfrak{g} \times_{\mathfrak{c}} \mathfrak{c}^{\mathsf{reg}})(\mathbb{C})$. Suppose that v is regular semisimple, then its image under the map $\mathfrak{b} \to \mathsf{h}$ is contained in $\mathsf{h}^{\mathsf{reg}}$. Hence its image in $\mathfrak{c}$ is contained in $\mathfrak{c}^{\mathsf{reg}}$ as desired.

We now show that elements in $(\mathfrak{g} \times_{\mathfrak{c}} \mathfrak{c}^{\mathsf{reg}})(\mathbb{C})$ are regular semisimple. Suppose that the image of v in $\mathfrak{c}$ is contained in $\mathfrak{c}^{\mathsf{reg}}$. Then its image under $\mathfrak{b} \to \mathsf{h}$ is contained in $\mathsf{h}^{\mathsf{reg}}$. Let $v = v_s + v_n$ be the Jordan decomposition of v, such that v_s is semisimple, v_n is nilpotent and $[v_s, v_n] = 0$. The assumption on v implies that v_s is regular. Then the centralizer $\mathfrak{g}^{v_s}$ is the unique Cartan subalgebra that contains v_s. Since $v_n \in \mathfrak{g}^{v_s}$, we obtain $v_n = 0$. Hence $v = v_s$ is regular semisimple.

□

Definition 5.4.19. The *regular semisimple locus* of $\mathfrak{g}$ is defined to be the following open subscheme:

$$\mathfrak{g}^{\mathsf{rss}} := \mathfrak{g} \underset{\mathfrak{c}}{\times} \mathfrak{c}^{\mathsf{reg}} \subseteq \mathfrak{g}$$

The *regular semisimple locus* of $\widetilde{\mathfrak{g}}$ is defined to be the following open subscheme:

$$\widetilde{\mathfrak{g}}^{\mathsf{rss}} := \widetilde{\mathfrak{g}} \underset{\mathfrak{g}}{\times} \mathfrak{g}^{\mathsf{rss}} \subseteq \widetilde{\mathfrak{g}}.$$

5.5. **Regular loci II.** In this subsection we prove Theorem 5.4.11. We first show the fibers of the horizontal morphisms in (5.11) are of the same size.

Proposition 5.5.1. *Let* v *be a regular element in* $\mathfrak{g}$ *and* $\overline{v}$ *be its image in* $\mathfrak{c}$. *We have*

$$|\mathsf{h} \underset{\mathfrak{c}}{\times} \overline{v}| = |\widetilde{\mathfrak{g}} \underset{\mathfrak{g}}{\times} v|.$$

Example 5.5.2. When v is a regular nilpotent element, the proposition follows immediately from Lemma 5.4.8.

Exercise 5.5.3 (Problem 5, Homework 8)**.** Verify Proposition 5.5.1 for $\mathfrak{sl}_n$ by calculating both sides in terms of the Jordan blocks of v.

Exercise 5.5.4 (Problem 6, Homework 8)**.** Verify Proposition 5.5.1 for regular semisimple elements by using Exercise 5.4.15.

Sketch of Proposition 5.5.1. Let $v = v_s + v_n$ be the Jordan decomposition of v.

Let $\mathfrak{l} \subseteq \mathfrak{g}$ be the centralizer of v_s, and $L \subseteq G$ be the corresponding connected subgroup[11]. By the theory of root systems, it is easy to see $\mathfrak{l}$ is a *reductive* Lie algebra. In other words,

$$\mathfrak{l} = \mathfrak{z}(\mathfrak{l}) \oplus \mathfrak{l}',$$

where $\mathfrak{z}(\mathfrak{l})$ is the center of $\mathfrak{l}$, and $\mathfrak{l}' := [\mathfrak{l},\mathfrak{l}]$ is a semisimple Lie algebra.

Note that $v_s \in \mathfrak{z}(\mathfrak{l})$ and $v_n \in \mathfrak{l}'$. Moreover, v being regular in $\mathfrak{g}$ implies v_n being regular in $\mathfrak{l}'$. By Lemma 5.4.8, there is a unique Borel subalgebra $\mathfrak{b}_0 \subseteq \mathfrak{l}'$ that contains v_n.

We claim that a Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ contains v iff

$$\mathfrak{b} \cap \mathfrak{l} = \mathfrak{z}(\mathfrak{l}) \oplus \mathfrak{b}_0. \tag{5.12}$$

The "if" statement is obvious. For the "only if" statement, suppose $v \in \mathfrak{b}$. The uniqueness of the Jordan decomposition implies $v_s \in \mathfrak{b}$ and $v_n \in \mathfrak{b}$. Let $\mathfrak{h} \subseteq \mathfrak{b}$ be a Cartan subalgebra that contains v_s. By definition, we have $\mathfrak{h} \subseteq \mathfrak{l}$, hence $\mathfrak{h}$ is also a Cartan subalgebra of $\mathfrak{l}$. In particular, $\mathfrak{z}(\mathfrak{l}) \subseteq \mathfrak{h} \subseteq \mathfrak{b}$. Since $\mathfrak{b}$ and $\mathfrak{l}$ have a common Cartan subalgebra, the intersection $\mathfrak{b} \cap \mathfrak{l}$ is also a Borel subalgebra of $\mathfrak{l}$. Hence $\mathfrak{b} \cap \mathfrak{l}'$ is a Borel subalgebra of $\mathfrak{l}'$. Then $v_n \in \mathfrak{b} \cap \mathfrak{l}'$ implies that $\mathfrak{b}_0 = \mathfrak{b} \cap \mathfrak{l}'$. In summary, we have obtained the desired identity (5.12).

It remains to show that the number of Borel subalgebras $\mathfrak{b} \subseteq \mathfrak{g}$ satisfying (5.12) is equal to $|\mathsf{h} \times_{\mathsf{c}} \overline{v}|$.

Choose a Cartan subalgebra $\mathfrak{h}_0$ of $\mathfrak{l}'$ inside $\mathfrak{b}_0$. Note that $\mathfrak{h} := \mathfrak{z}(\mathfrak{l}) \oplus \mathfrak{h}_0$ is a Cartan subalgebra of $\mathfrak{g}$. Let $H_0 \subseteq L'$ and $H \subseteq G$ be the corresponding Cartan subgroups. Let

$$W_L := N_{L'}(H_0)/H_0, \;\; W := N_G(H)/H$$

be the corresponding Weyl groups. Note that

$$W_L \simeq N_L(H)/H$$

can be viewed as a subgroup of W. It is easy to see that this subgroup is equal to $\mathsf{Stab}_W(v_s)$. Unwinding the definitions, we have a bijection $\mathsf{h} \times_{\mathsf{c}} \overline{v} \simeq W/\mathsf{Stab}_W(v_s)$. Hence we only need to show that the number of Borel subalgebras $\mathfrak{b} \subseteq \mathfrak{g}$ satisfying (5.12) is equal to the index of $[W : W_L]$. This is a standard fact and can be proved by identifying Borel subalgebras containing $\mathfrak{h}$ with Weyl chambers.

□[Proposition 5.5.1]

Now consider the morphism

$$f : \widetilde{\mathfrak{g}}^{\mathsf{reg}} \to \mathsf{h} \underset{\mathsf{c}}{\times} \mathfrak{g}^{\mathsf{reg}}$$

induced by (5.11).

Lemma 5.5.5. *The morphism* $f : \widetilde{\mathfrak{g}}^{\mathsf{reg}} \to \mathsf{h} \times_{\mathsf{c}} \mathfrak{g}^{\mathsf{reg}}$ *is proper.*

Proof. Follows from the fact that both the source and the target are proper over $\mathfrak{g}^{\mathsf{reg}}$.

□

[11]This is known as the *Levi subgroup* corresponding to v_s.

Lemma 5.5.6. *The scheme* $\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}$ *is of equidimension* $\dim(\mathfrak{g})$.

Proof. The morphisms $\mathfrak{g}^{\mathrm{reg}}\to\mathfrak{c}$ and $\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}\to\mathsf{h}$ are flat. Then the claim follows from the dimension formula for flat morphisms (see [EGA4, Proposition 6.1.1]). □

Lemma 5.5.7. *The scheme* $\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}$ *is reduced.*

Proof. The morphism $\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}\to\mathfrak{g}^{\mathrm{reg}}$ is Cohen–Macaulay because it is finite and flat (see [EGA4, Definition 6.8.1]). Since $\mathfrak{g}^{\mathrm{reg}}$ is smooth, it is Cohen–Macaulay. It follows that Y is Cohen–Macaulay (see [EGA4, Corollary 6.3.5]).

By Serre's criterion (see [EGA4, Proposition 5.8.5]), we only need to show $\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}$ satisfies (R0), i.e., $\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}$ is regular in codimension 0. Note that

$$\mathsf{h}^{\mathrm{reg}}\underset{\mathfrak{c}}{\times}\mathfrak{g}^{\mathrm{reg}}\subseteq\mathsf{h}\underset{\mathfrak{c}}{\times}\mathfrak{g}^{\mathrm{reg}}$$

is a smooth open subscheme. Now the claim follows from the fact that the complement closed subscheme

$$(\mathsf{h}\smallsetminus\mathsf{h}^{\mathrm{reg}})\underset{\mathfrak{c}}{\times}\mathfrak{g}^{\mathrm{reg}}$$

is of equidimension

$$\dim(\mathfrak{g}^{\mathrm{reg}})+\dim(\mathsf{h}\smallsetminus\mathsf{h}^{\mathrm{reg}})-\dim(\mathfrak{c})=\dim(\mathfrak{g})-1.$$

□

Remark 5.5.8. Lemma 5.5.7 reflects the phenomenon that it is easier to study $\widetilde{\mathfrak{g}}\to\mathfrak{g}$ than $\widetilde{\mathcal{N}}\to\mathcal{N}$. Namely, it is much easier to verify (R0) for $\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}$ than $\mathcal{N}^{\mathrm{reg}}:=0\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}$.

Lemma 5.5.9. *The morphism* $f:\widetilde{\mathfrak{g}}^{\mathrm{reg}}\to\mathsf{h}\times_{\mathfrak{c}}\mathfrak{g}^{\mathrm{reg}}$ *is unramified*[12]*. In particular,* f *is quasi-finite*[13].

Proof. We only need to show $\widetilde{\mathfrak{g}}^{\mathrm{reg}}\to\mathsf{h}\times\mathfrak{g}^{\mathrm{reg}}$ is unramified (see [EGA4, Proposition 17.3.3(v)]). Since both $\widetilde{\mathfrak{g}}^{\mathrm{reg}}$ and $\mathsf{h}\times\mathfrak{g}^{\mathrm{reg}}$ is smooth, we only need to show f induces injections between the tangent spaces (see [EGA4, Theorem 17.4.1]). We sketch the proof of this claim in the following exercise.

Exercise 5.5.10 (Problem 7, Homework 8)**.** Let (x,v) be a closed point of $\widetilde{\mathfrak{g}}\subseteq\mathsf{Fl}_G\times\mathfrak{g}$ and y be its image in $\mathfrak{g}\times\mathsf{h}$. Write $\mathfrak{h}_x:=\mathfrak{b}_x/[\mathfrak{b}_x,\mathfrak{b}_x]$.

(1) Identify $\mathcal{T}_{\widetilde{\mathfrak{g}}}|_{(x,v)}$ with the cokernel of the map

$$\mathfrak{b}_x\to\mathfrak{g}\oplus\mathfrak{b}_x,\ \ u\mapsto(u,[v,u]).$$

(2) Identify $\mathcal{T}_{\mathfrak{g}\times\mathsf{h}}|_y$ with $\mathfrak{g}\oplus\mathfrak{h}_x$.

(3) Show that via the identifications in (1) and (2), the map

$$\mathcal{T}_{\widetilde{\mathfrak{g}}}|_{(x,v)}\to\mathcal{T}_{\mathfrak{g}\times\mathsf{h}}|_y \tag{5.13}$$

is induced by the map

$$\mathfrak{g}\oplus\mathfrak{b}_x\to\mathfrak{g}\oplus\mathfrak{h}_x,\ (a,b)\mapsto([a,v]+b,\bar{b}).$$

(4) Show that if v is regular, the map (5.13) is injective.

□[Lemma 5.5.9]

[12]See [EGA4, Section 17.3] for what this means.
[13]See [EGA2, Section 6.2] for what this means.

Lemma 5.5.11. *The following base-change of f is an isomorphism*

$$f' : \widetilde{\mathfrak{g}}^{\mathsf{rss}} \to \mathsf{h} \underset{\mathfrak{c}}{\times} \mathfrak{g}^{\mathsf{rss}}.$$

Proof. Note that we have

$$\mathsf{h} \underset{\mathfrak{c}}{\times} \mathfrak{g}^{\mathsf{rss}} \simeq \mathsf{h}^{\mathsf{reg}} \underset{\mathfrak{c}^{\mathsf{reg}}}{\times} \mathfrak{g}^{\mathsf{rss}}. \tag{5.14}$$

Hence $\mathsf{h} \times_{\mathfrak{c}} \mathfrak{g}^{\mathsf{rss}}$ is smooth because $\mathsf{h}^{\mathsf{reg}} \to \mathfrak{c}^{\mathsf{reg}}$ is smooth.

The morphism f' is proper and unramified by Lemma 5.5.5, Lemma 5.5.9. Hence f' is a surjective local isomorphism. To show f' is an isomorphism, we only need to show f' is injective. Note that (5.14) is a Galois covering over $\mathfrak{g}^{\mathsf{rss}}$ of degree $|\mathsf{W}|$ because $\mathfrak{c}^{\mathsf{reg}} \simeq \mathsf{h}^{\mathsf{reg}}/\mathsf{W}$. Hence we only need to show the fibers of $\widetilde{\mathfrak{g}}^{\mathsf{rss}} \to \mathfrak{g}^{\mathsf{rss}}$ is of degree $|\mathsf{W}|$. But this is a particular case of Proposition 5.5.1.

□

Proof of Lemma 5.3.4. Lemma 5.5.11 implies that $\widetilde{\mathfrak{g}}^{\mathsf{rss}} \to \mathfrak{g}^{\mathsf{rss}}$ is a base-change of $\mathsf{h}^{\mathsf{reg}} \to \mathfrak{c}^{\mathsf{reg}} \simeq \mathfrak{c}^{\mathsf{reg}}/W$.

□[Lemma 5.3.4]

Lemma 5.5.12. *The morphism $f : \widetilde{\mathfrak{g}}^{\mathsf{reg}} \to \mathsf{h} \times_{\mathfrak{c}} \mathfrak{g}^{\mathsf{reg}}$ is bijective on $\mathbb{C}$-points.*

Proof. Proposition 5.5.1 implies that the fibers of the source and the target over a $\mathbb{C}$-point $v \in \mathfrak{g}^{\mathsf{reg}}$ have the same size. Hence we only need to show f is surjective. Since f is proper, we only need to show the image of f is dense. By Lemma 5.5.11, this image contains the open subset

$$\mathsf{h} \underset{\mathfrak{c}}{\times} \mathfrak{g}^{\mathsf{rss}} \simeq \mathsf{h}^{\mathsf{reg}} \underset{\mathfrak{c}}{\times} \mathfrak{g}^{\mathsf{reg}},$$

whose complement is of codimension 1 (see the proof of Lemma 5.5.7). This implies the desired claim.

□

Now Theorem 5.4.11 follows from the following general fact.

Proposition 5.5.13. *Let $f : Y \to Z$ be a morphism between finite type $\mathbb{C}$-schemes satisfying the following conditions:*

- *The scheme Z is reduced;*
- *The morphism f is proper and unramified;*
- *The morphism f induces a bijection between $\mathbb{C}$-points.*

Then f is an isomorphism.

Proof. By Zariski's Main Theorem, f is finite (see [EGA4, Remark 8.11.3]) and therefore affine. Hence we only need to show $\mathcal{O}_Z \to f_*\mathcal{O}_Y$ is an isomorphism. In fact, we only need to show it is a surjection because this would imply that f is a dominant closed immersion into a reduced scheme, which must be an isomorphism.

To show that $\mathcal{O}_Z \to f_*\mathcal{O}_Y$ is surjective, we only need to show that $\mathcal{O}_{Z,z} \to \mathcal{O}_{Y,y}$ is surjective for any closed point $y \in Y$ and $z = f(y)$. Since f is unramified, we have $\mathfrak{m}_z\mathcal{O}_{Y,y} = \mathfrak{m}_y$ (see [EGA4, Theorem 17.4.1(d")]). Since f is finite, $\mathcal{O}_{Y,y}$ is a finitely generated $\mathcal{O}_{Z,z}$-module. Now Nakayama's lemma implies $\mathcal{O}_{Z,z} \to \mathcal{O}_{Y,y}$ is surjective as desired.

□

5.6. **Nilpotent orbits I.**

Definition 5.6.1. An *adjoint orbit* $\mathbb{O}$ of $\mathfrak{g}$ is a G-orbit of $\mathcal{N}$ with respect to the adjoint action.

Let $\mathbb{O}$ be an adjoint orbit. We say $\mathbb{O}$ is nilpotent (resp. regular, irregular) if it consists of nilpotent (resp. regular, irregular) elements of $\mathfrak{g}$.

The main goal of this subsection is to prove the following two results.

Proposition 5.6.2. *The dimension of an adjoint orbit of $\mathfrak{g}$ is even.*

Theorem 5.6.3. *There are finitely many nilpotent orbits of $\mathfrak{g}$.*

We first deduce Proposition 5.4.14 from these two results.

Proof of Proposition 5.4.14. Recall that

$$\dim(\mathcal{N}) = \dim(\mathfrak{g}) - \dim(\mathfrak{c}) = \dim(\mathfrak{g}) - \dim(\mathsf{h}) = 2\dim(\mathsf{Fl}_G).$$

Let $\mathbb{O}$ be an irregular nilpotent orbit and $v \in \mathbb{O}$ be an irregular nilpotent element in it. By definition,

$$\dim(\mathbb{O}) = \dim(G) - \dim(G^v) < \dim(G) - \dim(\mathsf{h}).$$

Then Proposition 5.6.2 implies

$$\dim(\mathbb{O}) \le \dim(\mathcal{N}) - 2.$$

On the other hand, Theorem 5.6.3 implies that $\mathcal{N} \smallsetminus \mathcal{N}^{\mathsf{reg}}$ is a union of finitely many irregular nilpotent orbits. It follows that

$$\dim(\mathcal{N} \smallsetminus \mathcal{N}^{\mathsf{reg}}) \le \dim(\mathcal{N}) - 2$$

as desired.

□[Proposition 5.4.14]

We will prove Proposition 5.6.2 by constructing a canonical *symplectic structure* on an adjoint orbit.

Definition 5.6.4. Let X be a smooth scheme. A *symplectic form* on X is a global nondegenerate closed 2-form, i.e., a section $\omega \in \Omega^2(X)$ such that

- $\mathsf{d}\omega = 0$;
- pairing with ω induces an isomorphism $\mathcal{T}_X \xrightarrow{\simeq} \Omega^1_X$.

A *symplectic scheme* (X, ω) is a smooth scheme equipped with a symplectic form.

The following result is well-known.

Theorem 5.6.5. *Each connected component of a symplectic scheme is of even dimension. Moreover,*

$$\wedge^{\dim(X)/2}(\omega) \in \mathcal{K}(X)$$

induces an isomorphism $\mathbb{O}_X \xrightarrow{\simeq} \mathcal{K}_X$.

Construction 5.6.6. Let Y be a smooth scheme. The cotangent bundle T^*Y has a canonical symplectic form constructed as follows.

Let $\pi : \mathsf{T}^*Y \to Y$ be the projection map. We have

$$\mathcal{T}_{\mathsf{T}^*Y} \simeq \pi^*(\mathcal{T}_Y) \oplus \pi^*(\Omega^1_Y),$$

where $\pi^*(\mathcal{T}_Y)$ is the subsheaf of horizontal vector fields, while $\pi^*(\Omega^1_Y)$ is that of vertical vector fields. Since $\mathcal{T}_Y$ and Ω^1_Y are dual to each other, we obtain isomorphisms

$$\mathcal{T}_{\mathsf{T}^*Y} \simeq (\mathcal{T}_{\mathsf{T}^*Y})^\vee \simeq \Omega^1_{\mathsf{T}^*Y}.$$

One can check that this corresponds to an anti-symmetric global section of $(\Omega^1_{\mathsf{T}^*Y})^{\otimes 2}$, and the corresponding 2-form $\omega \in \Omega^2(\mathsf{T}^*Y)$ is symplectic.

Construction 5.6.7. Let K be a connected algebraic group and $\mathfrak{k}$ be its Lie algebra. Each *coadjoint orbit* of $\mathfrak{k}^*$ has a canonical symplectic form constructed as follows.

Let $v \in \mathfrak{k}^*$ be a vector. We identify the coadjoint orbit $\mathbb{O}$ containing v with K/K^v, where $K^v \subseteq K$ is the stablizer subgroup at v. Via this identification, we have

$$\mathcal{T}_{\mathbb{O}} \simeq (\mathfrak{k}/\mathfrak{k}^v)_{K/K^v},$$

where $\mathfrak{k}/\mathfrak{k}^v$ is equipped with the adjoint K^v-action. Consider the antisymmetric bilinear form

$$\mathfrak{k} \otimes \mathfrak{k} \to \mathbb{C}, \;\; x \otimes y \mapsto \langle [x,y], v\rangle.$$

One can check that it induces an antisymmetric *nondegenerate* bilinear form on $\mathfrak{k}/\mathfrak{k}^v$ that is compatible with the adjoint K^v-action. This induces an antisymmetric perfect pairing

$$\mathcal{T}_{\mathbb{O}} \underset{\mathcal{O}_{\mathbb{O}}}{\otimes} \mathcal{T}_{\mathbb{O}} \to \mathcal{O}_{\mathbb{O}}.$$

By duality, we obtain a nondegenerate 2-form $\omega \in \Omega^2(\mathbb{O})$. We call it the *Kirillov–Kostant–Souriau* form on $\mathbb{O}$.

Exercise 5.6.8 (Problem 1, Homework 9)**.** In above, $(\mathbb{O}, \omega)$ is a symplectic scheme, i.e., $\mathsf{d}\omega = 0$.

Corollary 5.6.9. *The dimension of a coadjoint orbit of an algebraic Lie algebra is even.*

Proof of Proposition 5.6.2. Follows from Corollary 5.6.9 and the G-linear isomorphism $\mathfrak{g} \simeq \mathfrak{g}^*$.

□[Proposition 5.6.2]

Definition 5.6.10. Let (X, ω) be a symplectic scheme. Let $Z \subseteq X$ be a (*not necessarily smooth*) reduced subscheme. For a smooth closed point $z \in Z$, we view $\mathcal{T}_Z|_z$ as a subspace of $\mathcal{T}_X|_z$. Write

$$\mathcal{T}_Z|_z^{\perp,\omega} \subseteq \mathcal{T}_X|_z$$

for the subspace orthogonal to $\mathcal{T}_Z|_z$ with respect to the perfect paring $\omega(-,-)|_z$.

- We say Z is *isotropic* if $\mathcal{T}_Z|_z \subseteq \mathcal{T}_Z|_z^{\perp,\omega}$ for any smooth closed point $z \in Z$.
- We say Z is *coisotropic* if $\mathcal{T}_Z|_z \supseteq \mathcal{T}_Z|_z^{\perp,\omega}$ for any smooth closed point $z \in Z$.
- We say Z is *Langrangian* if $\mathcal{T}_Z|_z = \mathcal{T}_Z|_z^{\perp,\omega}$ for any smooth closed point $z \in Z$.

We have the following easy facts.

Lemma 5.6.11. *Let (X, ω) be a symplectic scheme and $Z \subseteq X$ be a reduced subscheme.*

(1) *If Z is isotropic, then* $\dim(Z) \le \dim(X)/2$*;*
(2) *If Z is coisotropic, then* $\dim(Z) \ge \dim(X)/2$*;*
(3) *If Z is Langrangian, then* $\dim(Z) = \dim(X)/2$.

Example 5.6.12. Let Y be a smooth scheme and $Z \subseteq Y$ be a smooth subscheme. Then the conormal bundle $\mathsf{T}^*_Z Y$ is a Langrangian subscheme of $\mathsf{T}^* Y$.

Proposition 5.6.13. *Let $\mathfrak{b} \subseteq \mathfrak{g}$ be a Borel subalgebra of $\mathfrak{g}$, and $\mathbb{O} \subseteq \mathfrak{g}$ be a nilpotent orbit. Then $(\mathbb{O} \cap \mathfrak{n})_{\mathsf{red}}$ is a Langrangian subscheme of $\mathbb{O}$.*

Remark 5.6.14. The scheme $\mathbb{O} \cap \mathfrak{n}$ does not depend on the choice of $\mathfrak{b}$ because Borel subalgebras are conjugate to each other.

Proposition 5.6.13 and Theorem 5.6.3 will be deduced from the following two results.

Proposition 5.6.15. *The subscheme $(\mathbb{O} \cap \mathfrak{n})_{\mathsf{red}} \subseteq \mathbb{O}$ is coisotropic.*

Lemma 5.6.16. *The* Steinberg scheme

$$\mathsf{St} := \widetilde{\mathcal{N}} \underset{\mathcal{N}}{\times} \widetilde{\mathcal{N}}.$$

is of equidimension $\dim(\mathfrak{g}) - \dim(\mathfrak{h})$.

Lemma 5.6.16 will be proved at the end of this subection, while Proposition 5.6.15 will be proved in §5.7.

Proof of Proposition 5.6.13 and Theorem 5.6.3. Write

$$\widetilde{\mathbb{O}} := \mathbb{O} \underset{\mathcal{N}}{\times} \widetilde{\mathcal{N}}.$$

By definition, a closed point (v, x) of $\mathfrak{g} \times \mathsf{Fl}_G$ belongs to $\widetilde{\mathbb{O}}$ iff $v \in \mathfrak{n}_x \cap \mathbb{O}$. By Proposition 5.6.15, all the fibers of the morphism $\widetilde{\mathbb{O}} \to \mathsf{Fl}_G$ have dimension $\geq \dim(\mathbb{O})/2$. This implies that

$$\dim(\widetilde{\mathbb{O}}) \geq \dim(\mathsf{Fl}_G) + \dim(\mathbb{O})/2.$$

This implies that

$$\dim(\widetilde{\mathbb{O}} \underset{\mathbb{O}}{\times} \widetilde{\mathbb{O}}) \geq 2\dim(\widetilde{\mathbb{O}}) - \dim(\mathbb{O}) \geq 2\dim(\mathsf{Fl}_G) = \dim(\mathfrak{g}) - \dim(\mathsf{h}) = \dim(\mathsf{St}),$$

where the last equality is due to Lemma 5.6.16. Since $\widetilde{\mathbb{O}} \times_{\mathbb{O}} \widetilde{\mathbb{O}}$ is a subscheme of St, all the inequalities in above must be equalities. This implies the following results:

(i) The coisotropic subscheme $(\mathbb{O} \cap \mathfrak{n})_{\mathsf{red}} \subseteq \mathbb{O}$ is of dimension $\dim(\mathbb{O})/2$.
(ii) The subscheme $\widetilde{\mathbb{O}} \times_{\mathbb{O}} \widetilde{\mathbb{O}}$ contains the generic point of at least one irreducible component of St.

Now (i) implies that $(\mathbb{O} \cap \mathfrak{n})_{\mathsf{red}} \subseteq \mathbb{O}$ is Langrangian, while (ii) implies that the number of $\mathbb{O}$'s is not greater than the number of irreducible components of St.

□[Proposition 5.6.13 and Theorem 5.6.3]

To prove Lemma 5.6.16, we study the fibers of the canonical morphism

$$\mathsf{St} = \widetilde{\mathcal{N}} \underset{\mathcal{N}}{\times} \widetilde{\mathcal{N}} \to \mathsf{Fl}_G \times \mathsf{Fl}_G.$$

We first define a stratification on $\mathsf{Fl}_G \times \mathsf{Fl}_G$ such that the fibers over each stratum have the same dimension.

Write

$$|(\mathsf{Fl}_G \times \mathsf{Fl}_G)/G|$$

for the set of orbits for the diagonal G-action on $\mathsf{Fl}_G \times \mathsf{Fl}_G$.

Exercise 5.6.17 (Problem 2, Homework 9)**.** Let (B, H) be a Borel pair and $W := N_G(H)/H$ be the corresponding Weyl group.

(1) Use the Bruhat decomposition to construct a bijection

$$W \xrightarrow{\simeq} |(\mathsf{Fl}_G \times \mathsf{Fl}_G)/G|$$

- Let

$$(\mathsf{Fl}_G \times \mathsf{Fl}_G)^{=w} \subseteq \mathsf{Fl}_G \times \mathsf{Fl}_G$$

be the G-orbit corresponding to an element $w \in W$.

(2) Show that

$$\dim((\mathsf{Fl}_G \times \mathsf{Fl}_G)^{=w}) = \dim(\mathsf{Fl}_G) + \ell(w).$$

(3) Show that

$$(\mathsf{Fl}_G \times \mathsf{Fl}_G)^{=w'} \subseteq \overline{(\mathsf{Fl}_G \times \mathsf{Fl}_G)^{=w}}$$

iff $w' \leq w$.

We include the following exercise for aesthetic reasons.

Exercise 5.6.18. Consider the abstract Weyl group W. Show that the composition

$$\mathsf{W} \xrightarrow{\mathfrak{r}_{(B,H)}} W \xrightarrow{\simeq} |(\mathsf{Fl}_G \times \mathsf{Fl}_G)/G|$$

does not depend on the choice of the Borel pair (B,H). Deduce that the length function $\ell(-)$ and the Bruhat ordering $\leq$ are well-defined on W.

Exercise 5.6.19 (Problem 3, Homework 9)**.** Let (B,H) be a Borel pair and $w \in W := N_G(H)/H$ be an element in the corresponding Weyl group. Note that $B^w := \mathsf{Ad}_w(B)$ is well-defined.

(1) Construct an isomorphism

$$(\mathsf{Fl}_G \times \mathsf{Fl}_G)^{=w} \simeq G/(B \cap B^w).$$

(2) Write

$$\mathsf{St}^{=w} := \mathsf{St} \underset{\mathsf{Fl}_G \times \mathsf{Fl}_G}{\times} (\mathsf{Fl}_G \times \mathsf{Fl}_G)^{=w}.$$

Construct an isomorphism

$$\mathsf{St}^{=w} \simeq (G \times (\mathfrak{n} \underset{\mathcal{N}}{\times} \mathfrak{n}^w))/(B \cap B^w).$$

Proof of Lemma 5.6.16. By Exercise 5.6.17, we have a stratification of St labelled by the partially ordered set $(W, \leq)$. By Exercise 5.6.19, each reduced stratum $(\mathsf{St}^{=w})_{\mathsf{red}}$ is a vector bundle over $G/(B \cap B^w)$ with dimension equal to

$$\dim(G) + \dim(\mathfrak{n} \cap \mathfrak{n}^w) - \dim(B \cap B^w) = \dim(\mathfrak{g}) - \dim(\mathfrak{h}).$$

This implies that irreducible components of St are given by $(\overline{\mathsf{St}^{=w}})_{\mathsf{red}}$, and each of them is of dimension $\dim(\mathfrak{g}) - \dim(\mathfrak{h})$.

$\square$[Lemma 5.6.16]

Exercise 5.6.20 (Problem 4, Homework 9)**.** For $\mathfrak{g} = \mathfrak{sl}_n$, show that the following sets can be canonically identified:

- The set of nilpotent orbits in $\mathfrak{g}$;
- The set of the adjoint orbits of the abstract Weyl group W.

Exercise 5.6.21 (Problem 5, Homework 9)**.** Let $\mathfrak{b} \subseteq \mathfrak{g}$ be a Borel subalgebra and $\mathfrak{n} := [\mathfrak{b}, \mathfrak{b}]$ be its nilpotent radical.

(1) Show that $\mathcal{N}^{\mathsf{reg}} \cap \mathfrak{n}$ is nonempty.

(2) Let $v \in \mathfrak{n}$ be a regular element. Show that the orbit Nv is dense in the fiber of $\mathfrak{n} \to \mathfrak{n}/[\mathfrak{n}, \mathfrak{n}]$.
(3) Let X be an affine scheme equipped with an N-action. Show that any N-orbit is closed in X. Deduce that the orbit Nv is equal to the fiber $\mathfrak{n} \to \mathfrak{n}/[\mathfrak{n}, \mathfrak{n}]$.
(4) Deduce that the converse of Lemma 5.4.6 is also true.

5.7. **Nilpotent orbit II.** In this subsection we prove Proposition 5.6.15. We need some general notions in symplectic geometry.

Definition 5.7.1. Let X be a smooth scheme. A *Poisson bracket* on X is a $\mathbb{C}$-linear morphism

$$\{-,-\} : \mathcal{O}_X \underset{\mathbb{C}}{\otimes} \mathcal{O}_X \to \mathcal{O}_X$$

such that:

- The binary operator $\{-,-\}$ is a Lie bracket on $\mathcal{O}_X$, i.e., it is antisymmetric and satisfies the Jacobi identity.
- The binary operator $\{-,-\}$ is a biderivation, i.e., it satisfies the Lebniz rule for both factors.

A *Poisson scheme* $(X, \{-,-\})$ is a smooth scheme equipped with a Poisson bracket.

Remark 5.7.2. Knowing a biderivation

$$\{-,-\} : \mathcal{O}_X \underset{\mathbb{C}}{\otimes} \mathcal{O}_X \to \mathcal{O}_X$$

is equivalent to knowing a derivation

$$\mathcal{O}_X \to \mathcal{D}er(\mathcal{O}_X, \mathcal{O}_X) \simeq \mathcal{T}_X,$$

which is equivalent to knowing an $\mathcal{O}_X$-linear morphism

$$\Omega^1_X \to \mathcal{T}_X.$$

Exercise 5.7.3. Let $(X, \{-,-\})$ be a Poisson scheme and $\Omega^1_X \to \mathcal{T}_X$ be the $\mathcal{O}_X$-linear morphism that corresponds to the Poisson bracket. Show that the composition

$$\mathcal{O}_X \xrightarrow{\mathsf{d}} \Omega^1_X \to \mathcal{T}_X$$

is compatible with the Poisson bracket on $\mathcal{O}_X$ and the usual Lie bracket on $\mathcal{T}_X$.

Construction 5.7.4. A symplectic scheme (X, ω) has a canonical Poisson structure constructed as follows.

By definition, pairing with ω gives an $\mathcal{O}_X$-linear isomorphism $\mathcal{T}_X \to \Omega^1_X$. Consider the inverse $\mathcal{O}_X$-linear morphism

$$\Omega^1_X \to \mathcal{T}_X = \mathcal{D}er(\mathcal{O}_X, \mathcal{O}_X).$$

By Remark 5.7.2, we obtain a biderivation

$$\{-,-\} : \mathcal{O}_X \otimes \mathcal{O}_X \to \mathcal{O}_X.$$

One can check that ω belonging to $\wedge^2(\Omega^1_X)$ implies that $\{-,-\}$ is antisymmetric, while $\mathsf{d}\omega = 0$ implies that $\{-,-\}$ satisfies the Jacobi identity. In other words, $\{-,-\}$ is indeed a Poisson bracket.

Exercise 5.7.5 (Problem 6, Homework 9)**.** Let $\mathfrak{k}$ be a finite dimensional Lie algebra. Show that the scheme $\mathfrak{k}^*$ has a unique Poisson bracket

$$\{-,-\} : \mathcal{O}(\mathfrak{k}^*) \otimes \mathcal{O}(\mathfrak{k}^*) \to \mathcal{O}(\mathfrak{k}^*)$$

such that for $\phi, \varphi \in \mathfrak{k} \subseteq \mathcal{O}(\mathfrak{k}^*)$ and $v \in \mathfrak{k}^*$

$$\{\phi, \varphi\}(v) = \langle [\mathsf{d}\phi|_v, \mathsf{d}\varphi|_v], v\rangle,$$

where $\mathsf{d}\phi|_v$ and $\mathsf{d}\varphi|_v$ are viewed as vectors in $\mathfrak{k}$ via the canonical isomorphism $\Omega^1_{\mathfrak{k}^*}|_v \simeq \mathfrak{k}$.

Exercise 5.7.6 (Problem 7, Homework 9)**.** Let $\mathfrak{k}$ be a finite dimensional Lie algebra and $\mathbb{O} \subseteq \mathfrak{k}$ be a coadjoint orbit. Show that $\mathbb{O} \to \mathfrak{k}$ is a morphism between Poisson schemes.

Definition 5.7.7. Let $(X, \{-,-\})$ be a Poisson scheme and $Z \subseteq X$ be a reduced subscheme. We say Z is *coisotropic* if for any smooth closed point $z \in Z$, the morphism $\Omega^1_X|_z \to \mathcal{T}_X|_z$ sends the subspace

$$\mathcal{T}_Z|_z^{\perp} \subseteq (\mathcal{T}_X|_z)^* \simeq \Omega^1_X|_z$$

into the subspace

$$\mathcal{T}_Z|_z \subseteq \mathcal{T}_X|_z.$$

The following result can be proved by unwinding the definitions.

Lemma 5.7.8. *Let (X, ω) be a symplectic scheme and $Z \subseteq X$ be a reduced subscheme. Then Z is coisotropic with respect to the symplectic structure of X iff Z is coisotropic with respect to the Poisson structure of X.*

Lemma 5.7.9. *Let $(X, \{-,-\})$ be a Poisson scheme and $Z \subseteq X$ be a reduced closed subscheme. Let $\mathcal{I} \subseteq \mathcal{O}_X$ be the corresponding ideal sheaf. Then Z is coisotropic iff $\{\mathcal{I}, \mathcal{I}\} \subseteq \mathcal{I}$.*

Proof. We can replace X by a dense open subscheme and therefore assyme Z is smooth. Then Z is coisotropic iff

$$\Omega^1_X|_Z \to \mathcal{T}_X|_Z \simeq \mathcal{D}er(\mathcal{O}_X, \mathcal{O}_X/\mathcal{I})$$

sends

$$\mathcal{I}/\mathcal{I}^2 \simeq \mathcal{T}_Z^{\perp} \subseteq \Omega^1_X|_Z$$

into

$$\mathsf{Der}(\mathcal{O}_X/\mathcal{I}, \mathcal{O}_X/\mathcal{I}) \subseteq \mathcal{D}er(\mathcal{O}_X, \mathcal{O}_X/\mathcal{I})$$

Unwinding the definitions, for a local section ϕ of $\mathcal{I}$, the composition

$$\mathcal{I}/\mathcal{I}^2 \to \Omega^1_X|_Z \to \mathcal{D}er(\mathcal{O}_X, \mathcal{O}_X/\mathcal{I})$$

sends $\overline{\phi}$ to the derivation $\{\phi, -\}$. Hence Z is coisotropic iff for any such ϕ,

$$\{\phi, \mathcal{I}\} \subseteq \mathcal{I}.$$

□

Remark 5.7.10. Lemma 5.7.9 would imply Proposition 5.6.15 if $\mathbb{O} \cap \mathfrak{n}$ was reduced. Namely, we have

$$\mathbb{O} \cap \mathfrak{n} \simeq \mathbb{O} \underset{\mathfrak{n}^*}{\times} 0,$$

where the morphism $\mathbb{O} \to \mathfrak{n}^*$ is the composition

$$\mathbb{O} \to \mathfrak{g} \simeq \mathfrak{g}^* \to \mathfrak{n}^*.$$

Note that this composition is a morphism between Poisson schemes. It is obvious that $0 \subseteq \mathfrak{n}^*$ is coisotropic, therefore its ideal sheaf is closed under the Poisson bracket for $\mathfrak{n}^*$. This implies that the ideal sheaf $\mathcal{I}$ for $\mathbb{O} \cap \mathfrak{n}$ is closed under the Poisson bracket for $\mathbb{O}$. However, to show that $(\mathbb{O} \cap \mathfrak{n})_{\mathsf{red}}$ is coisotropic, we need to show $\sqrt{\mathcal{I}}$ is closed under the Poisson bracket, which does *not* follow from the corresponding claim for $\mathcal{I}$.

To prove Proposition 5.6.15, we need more definitions.

Definition 5.7.11. Let (X, ω) be a symplectic scheme. Consider the composition

$$\mathcal{O}(X) \xrightarrow{\mathsf{d}} \Omega^1(X) \simeq \mathcal{T}(X),$$

where the last isomorphism is provided by the symplectic structure on X. Let $f \in \mathcal{O}(X)$ be a global function on X. We call its image $\xi_f \in \mathcal{T}(X)$ the *Hamiltonian vector field* for f.

Definition 5.7.12. Let (X, ω) be a symplectic scheme and K be an algebraic group.

A *symplectic K-action* on (X, ω) is a K-action on the scheme X such that the 2-form ω is K-invariant.

A *Hamiltonian K-action* on (X, ω) is a symplectic K-action equipped with a Lie algebra homomorphism $H : \mathfrak{k} \to \mathcal{O}(X)$ such that

$$\xi_{H(v)} = a(v),$$

where $\mathcal{O}(X)$ is viewed as a Lie algebra via the Poisson bracket induced by ω, and

$$a : \mathfrak{k} \to \mathcal{T}(X)$$

is the differentiation of the K-action on X.

Construction 5.7.13. Let (X, ω) be a symplectic scheme equipped with a Hamiltonian K-action. The map $H : \mathfrak{k} \to \mathcal{O}(X)$ induces a ring homomorphism

$$\mathsf{Sym}(\mathfrak{k}) \to \mathcal{O}(X),$$

which corresponds to a morphism

$$\mu : X \to \mathsf{Spec}(\mathsf{Sym}(\mathfrak{k})) \simeq \mathfrak{k}^*.$$

We call $\mu : X \to \mathfrak{k}^*$ the *momentum map* for the Hamiltonian K-action on (X, ω).

The following results can be proved by unwinding the definitions.

Lemma 5.7.14. *Let (X, ω) be a symplectic scheme equipped with a Hamiltonian K-action. Consider the momentum map $\mu : X \to \mathfrak{k}^*$.*

(1) *The morphism μ is compatible with the Poisson structures on X and $\mathfrak{k}^*$.*
(2) *The morphism μ is compatible with the K-actions on X and $\mathfrak{k}^*$.*

Example 5.7.15. Let Y be a smooth scheme equipped with a K-action. By functoriality, the cotangent bundle T^*Y has a natural symplectic K-action. Consider the map

$$\mathcal{T}(Y) \to \mathcal{O}(\mathsf{T}^*Y)$$

given by pairing a tangent field with a cotangent vector. One can check that the composition

$$H : \mathfrak{k} \xrightarrow{a} \mathcal{T}(Y) \to \mathcal{O}(\mathsf{T}^*Y)$$

defines a Hamiltonian K-action on T^*Y. Moreover, the moment map

$$\mu : \mathsf{T}^*Y \to \mathfrak{k}^*$$

is the unique morphism such that for any closed point $y \in Y$, its restriction $\mu_y : \mathsf{T}^*_y Y \to \mathfrak{k}^*$ is the linear morphism that is dual to the morphism

$$a_y : \mathfrak{k} \to \mathsf{T}_y Y.$$

In particular, we have recovered Construction 4.4.5.

Example 5.7.16. Let $\mathbb{O} \subseteq \mathfrak{k}^*$ be a coadjoint orbit. It is easy to see the K-action on $\mathbb{O}$ fixes the KKS symplectic form. One can check that

$$H : \mathfrak{k} \xrightarrow{\subseteq} \mathcal{O}(\mathfrak{k}^*) \to \mathcal{O}(\mathbb{O})$$

defines a Hamiltonian K-action on $\mathbb{O}$. Moreover, the momentum map

$$\mu : \mathbb{O} \to \mathfrak{k}^*$$

is just the obvious embedding.

Theorem 5.7.17 ([CG, Theorem 1.5.7])**.** *Let A be a solvable group and $\mathfrak{a}$ be its Lie algebra. Let (X, ω) be a symplectic scheme equipped with a Hamiltonian A-action. Let $\mu : X \to \mathfrak{a}^*$ be the momentum map. Then for any coadjoint orbit $\mathbb{O} \subseteq \mathfrak{a}^*$, the subscheme*

$$(X \underset{\mathfrak{a}^*}{\times} \mathbb{O})_{\mathsf{red}} \to X$$

is coisotropic.

Warning 5.7.18. The assumption on solvability is necessary. The standard SL_2-action on $(\mathbb{A}^2, \mathsf{d}x \wedge \mathsf{d}y)$ is a counterexample. Namely, the reduced preimage of the zero orbit $0 \subseteq \mathfrak{sl}_2^*$ along the momentum map $\mathbb{A}^2 \to \mathfrak{sl}_2^*$ is equal to the origin of $\mathbb{A}^2$, which is not coisotropic.

Proof of Proposition 5.6.15. Follows by applying Theorem 5.7.17 to the Hamiltonian B-action on the nilpotent orbit $\mathbb{O} \subseteq \mathfrak{g}$ and the zero coadjoint orbit $0 \subseteq \mathfrak{b}^*$.
□[Proposition 5.6.15]

Sketch of Theorem 5.7.17. We prove by induction on $\dim(A)$. The claim is obvious when $\dim(A) = 0$. When $\dim(A) > 0$, choose a normal subgroup $A_0 \subseteq A$ such that A/A_0 is 1-dimensional. Note that A/A_0 is abelian. Write $\pi : \mathfrak{a}^* \to \mathfrak{a}_0^*$ for the canonical morphism. Consider the composition

$$\mu_0 : X \xrightarrow{\mu} \mathfrak{a}^* \xrightarrow{\pi} \mathfrak{a}_0^*,$$

which is just the momentum map for the Hamiltonian A_0-action on (X, ω).

There are two two cases:

(i) If $\dim(\pi(\mathbb{O})) = \dim(\mathbb{O}) - 1$, then $\mathbb{O}$ is open in $\pi^{-1}\pi(\mathbb{O})$.
(ii) If $\dim(\pi(\mathbb{O})) = \dim(\mathbb{O})$, then $\pi(\mathbb{O})$ is a coadjoint orbit, and $\mathcal{O}$ is of codimensional 1 in $\pi^{-1}\pi(\mathbb{O})$.

For case (i), the induction hypothesis says that for any coadjoint orbit $\mathbb{O}_0 \subseteq \pi(\mathbb{O}) \subseteq \mathfrak{a}_0^*$,

$$(X \underset{\mathfrak{a}_0^*}{\times} \mathbb{O}_0)_{\mathsf{red}}$$

is coisotropic. Since $\pi(\mathbb{O})$ is a union of such orbits,

$$(X \underset{\mathfrak{a}^*}{\times} \pi^{-1}\pi(\mathbb{O}))_{\mathsf{red}} = (X \underset{\mathfrak{a}_0^*}{\times} \pi(\mathbb{O}))_{\mathsf{red}}$$

This implies the open subscheme

$$(X \underset{\mathfrak{a}^*}{\times} \mathbb{O})_{\mathsf{red}}$$

is coisotropic as desired.

For case (ii), write

$$Y := (X \underset{\mathfrak{a}^*}{\times} \pi^{-1}\pi(\mathbb{O}))_{\mathsf{red}}, \;\; Z := (X \underset{\mathfrak{a}^*}{\times} \mathbb{O})_{\mathsf{red}}.$$

Then the codimension of Z in Y is at most 1. Note that the induction hypothesis implies that Y is coisotropic. Let $z \in Z$ be any smooth closed point. Let $Z' \subseteq Z$ be the irreducible component containing z. We only need to show that $\mathcal{T}_{Z'}|_z$ is a coisotropic subspace of $\mathcal{T}_X|_z$.

Let $Y' \subseteq Y$ be an irreducible component containing Z'. Note that Y' is coisotropic because Y is so. If Z' is dense in Y', the proof is finished. Otherwise Z' is of codimensional 1 in Y. Choose a function F defined on neighborhood of $\mu(z)$ in $\mathfrak{a}^*$ such that the 1-codimensional subscheme $\mathbb{O} \subseteq \pi^{-1}\pi(\mathbb{O})$ is locally given by the equation $F = 0$. By restriction, we obtain a function f defined on a neighborhood of z such that locally Z' is the *reduced* subscheme of the subscheme of Y' given by the equation $f = 0$.

For any complex number $c \in \mathbb{A}^1(\mathbb{C})$, let $\Sigma_c \subseteq Y'$ be the subscheme given by the equation $f = c$. It is easy to see that for almost all complex numbers c, Σ_c is generically reduced (i.e., satisfies (R0)). We claim for such a complex number c, Σ_c is coisotropic.

Assuming the claim, one can use tangent spaces of Σ_c (at smooth points) to approximate the tangent spaces of $Z' = (\Sigma_0)_{\mathsf{red}}$ at closed points in a Zariski dense open subset of it, and deduce that the latter tangent spaces are also coisotropic[14].

It remains to prove the claim. Note that locally Σ_c is given by the ideal sheaf $\mathcal{I} + (f - c)$, where $\mathcal{I}$ is the ideal sheaf for Y'. We only need to show that $\mathcal{I} + (f - c)$ is closed under the Poisson bracket for X. Since Y' is coisotropic, $\mathcal{I}$ is closed under the Poisson bracket. Also note that $\{f - c, f - c\} = 0$ because the Poisson bracket is antisymmetric. Hence we only need to show that $\{f - c, -\}$ preserves the ideal

[14] This is the content of [CG, Lemma 1.5.12]. The authors of *loc.cit.* stated and proved this result using the complex topology, but it is not hard to translate it to the following result, which can be proved using similar method (i.e. normalization).

Lemma: Let N be an integral subscheme in a smooth scheme M and $f : N \to \mathbb{A}^1$ be a flat morphism. Consider the relative tangent scheme

$$\mathsf{T}(N/\mathbb{A}^1) := \mathsf{Spec}_N(\mathsf{Sym}_{\mathcal{O}_N}\Omega^1_{N/\mathbb{A}^1}),$$

viewed as a closed subscheme of $\mathsf{T}M \times_M N$. Consider the open subscheme

$$U := \mathsf{T}(N/\mathbb{A}^1) \underset{\mathbb{A}^1}{\times} (\mathbb{A}^1 \smallsetminus 0) \subseteq \mathsf{T}(N/\mathbb{A}^1)$$

and its scheme theoretic closure $\overline{U}$ in $\mathsf{T}M \times_M N$. Then there exists a Zariski dense open subset $V \subseteq (N \times_{\mathbb{A}^1} 0)_{\mathsf{red}}$ such that

$$\overline{U} \underset{N}{\times} V = \mathsf{T}V,$$

where we view both sides as subschemes of $\mathsf{T}M \times_M V$.

sheaf $\mathcal{I}$. Unwinding the definitions, this is equivalent to the condition that the Hamiltonian vector field ξ_{f-c} is tangent to the subscheme Y'. Recall that the local function $f-c$ on X is the restriction of the local function $F-c$ on $\mathfrak{a}^*$. Moreover, for any closed point $x \in X$, the tangent vector $\xi_{f-c}|_x$ is the image of

$$\mathsf{d}(F-c)|_{\mu(x)} \in \mathfrak{a}$$

under the map $a_x : \mathfrak{a} \to \mathcal{T}_X|_x$. Hence we only need to show that $Y' \subseteq X$ is stablized by the A-action. But this follows obviously from the definition of Y'.

□[Theorem 5.7.17]

6. The category of D-modules

Let X be a smooth scheme. Recall that we have the abelian categories

$$\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}} \text{ and } \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}$$

of left and right D-modules on X. Let

$$\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{c}} \subseteq \mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$$

be the full subcategory of *coherent* $\mathcal{D}_X$-modules. Let

$$\mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}) \text{ and } \mathsf{D}^b(\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{c}})$$

be the unbounded derived category (resp. bounded derived category) of $\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$ of $\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{c}}$.

6.1. First properties.

Let

$$\mathsf{oblv}^{l/r} : \mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}} \to \mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc}}$$

be the forgetful functors given by restricting along the homomorphism $\mathcal{O}_X \to \mathcal{D}_X$. The following result is obvious.

Lemma 6.1.1. *The functor* $\mathsf{oblv}^{l/r}$ *admits a left adjoint*

$$\mathsf{Ind}^{l/r} : \mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc}} \to \mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$$

given by

$$\mathsf{Ind}^l(\mathcal{F}) \coloneqq \mathcal{D}_X \underset{\mathcal{O}_X}{\otimes} \mathcal{F},\ \ \mathsf{Ind}^r(\mathcal{F}) \coloneqq \mathcal{F} \underset{\mathcal{O}_X}{\otimes} \mathcal{D}_X.$$

Moreover, both $\mathsf{oblv}^{l/r}$ *and* $\mathsf{Ind}^{l/r}$ *are exact.*

Lemma 6.1.2. *The category* $\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$ *has enough injective objects. In fact, any object in this category can be embedded into an injective object that is also a flabby sheaf on* X*.*

Proof. See [HTT, Proposition 1.4.14]. □

Lemma 6.1.3. *The category* $\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$ *has enough locally projective objects.*

Proof. Follows from Lemma 6.1.1 and the well-known fact that $\mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc}}$ has enough locally projective objects. □

Lemma 6.1.4. *Suppose that* X *is quasi-projective. Then any D-module on* X *admits a resolution by locally projective objects such that the length of this resolution is not greater than* $2\dim(X)$*.*

Proof. See [HTT, Corollary 1.4.20]. □

Lemma 6.1.5. *Suppose that* X *is quasi-compact. Then the full subcategory*

$$\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{c}} \subseteq \mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$$

contains exactly the compact objects in $\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$*. Similarly, the obvious functor*

$$\mathsf{D}^b(\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{c}}) \to \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}})$$

is fully faithful and its image contains exactly the compact objects in the target.

Proof. It is easy to reduce to the case when X is affine. The claim for abelian categories follows from the fact that $\mathcal{D}(X)$ is Noetherian. The claim for derived categories follows from Lemma 6.1.4.

□

6.2. **Tensor and Hom.**

Construction 6.2.1. Let $\mathcal{M},\mathcal{N} \in \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ and $\mathcal{M}',\mathcal{N}' \in \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}$. We can define the following objects using the *signed Lebniz rules*:

- $\mathcal{M} \otimes_{\mathcal{O}_X} \mathcal{N} \in \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ defined by $\partial \cdot (m \otimes n) = (\partial \cdot m) \otimes n + m \otimes (\partial \cdot n)$;
- $\mathcal{M}' \otimes_{\mathcal{O}_X} \mathcal{N} \in \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}$ defined by $(m' \otimes n) \cdot \partial = (m' \cdot \partial) \otimes n - m' \otimes (\partial \cdot n)$;
- $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{M},\mathcal{N}) \in \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ defined by $(\partial \cdot \phi)(m) = \partial \cdot \phi(m) - \phi(\partial \cdot m)$;
- $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{M}',\mathcal{N}') \in \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ defined by $(\partial \cdot \phi)(m') = -\phi(m') \cdot \partial + \phi(m \cdot \partial)$;
- $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{M},\mathcal{N}') \in \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}$ defined by $(\phi \cdot \partial)(m) = \phi(m) \cdot \partial + \phi(m \cdot \partial)$.

Remark 6.2.2. One way to memorize the above rules for left versus right is testing with the objects $\mathcal{O}_X \in \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ and $\mathcal{K}_X \in \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}$. For example, $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{K}_X,\mathcal{K}_X) \simeq \mathcal{O}_X$ has a left $\mathcal{D}$-module structure, while $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{K}_X,\mathcal{O}_X) \simeq \mathcal{K}_X^{-1}$ in general has no $\mathcal{D}$-module structures.

Remark 6.2.3. One way to memorize the signed Lebniz rules: (i) put a minus sign when acting on the source object in $\mathcal{H}om_{\mathcal{O}_X}(-,-)$; (ii) put another minus sign when moving ∂ from the one side of $\cdot$ to the other side.

Remark 6.2.4. One can upgrade $\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ to a symmetric monoidal category such that the forgetful functor oblv^l is naturally symmetric monoidal. Using left derived functors, we obtain a symmetric monoidal structure on $\mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}})$.

The following result is obvious.

Lemma 6.2.5. *Let $\mathcal{M} \in \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ and $\mathcal{M}' \in \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}$. We have adjoint functors*

$$\mathcal{M} \underset{\mathcal{O}_X}{\otimes} - : \mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}} \rightleftarrows \mathcal{D}_X\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}} : \mathcal{H}om_{\mathcal{O}_X}(\mathcal{M},-)$$
$$\mathcal{M}' \underset{\mathcal{O}_X}{\otimes} - : \mathcal{D}_X\text{-}\mathsf{mod}^{l}_{\mathsf{qc}} \rightleftarrows \mathcal{D}_X\text{-}\mathsf{mod}^{r}_{\mathsf{qc}} : \mathcal{H}om_{\mathcal{O}_X}(\mathcal{M}',-)$$

compatible with the similar adjunctions for $\mathcal{O}_X$-modules.

Construction 6.2.6. Let X_1 and X_2 be smooth schemes and $\mathcal{M}_i \in \mathcal{D}_{X_i}\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$. There are natural objects

$$\mathcal{M}_1 \boxtimes \mathcal{M}_2 \coloneqq \mathcal{M}_1 \underset{\mathbb{C}}{\otimes} \mathcal{M}_2 \in \mathcal{D}_{X_1 \times X_2}\text{-}\mathsf{mod}^{l/r}_{\mathsf{qc}}$$

induced by the obvious isomorphism $\mathcal{D}_{X_1} \otimes_{\mathbb{C}} \mathcal{D}_{X_2} \simeq \mathcal{D}_{X_1 \times X_2}$. This is called the *external tensor product* of $\mathcal{M}_1$ and $\mathcal{M}_2$.

Note that the functor $- \boxtimes -$ is exact in both factors, and we can define a similar functor for the derived categories.

6.3. **left versus right.** Since $\mathcal{K}_X$ is invertible, we obtain the following result.

Corollary 6.3.1. *The following functors are inverse to each other:*

$$\mathcal{K}_X \underset{\mathcal{O}_X}{\otimes} - : \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}} \rightleftarrows \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}} : \mathcal{H}om_{\mathcal{O}_X}(\mathcal{K}_X, -).$$

When identifying the *derived* categories of left and right $\mathcal{D}$-modules, it is more convenient to use the *dualizing complex*

$$\omega_X := \pi^!(\mathbb{C}) \simeq \mathcal{K}_X[d_X].$$

In other words, we use the equivalences[15]

$$\omega_X \underset{\mathcal{O}_X}{\otimes} - : \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}) \rightleftarrows \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}) : \mathcal{H}om_{\mathcal{O}_X}(\omega_X, -).$$

to identify these derived categories. Note that these functors are t-exact up to a shift.

Construction 6.3.2. As a consequence, the symmetric monoidal structure on $\mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}})$ induces a symmetric monoidal structure on $\mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}})$. We denote the multiplication functor by

$$(\mathcal{M}', \mathcal{N}') \mapsto \mathcal{M}' \overset{!}{\otimes} \mathcal{N}', \ \ \mathcal{M}', \mathcal{N}' \in \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}).$$

Note that the underlying $\mathcal{O}_X$-module of $\mathcal{M}' \otimes^! \mathcal{N}'$ is given by

$$\mathcal{M}' \overset{!}{\otimes} \mathcal{N}' \simeq \mathcal{M}' \underset{\mathcal{O}_X}{\otimes} \mathcal{N}' \underset{\mathcal{O}_X}{\otimes} \mathcal{K}_X^{-1}[-\dim(X)].$$

6.4. **Pullback.** Let $\phi : Y \to X$ be a map between smooth schemes.

Construction 6.4.1. We construct a commutative diagram

$$\begin{tikzcd} \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}} \arrow[r, dotted, "\phi^*"] \arrow[d, "\mathsf{oblv}^l"] & \mathcal{D}_Y\text{-}\mathsf{mod}^l_{\mathsf{qc}} \arrow[d, "\mathsf{oblv}^l"] \\ \mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc}} \arrow[r, "\phi^*"] & \mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}} \end{tikzcd}$$

as follows.

Let $\mathcal{M}$ be a left D-module on X. We equip

$$\phi^*(\mathcal{M}) := \mathcal{O}_Y \underset{\phi^{-1}\mathcal{O}_X}{\otimes} \phi^{-1}\mathcal{M}$$

with the left D-module structure according to the following Lebniz rule

$$\partial \cdot (f \otimes s) := \partial(f) \otimes s + f \mathsf{d}\phi(\partial) \cdot s,$$

where

- ∂ is a local section of $\mathcal{T}_Y$ and $\mathsf{d}\phi(\partial)$ is its image under the map
$$\mathsf{d}\phi : \mathcal{T}_Y \to \phi^*\mathcal{T}_X = \mathcal{O}_Y \underset{\phi^{-1}\mathcal{O}_X}{\otimes} \phi^{-1}\mathcal{T}_X;$$
- f is a local section of $\mathcal{O}_Y$;
- s is a local section of $\phi^{-1}\mathcal{M}$, and $\mathsf{d}\phi(\partial) \cdot s$ is defined using the action of $\phi^{-1}\mathcal{T}_X$ on $\phi^{-1}\mathcal{M}$.

[15]In these notes, we omit the decorations "L" and "R" from the notations for derived functors.

We call

$$\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}} \to \mathcal{D}_Y\text{-}\mathsf{mod}^l_{\mathsf{qc}}$$

the *(usual) pullback functor for left D-modules.*

Remark 6.4.2. It is easy to show that the pullback functors for left $\mathcal{D}$-modules are compatible with compositions.

Example 6.4.3. The pullback of the object $\mathcal{O}_X \in \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ is $\mathcal{O}_Y \in \mathcal{D}_Y\text{-}\mathsf{mod}^l_{\mathsf{qc}}$.

Construction 6.4.4. Note that the functor $\phi^* : \mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}} \to \mathcal{D}_Y\text{-}\mathsf{mod}^l_{\mathsf{qc}}$ is right exact. Let

$$\phi^* : \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}) \to \mathsf{D}(\mathcal{D}_Y\text{-}\mathsf{mod}^l_{\mathsf{qc}})$$

be the left derived functor of it. Note that we have a commutative diagram

$$\begin{tikzcd}
\mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}) \arrow[r, "\phi^*"] \arrow[d, "\mathsf{oblv}^l"] & \mathsf{D}(\mathcal{D}_Y\text{-}\mathsf{mod}^l_{\mathsf{qc}}) \arrow[d, "\mathsf{oblv}^l"] \\
\mathsf{D}(\mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc}}) \arrow[r, "\phi^*"] & \mathsf{D}(\mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}}).
\end{tikzcd}$$

Remark 6.4.5. We have:

- If ϕ is flat, then ϕ^* is t-exact, i.e., preserves the heart.
- If ϕ is a closed embedding, then ϕ^* has cohomological amplitude $[-\dim(X)+\dim(Y), 0]$.

Indeed, both claims follow from the corresponding claim for $\mathcal{O}$-modules.

Warning 6.4.6. In general, for a right $\mathcal{D}_X$-module $\mathcal{M}$, the $\mathcal{O}$-module pullback $\phi^*\mathcal{M}$ has no D-module structure.

Construction 6.4.7. Let $\phi^!$ be the unique functor that makes the following diagram commute

$$\begin{tikzcd}
\mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}}) \arrow[r, "\phi^*"] \arrow[d, "\omega_X"', "\simeq"] & \mathsf{D}(\mathcal{D}_Y\text{-}\mathsf{mod}^l_{\mathsf{qc}}) \arrow[d, "\simeq"', "\omega_Y"] \\
\mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}) \arrow[r, dotted, "\phi^!"] & \mathsf{D}(\mathcal{D}_Y\text{-}\mathsf{mod}^r_{\mathsf{qc}}),
\end{tikzcd}$$

We call it the *!-pullback functor for right D-modules.*

Remark 6.4.8. Note that for $\mathcal{M} \in D(\mathcal{D}_X\text{-}\mathsf{mod}^l_{\mathsf{qc}})$, the underlying $\mathcal{O}_Y$-module of $\phi^!\mathcal{M}$ is given by

$$\phi^!\mathcal{M} \simeq \phi^*(\mathcal{M} \underset{\mathcal{O}_X}{\otimes} \omega_X^{-1}) \underset{\mathcal{O}_Y}{\otimes} \omega_Y \simeq \phi^*(\mathcal{M}) \underset{\mathcal{O}_Y}{\otimes} \omega_{Y/X},$$

where $\omega_{Y/X}$ is the relative dualizing complex for the morphism $Y \to X$. In other words, the underlying $\mathcal{O}_Y$-module of $\phi^!\mathcal{M}$ is indeed the !-pullback of the $\mathcal{O}_X$-module[16] $\mathcal{M}$. In other words, we have a commutative diagram

$$\begin{tikzcd} \mathsf{D}(\mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}}) & \mathsf{D}(\mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc}}) \arrow[l, "\phi^!"'] \\ \mathsf{D}(\mathcal{D}_Y\text{-}\mathsf{mod}^r_{\mathsf{qc}}) \arrow[u, "\mathsf{oblv}^r"'] & \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}) \arrow[l, "\phi^!"] \arrow[u, "\mathsf{oblv}^r"'] \end{tikzcd}$$

Example 6.4.9. The pullback of the object $\omega_X \in \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}})$ is $\omega_Y \in \mathsf{D}(\mathcal{D}_Y\text{-}\mathsf{mod}^r_{\mathsf{qc}})$.

Example 6.4.10. Let $j : U \to X$ be an open immersion. Then the functor

$$j^! : \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}) \to \mathsf{D}(\mathcal{D}_U\text{-}\mathsf{mod}^r_{\mathsf{qc}})$$

is just the restriction functor along j. Indeed, this follows from the fact that $\omega_X|_U \simeq \omega_U$. In this case, we also write $j^! = j^*$.

Lemma 6.4.11. *Let $i : Y \to X$ be a closed embedding. The functor*

$$i^! : \mathsf{D}(\mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}}) \to \mathsf{D}(\mathcal{D}_Y\text{-}\mathsf{mod}^r_{\mathsf{qc}})$$

is isomorphic to the right derived functor of

$$i^{!,\heartsuit} := \mathsf{H}^0 \circ i^! : \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}} \to \mathcal{D}_Y\text{-}\mathsf{mod}^r_{\mathsf{qc}}.$$

Proof. See [HTT, Proposition 1.5.16]. □

Remark 6.4.12. The left-exact functor

$$i^{!,\heartsuit} : \mathcal{D}_X\text{-}\mathsf{mod}^r_{\mathsf{qc}} \to \mathcal{D}_Y\text{-}\mathsf{mod}^r_{\mathsf{qc}}$$

can be described as follows (see the proof of [HTT, Proposition 1.5.16]).

Let

$$\partial \in \Gamma(V, \mathcal{T}_Y),$$

be a local vector field on Y. We can extend it to a section $\widetilde{\partial} \in \Gamma(U, \mathcal{T}_X)$, where $U \subseteq X$ is some open subset such that $V = U \cap Y$. For a right $\mathcal{D}_X$-module $\mathcal{M}$, by definition, a section

$$m \in \Gamma(V, \phi^{!,\heartsuit}(\mathcal{M}))$$

is given by a section $\widetilde{m} \in \Gamma(U, \mathcal{M})$ such that $\mathcal{I}_Y(U) \cdot \widetilde{m} = 0$, where $\mathcal{I}_Y \subseteq \mathcal{O}_X$ is the ideal sheaf for $i : Y \to X$. Now the section

$$m \cdot \partial \in \Gamma(V, \phi^{!,\heartsuit}(\mathcal{M}))$$

is determined by the formula

$$\widetilde{m \cdot \partial} = \widetilde{m} \cdot \widetilde{\partial}.$$

Warning 6.4.13. For a general morphism $\phi : Y \to X$, the functor $\phi^!$ is not a derived functor.

---

[16]One can define a functor

$$\phi^! : \mathsf{D}(\mathcal{O}_X\text{-}\mathsf{mod}_{\mathsf{qc}}) \to \mathsf{D}(\mathcal{O}_Y\text{-}\mathsf{mod}_{\mathsf{qc}})$$

for any finitely presented morphism $f : Y \to X$ between schemes such that:

- If ϕ is proper, the functor $\phi^!$ is right adjoint to the direct image functor ϕ_*;
- If ϕ is an open immersion, the functor $\phi^!$ is isomorphic to the left adjoint ϕ^* of ϕ_*.

Remark 6.4.14. The functor $\phi^!$ has cohomological amplitude $[-\dim(Y), \dim(X)-\dim(Y)]$. Moreover,

- If ϕ is flat, then $\phi^![\dim(X)-\dim(Y)]$ is t-exact.
- If ϕ is a closed embedding, then $\phi^!$ has cohomological amplitude $[0, \dim(X)-\dim(Y)]$.

Exercise 6.4.15 (Problem 1, Homework 10)**.** Let $\Delta : X \to X \times X$ be the diagonal morphism.

(1) For $\mathcal{M}, \mathcal{N} \in \mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^l_{\mathsf{qc}})$, show that

$$\mathcal{M} \underset{\mathcal{O}_X}{\otimes} \mathcal{N} \simeq \Delta^*(\mathcal{M} \boxtimes \mathcal{N})$$

as objects in $\mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^l_{\mathsf{qc}})$.

(2) For $\mathcal{M}, \mathcal{N} \in \mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}})$, show that

$$\mathcal{M} \overset{!}{\otimes} \mathcal{N} \simeq \Delta^!(\mathcal{M} \boxtimes \mathcal{N})$$

as objects in $\mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}})$.

6.5. **Pushforward.** Let $\phi : Y \to X$ be a map between smooth schemes.

Construction 6.5.1. We write:

$$\mathcal{D}_{Y\to X} := \phi^*\mathcal{D}_X \simeq \mathcal{O}_Y \underset{\phi^{-1}\mathcal{O}_X}{\otimes} \phi^{-1}\mathcal{D}_X$$

and call it the *transfer D-module* for the morphism ϕ.

Construction 6.4.1 gives a left $\mathcal{D}_Y$-module structure on $\mathcal{D}_{Y\to X}$. On the other hand, there is an obvious *right* $\phi^{-1}\mathcal{D}_X$-module structure on $\mathcal{D}_{Y\to X}$. One can show there two actions commute. In other words, $\mathcal{D}_{Y\to X}$ is a $(\mathcal{D}_Y, \phi^{-1}\mathcal{D}_X)$-bimodule.

Remark 6.5.2. Note that for $\mathcal{M} \in \mathcal{D}_X\text{–}\mathsf{mod}^l_{\mathsf{qc}}$, we have

$$\phi^*\mathcal{M} \simeq \mathcal{D}_{Y\to X} \underset{\phi^{-1}\mathcal{D}_X}{\otimes} \phi^{-1}\mathcal{M}.$$

Example 6.5.3. Let ϕ be a closed embedding. Since X and Y are smooth, ϕ is a regular immersion. For any closed point $p \in Y$, we can find an étale coordinate system $x_1, \cdots, x_m$ of X near p such that Y is locally cut out by the ideal $(x_{n+1}, \cdots, x_m)$. Let $y_1, \cdots, y_n$ be the restriction of $x_1, \cdots, x_n$ to Y. They form an étale coordinate system of Y near p. Then near the point $p \in Y$, we have

$$\mathcal{D}_{Y\to X} \simeq \mathcal{D}_Y \underset{k}{\otimes} k[\partial_{n+1}, \cdots, \partial_m]$$

as left $\mathcal{D}_Y$-modules. In particular, $\mathcal{D}_{Y\to X}$ is a locally free left $\mathcal{D}_Y$-module.

Construction 6.5.4. We define a functor

$$\phi_{*,\mathsf{dR}} : D(\mathcal{D}_Y\text{–}\mathsf{mod}^r_{\mathsf{qc}}) \to D(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}})$$

by the formula

$$\phi_{*,\mathsf{dR}}(\mathcal{N}) := \phi_*(\mathcal{N} \underset{\mathcal{D}_Y}{\otimes} \mathcal{D}_{Y\to X}),$$

where

- the (left derived) tensor product functor $\mathcal{N} \otimes_{\mathcal{D}_Y} \mathcal{D}_{Y\to X}$ is equipped with the right $\phi^{-1}\mathcal{D}_X$-module structure inherited from $\mathcal{D}_{Y\to X}$;
- the (right derived) direct image $\phi_*(\mathcal{N} \otimes_{\mathcal{D}_Y} \mathcal{D}_{Y\to X})$ is equipped with the right $\mathcal{D}_X$-module structure given by the homomorphism $\mathcal{D}_X \to \phi_*(\phi^{-1}\mathcal{D}_X)$.

We call $\phi_{*,\mathsf{dR}}$ the *de Rham pushforward functor* for right D-modules.

Let $\pi : X \to \mathsf{pt}$ be the projection morphism. We also write

$$\Gamma_{\mathsf{dR}}(X,-) := \pi_{*,\mathsf{dR}}(-).$$

The following result justifies the name *de Rham* pushforward.

Exercise 6.5.5 (Problem 2, Homework 10)**.** Let X be a smooth scheme and $\pi : X \to \mathsf{pt}$ be the projection morphism. Let $\mathcal{M} \in \mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}})$.

(1) Show that $\pi_{*,\mathsf{dR}}(\mathcal{M}) \simeq \pi_*(\mathcal{M} \otimes_{\mathcal{D}_X} \mathcal{O}_X)$.
(2) Construct a resolution of $\mathcal{O}_X$ by locally free left $\mathcal{D}_X$-modules:

$$0 \to \mathcal{D}_X \underset{\mathcal{O}_X}{\otimes} \wedge^{\dim(X)}\mathcal{T}_X \to \cdots \to \mathcal{D}_X \underset{\mathcal{O}_X}{\otimes} \wedge^2\mathcal{T}_X \to \mathcal{D}_X \underset{\mathcal{O}_X}{\otimes} \mathcal{T}_X \to \mathcal{D}_X \to \mathcal{O}_X \to 0.$$

Deduce that the cohomologies of $\pi_{*,\mathsf{dR}}(\mathcal{M})$ can be calculated as the hypercohomologies of the complex

$$0 \to \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \wedge^{\dim(X)}\mathcal{T}_X \to \cdots \mathcal{M} \underset{\mathcal{O}_X}{\otimes} \mathcal{T}_X \to \mathcal{M} \to 0. \tag{6.1}$$

(3) For $\mathcal{M} = \mathcal{K}_X$, identify (6.1) with the de Rham complex

$$0 \to \mathcal{O}_X \xrightarrow{\mathsf{d}} \Omega^1_X \xrightarrow{\mathsf{d}} \cdots \xrightarrow{\mathsf{d}} \Omega^{\dim(X)}_X \to 0.$$

Deduce that

$$\mathsf{H}^i(\pi_{*,\mathsf{dR}}(\mathcal{K}_X)) \simeq \mathsf{H}^{i-\dim(X)}_{\mathsf{dR}}(X)$$

Remark 6.5.6. Using the resolution in Exercise 6.5.5(2), one can show that the functor $\phi_{*,\mathsf{dR}}$ has cohomological amplitude $[-\dim(Y), \dim(Y)]$. Moreover,

- If ϕ is affine, then $\phi_{*,\mathsf{dR}}$ has cohomological amplitude $[-\dim(Y), 0]$;
- If ϕ is smooth, then $\phi_{*,\mathsf{dR}}$ has cohomological amplitude $[\dim(X)-\dim(Y), \dim(Y)]$.

Example 6.5.7. Let $j : U \to X$ be an open immersion. We have $\mathcal{D}_{U\to X} \simeq j^*\mathcal{D}_X \simeq \mathcal{D}_U$. It follows that $j_{*,\mathsf{dR}}\mathcal{M} \simeq j_*\mathcal{M}$.

Example 6.5.8. Let $x \in X$ be a closed point and $i : x \to X$ be the closed immersion. We have $i_{*,\mathsf{dR}}(\mathbb{C}) \simeq \delta_x$, where δ_x is the right δ-D-module defined in Exercise 4.4.3.

Lemma 6.5.9. *Let $Z \xrightarrow{\varphi} Y \xrightarrow{\phi} X$ be a chain of smooth schemes. We have a canonical isomorphism*

$$\phi_{*,\mathsf{dR}} \circ \varphi_{*,\mathsf{dR}} \simeq (\phi \circ \varphi)_{*,\mathsf{dR}}.$$

Proof. For $\mathcal{N} \in \mathsf{D}(\mathcal{D}_Z\text{–}\mathsf{mod}^r_{\mathsf{qc}})$, we have

$$\begin{aligned}
\phi_{*,\mathsf{dR}} \circ \varphi_{*,\mathsf{dR}}(\mathcal{N}) &\simeq \phi_*[\varphi_*(\mathcal{N} \underset{\mathcal{D}_Z}{\otimes} \mathcal{D}_{Z\to Y}) \underset{\mathcal{D}_Y}{\otimes} \mathcal{D}_{Y\to X}] \\
&\simeq \phi_* \circ \varphi_*(\mathcal{N} \underset{\mathcal{D}_Z}{\otimes} \mathcal{D}_{Z\to Y} \underset{\varphi^{-1}\mathcal{D}_Y}{\otimes} \varphi^{-1}\mathcal{D}_{Y\to X}) \\
&\simeq (\phi \circ \varphi)_*(\mathcal{N} \underset{\mathcal{D}_Z}{\otimes} \mathcal{D}_{Z\to X}) \\
&\simeq (\phi \circ \varphi)_{*,\mathsf{dR}}(\mathcal{N}),
\end{aligned}$$

where the second isomorphism is due to the projection formula for (ϕ^{-1}, ϕ_*), while the third isomorphism is due to the following exericse.

Exercise 6.5.10 (Problem 3, Homework 10)**.** Construct a canonical isomorphism

$$\mathcal{D}_{Z\to Y} \underset{\varphi^{-1}\mathcal{D}_Y}{\otimes} \varphi^{-1}\mathcal{D}_{Y\to X} \simeq \mathcal{D}_{Z\to X}$$

between $(\mathcal{D}_Z, (\phi\circ\varphi)^{-1}(\mathcal{D}_X))$-bimodules.

□[Proof of Lemma 6.5.9]

We now study the interaction between the de Rham pushforward functor and other operations on D-modules.

Exercise 6.5.11 (Problem 4, Homework 10)**.** Construct a commutative diagram

$$\begin{tikzcd} \mathsf{D}(\mathcal{O}_Y\text{–}\mathsf{mod}_{\mathsf{qc}}) \arrow[r, "\phi_*"] \arrow[d, "\mathsf{Ind}^r"'] & \mathsf{D}(\mathcal{O}_X\text{–}\mathsf{mod}_{\mathsf{qc}}) \arrow[d, "\mathsf{Ind}^r"'] \\ \mathsf{D}(\mathcal{D}_Y\text{–}\mathsf{mod}^r_{\mathsf{qc}}) \arrow[r, "\phi_{*,\mathsf{dR}}"] & \mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}}). \end{tikzcd}$$

Proposition 6.5.12. *Suppose that $\phi: Y\to X$ is a proper morphism. Then $\phi_{*,\mathsf{dR}}$ is left adjoint to $\phi^!$.*

Proof. See [HTT, Corollary 2.7.3].

□

Remark 6.5.13. The proof of Proposition 6.5.12 uses the Verdier duality for D-modules. In this remark, we sketch a direct proof for Proposition 6.5.12.

For $\mathcal{M}\in\mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}})$ and $\mathcal{N}\in\mathsf{D}(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}})$, we have

$$\mathsf{Hom}_{\mathcal{D}_X^r}(\phi_{*,\mathsf{dR}}(\mathcal{N}),\mathcal{M}) \quad\simeq\quad \mathsf{Hom}_{\mathcal{D}_X^r}(\underset{[n]\in\Delta^{\mathsf{op}}}{\mathsf{colim}}\,\phi_{*,\mathsf{dR}}(\mathcal{N}\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_Y^n),\mathcal{M}),$$

where

$$\mathcal{D}_Y^n := \mathcal{D}_Y \underset{\mathcal{O}_Y}{\otimes}\cdots\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_Y$$

is the tensor product of n-copies of $\mathcal{D}_Y$. It follows that

$$\begin{aligned}
\mathsf{Hom}_{\mathcal{D}_X^r}(\phi_{*,\mathsf{dR}}(\mathcal{N}),\mathcal{M}) &\simeq \lim_{[n]\in\Delta}\mathsf{Hom}_{\mathcal{D}_X^r}(\phi_*(\mathcal{N}\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_Y^{n-1}\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_{Y\to X}),\mathcal{M})\\
&\simeq \lim_{[n]\in\Delta}\mathsf{Hom}_{\mathcal{D}_X^r}(\phi_*(\mathcal{N}\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_Y^{n-1})\underset{\mathcal{O}_X}{\otimes}\mathcal{D}_X),\mathcal{M})\\
&\simeq \lim_{[n]\in\Delta}\mathsf{Hom}_{\mathcal{O}_X}(\phi_*(\mathcal{N}\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_Y^{n-1})),\mathcal{M})\\
&\simeq \lim_{[n]\in\Delta}\mathsf{Hom}_{\mathcal{O}_Y}(\mathcal{N}\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_Y^{n-1},\phi^!\mathcal{M}),
\end{aligned}$$

where the second isomorphism is given by the projection formula for (ϕ^*,ϕ_*), and the fourth isomorphism is given by the adjunction $(\phi_*,\phi^!)$ for $\mathcal{O}$-modules. Unwinding the definitions[17], the cosimplicial diagram in the RHS can be identified with the cosimplicial diagram in

$$\mathsf{Hom}_{\mathcal{D}_Y}(\mathcal{N},\phi^!\mathcal{M})\simeq\lim_{[n]\in\Delta}\mathsf{Hom}_{\mathcal{O}_Y}(\mathcal{N}\underset{\mathcal{O}_Y}{\otimes}\mathcal{D}_Y^{n-1},\phi^!\mathcal{M}.)$$

This establishes the desired adjunction $(\phi_{*,\mathsf{dR}},\phi^!)$.

[17]This is highly nontrivial!

Construction 6.5.14. Let $\phi^l_{*,\mathsf{dR}}$ be the unique functor that makes the following diagram commute

$$\begin{tikzcd} D(\mathcal{D}_Y\text{–}\mathsf{mod}^l_{\mathsf{qc}}) \arrow[r, dotted, "\phi^l_{*,\mathsf{dR}}"] \arrow[d, "\omega_Y"', "\simeq"] & D(\mathcal{D}_X\text{–}\mathsf{mod}^l_{\mathsf{qc}}) \arrow[d, "\simeq"', "\omega_X"] \\ D(\mathcal{D}_Y\text{–}\mathsf{mod}^r_{\mathsf{qc}}) \arrow[r, "\phi_{*,\mathsf{dR}}"] & D(\mathcal{D}_X\text{–}\mathsf{mod}^r_{\mathsf{qc}}). \end{tikzcd}$$

We call it the *de Rham pushforward functor for left D-modules.*

Exercise 6.5.15 (Problem 5, Homework 10)**.** Define $\mathcal{D}_{X\leftarrow Y} \coloneqq \phi^!\mathcal{D}_X$, where we view $\mathcal{D}_X$ as a right module over itself.

(1) Show that $\mathcal{D}_{X\leftarrow Y}$ is naturally a $(\phi^{-1}\mathcal{D}_X, \mathcal{D}_Y)$-bimodule.
(2) For $\mathcal{N} \in \mathsf{D}(\mathcal{D}_Y\text{–}\mathsf{mod}^l_{\mathsf{qc}})$, show that

$$\phi^l_{*,\mathsf{dR}}(\mathcal{N}) \simeq \phi_*(\mathcal{D}_{X\leftarrow Y} \underset{\mathcal{D}_Y}{\otimes} \mathcal{N})$$

as left $\mathcal{D}_X$-modules, where $\mathcal{D}_X$ acts on the RHS via the homomorphism $\mathcal{D}_X \to \phi_*\phi^{-1}\mathcal{D}_X$.

Lemma 6.5.16 (Duality between transfer modules)**.** *Suppose that $\phi : Y \to X$ is a smooth morphism. Then*

$$\mathsf{Hom}_{\mathcal{D}_Y^r}(\mathcal{D}_{X\leftarrow Y}[2(\dim(X) - \dim(Y))], \mathcal{N}) \simeq \mathcal{N} \underset{\mathcal{D}_Y}{\otimes} \mathcal{D}_{Y\to X} \tag{6.2}$$

as right $\phi^{-1}\mathcal{D}_X$-modules.

Sketch. The left $\mathcal{D}_Y$-module $\mathcal{D}_{Y\to X}$ can be calculated by a complex

$$\mathcal{D}_Y \underset{\mathcal{O}_Y}{\otimes} \wedge^{\dim(Y)}\mathcal{T}_{Y/X} \to \cdots \to \mathcal{D}_Y \underset{\mathcal{O}_Y}{\otimes} \mathcal{T}_{Y/X} \to \mathcal{D}_Y,$$

where we put $\mathcal{D}_Y$ in degree 0. The right $\mathcal{D}_Y$-module $\mathcal{D}_{X\leftarrow Y}[2(\dim(X) - \dim(Y))]$ can be calculated by a complex

$$\mathcal{D}_Y \to \Omega^1_{Y/X} \underset{\mathcal{O}_Y}{\otimes} \mathcal{D}_Y \to \cdots \to \Omega^{\dim(Y)-\dim(X)}_{Y/X} \underset{\mathcal{O}_Y}{\otimes} \mathcal{D}_Y,$$

where we put $\mathcal{D}_Y$ in degree 0. These provide an isomorphism (6.2) between abelian sheaves. A careful calculation shows that this isomorphism is compatible with the right $\phi^{-1}\mathcal{D}_X$ actions.

□

Proposition 6.5.17. *Suppose that $\phi : Y \to X$ is a smooth morphism. Then $\phi^![2(\dim(X) - \dim(Y))]$ is left adjoint to $\phi_{*,\mathsf{dR}}$.*

Exercise 6.5.18 (Problem 6, Homework 10)**.** Deduce Proposition 6.5.17 from Lemma 6.5.16.

Theorem 6.5.19 (Kashiwara's lemma)**.** *Let $i : Y \to X$ be a closed immersion between smooth schemes. Then the functor $i_{*,\mathsf{dR}}$ is fully faithful and t-exact. Moreover, its image contains exactly those right $\mathcal{D}_X$-modules that are set-theoretically supported on Y.*

Proof. See [HTT, Theorem 1.6.1]. □

One way to prove Kashiwara's lemma is to reduce to the case when the closed immersion is $0 \to \mathbb{A}^n$, which can be proved by the following direct calculation.

Exercise 6.5.20 (Problem 7, Homework 10)**.** Let $i : 0 \to \mathbb{A}^n$ be the obvious closed immersion. Show that $i^! \delta_0 \simeq \mathbb{C}$.

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